

TPN System Installation

SW20-600

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**Release Independent
SW20-600
March 2013**

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Revision 4.0 – March 2013

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About This Document

Abstract

This document provides instructions for use by the system hardware installation and checkout personnel.

References

The following Honeywell documents may be sources of reference for the material discussed in this document.

Document Title	Binder
<i>TPS System Site Planning</i>	System Site Planning
<i>LCN System Planning</i>	LCN System Installation
<i>LCN System Checkout</i>	LCN System Installation

Attention

With TPN R684, Enhanced Universal Control Network (EUCN) is introduced. In EUCN, the traditional coaxial cable system is no longer utilized and is replaced with a Fault Tolerant Ethernet (FTE) based communication network.

Throughout the document, unless specifically mentioned, the terms HPM, NIM, and UCN are interchangeably used with the terms EHPM, ENIM, and EUCN respectively.

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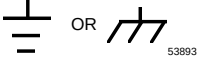
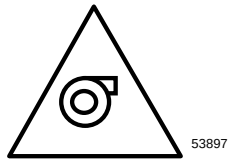
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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
ATTENTION	Notes inform the reader about information that is required, but not immediately evident.
CAUTION	Cautions tell the user that damage may occur to equipment if proper care is not exercised.
Warning	Warnings tell the reader that potent personal harm or serious economic loss may happen if instructions are not followed.
TIP	Identifies advice or hints for the user, often in terms of performing a task.
	Ground connection to building safety ground
	Ground Stake for building safety ground
	Electrical Shock Hazard—can be lethal
	Electrical Shock Hazard—can be lethal
	Rotating Fan—can cause personal injury

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1. Introduction

1.1 Overview

Scope

The *TPN System Installation* manual provides instructions for installing TPS TPN System equipment cabinets, Universal Stations, Scanner Application Module, and Universal Work Stations at the customer site. This manual, along with the TPS equipment installation publications, listed below, are required during the initial system installation and for modifying or expanding a TPS system that is already installed. The installation procedures assume that all intra-cabinet/console cabling and wiring have been made at the factory before the equipment is shipped.

The installation manual includes general instructions on how to move and position the system equipment, how to connect the power and grounding cables to the equipment, and how to install the Local Control Network (LCN) and Data Hiway (DH) cables that link the functional modules to the system. A successful power-up check concludes the installation phase and signals the beginning of the system-checkout phase.

Included in this publication is the cabling information needed to install cables to equipment located outside the console and cabinets.

1.2 Special Installation Tools and Test Equipment

Required

- Mobil lifter and/or Lift truck
- Set of standard service tools, including a multi-meter

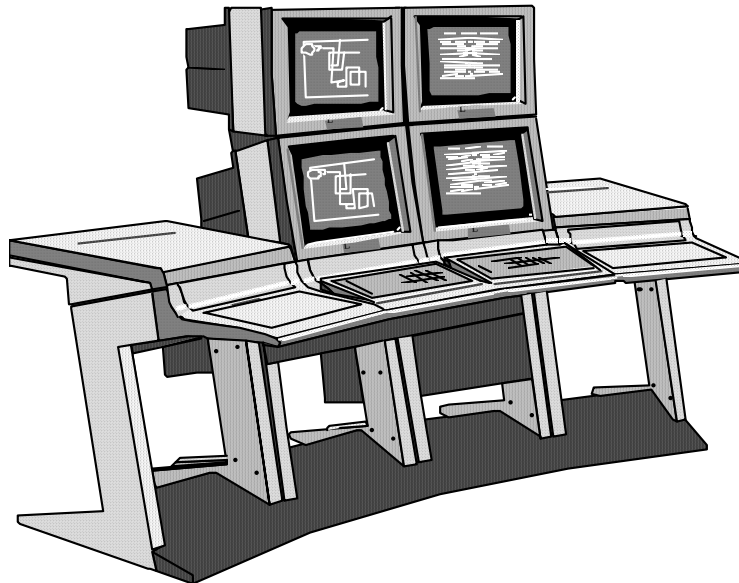
2. System Installation

2.1 Installation Overview

Scope

This section provides instructions for receiving, unpacking, and placement of the TPS LCN equipment cabinets, Global User Stations (GUSs), APP Nodes, and Universal Stations (USs); see Figure 2-1 and Figure 2-2. The instructions assume that all of the system LCN modules and all interconnecting LCN cables within the cabinets and consoles have been installed and connected at the factory as shown in Figure 2-3.

Z-Console furniture



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Figure 2-1 Z-Console Furniture

Classic Console furniture

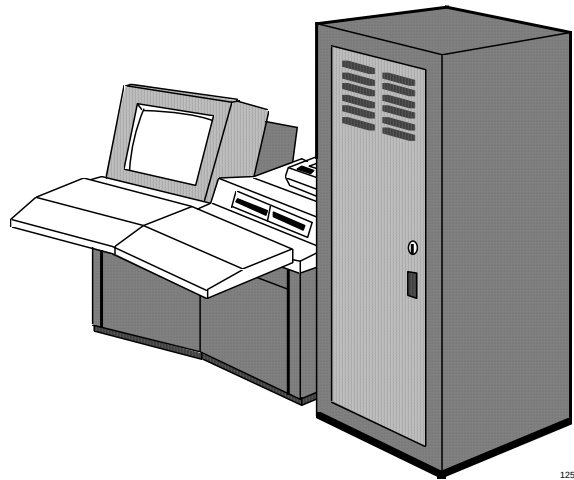


Figure 2-2 Classic Console and Utility Cabinet

Pre-wired cabinet

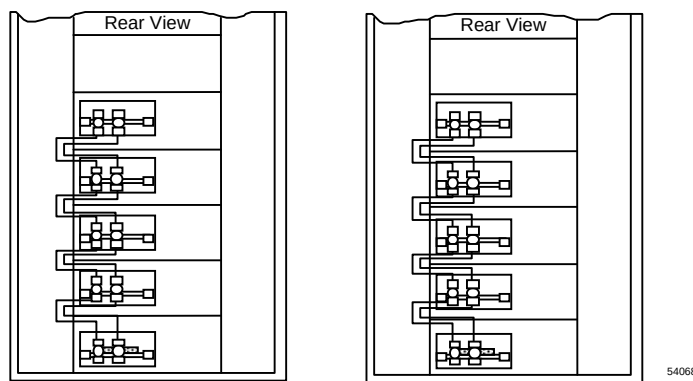


Figure 2-3 Utility Cabinets Pre-wired With Coax Cable

All LCN coaxial-cable assemblies required to interconnect cabinets and consoles are assembled at the factory to standard or specified lengths and shipped with the system equipment (see Section 3).

The desk-side Universal Work Station is protectively wrapped and shipped in a double-wall pack. Its peripherals are packaged separately.

2.2 Preliminary Site Inspection

Access

Inspect the area in which the equipment is to be installed—refer to the "Preparedness Checklist" in Section 6 of the *LCN Site Planning* manual. Check the route from the delivery point to the final destination for adequate space through hallways, doors, and around corners. If, for example, console stations cannot negotiate a narrow hallway or tight corner, it may be necessary to remove the keyboard/panel or table-top assembly before it can be moved into place. In addition, the trend-pen recorder option, mounted above the CRT shroud, adds approximately two inches to the overall width (front-to-back).

LCN equipment cabinets and the Universal Work Station do not have leveling pads and are designed to be installed on a level floor. Make sure the floor is level before moving the equipment into place. Note that LCN equipment cabinets are not designed for eyebolt lifting. If cabling is to be run under the false floor, do so before the cabinet is moved into place. Cabinets can be bolted to the floor, if required. Holes are provided in each corner of the base.

2.3 Storage

Attention

If LCN equipment cabinets, Universal Stations, Global User Stations, Universal Work Stations, and peripherals are to be placed in storage, the environmental constraints specified below must be followed, with particular emphasis on relative humidity and airborne contaminants.

Specification

Table 2-1 Environmental Conditions for Storage and Shipping

Environmental Condition	Specifications
Temperature	-35°C to +70° C
Humidity (R.H.) ¹ Maximum Wet Bulb	5% to 95% 32°
Vibration	1.0g
Shock ²	3.9 g
Altitude	35,000 Feet
Notes: 1. The temperature/relative humidity shall not be cycled such that moisture condensation occurs on the equipment. The rate of change shall be less than 6% per hour. 2. The total storage and shipping requirements are for one-year duration only, provided the equipment is properly packaged and contains an adequate amount of desiccant (moisture absorbing agent).	

3. Equipment Unpacking and Placement

3.1 Overview

Scope

This section provides instructions on how to unpack and move the equipment into place. In addition, LCN cable lengths and routing considerations are presented.

3.2 Equipment Unpacking

Following factory assembly, inspection, and test, the configured cabinets and consoles are protectively wrapped for domestic shipment, or wrapped and crated for overseas shipment. The LCN equipment cabinets are strapped to a wooden shipping skid or pallet that is blocked on all sides. The Universal Station (US), Universal Station^x, and Global User Station are bolted to a shipping skid and wrapped or crated in a similar manner as the cabinets. The USs can be optionally equipped with castors or leveler glides. Castors are available for use after the skid is removed.

All intra-cabinet/console cables and tabletop printers are boxed separately.

Attention

STATICIDE work area before unpacking equipment. Use anti-static spray on floors, chairs, workbench, mats, tools, etc., in the work area. This inhibits static generation that can damage electronic equipment.

Upon arrival at the system site, remove shipping crate and protective wrapping and carefully inspect the equipment for any physical damage. If damaged, immediately notify the carrier and your Honeywell sales representative as to extent and type of damage. Also, check each piece of equipment against the invoice list for any missing items.

3.3 Equipment Placement

The cabinets and consoles can be carefully moved into position, by use of lift trucks or mobile lifters. The Universal Stations can be rolled on their casters. Position the equipment according to the system-layout plan (refer to *LCN Site Planning* manual or *System Site Planning* manual). Verify that door openings allow access to LCN modules and controller card files for maintenance and checkout.

The shipping skid can be removed from the cabinet by first removing the wooden block on one side of the skid and then gently sliding the cabinet off the skid and onto the floor. Do not tilt the cabinet beyond 45° as this may cause damage to the doors and side skins. To remove the skid from the Universal Station, unbolt the wooden cleats that span the bottom-entry cutouts and slide the Universal Station off the skid as with the cabinet.

WARNING

Do not lift Z-Console stations by the keyboard tray.
--

3.4 Equipment Preparation

WARNING

The following safety procedures must be followed:

- | |
|---|
| <ol style="list-style-type: none">1. The power switch or breaker(s) on the system power control panel must be in the OFF position.2. All cabinet and console power entry (PE box) circuit breakers must be in the OFF position.3. If an optional cabinet is installed, its circuit breaker must be in the OFF position. |
|---|

- Set the LCN equipment cabinets in place according to the site configuration-layout plan. Do not exceed LCN cable-length limits; see Section 3.11, “LCN Cabling Rules” and Section 3.13, “Enhanced LCN Extender (XLCNE2 Cabling)”. Side skins are not removable; however, the front and rear doors can be removed. Cabinets can be placed side-to-side, as if complexed.
- Set the Universal Station in place according to the site configuration-layout plan. Do not exceed LCN cable-length limits; see Section 3.11, “LCN Cabling Rules” and Section 3.13, “Enhanced LCN Extender (XLCNE2 Cabling)”. Universal Station console bays can be complexed; see Sections 3.5 thru 3.8.
- Set the Universal Work Station (UWS) adjacent to a desk that holds the peripheral CRT, keyboard, mouse and optional printer. The UWS electronics package and each of its peripherals are equipped with 6-foot ac power cords. Provide two duplex outlets nearby for their connection. Do not exceed LCN cable length limits.

- Set the deskside Global User Station components on a tabletop or desktop. The electronics tower can be placed on the floor under the table or desk, directly under the Monitor and keyboard.

Attention

If the Universal Station (US) or Global user Station is located near an outside window, use shades or drapes to ensure that direct sunlight does not fall on the US CRT touch screen components. Direct sunlight can cause the touch screen (photocells) to activate and the beeper to sound. Be especially aware of early morning and late evening low sun positions relative to the CRT touch screen position.

- For Data Hiway coaxial cable-length limits and polled device-addressing constraints refer to Section 4 of the *Data Hiway Subsystem Site Planning* manual or to the *System Site Planning* manual.
- Open the front and rear door of the cabinet/console to gain access to the power-entry box and the cable-entry cutout area.
- The Z-Console station power connection is located under the footrest cover.
 - Remove the inside covers from each Z-Console leg.
 - Loosen the three bolts along the top rear edge.
 - Slide the footrest cover forward

3.5 Console Complexing—Classic Furniture

Attention

Ensure that all cable access ports located at the front bottom right and left sides are open before complexing. See Figure 3-1.

When the task of complexing the console is finished, close the cable access port on each end of the console.

Illustration

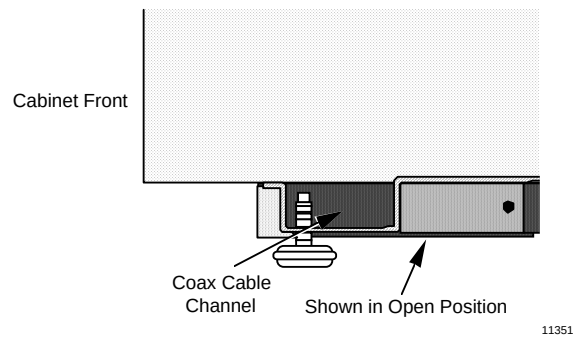


Figure 3-1 Location of Cable Access Ports in Classic Furniture

When Universal Stations (USs) are to be complexed together to form a console, use complexing kit P-ZCKY1. The number of kits required for a console is one less than the number of USs in the console. A complexing plate, 51303672, mounts on the bottom of the keyboard, connecting the two keyboards together. See Figure 3-2.

Illustration

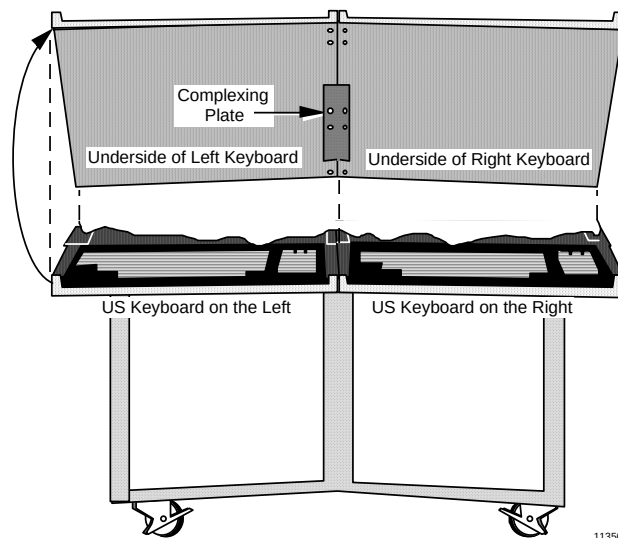
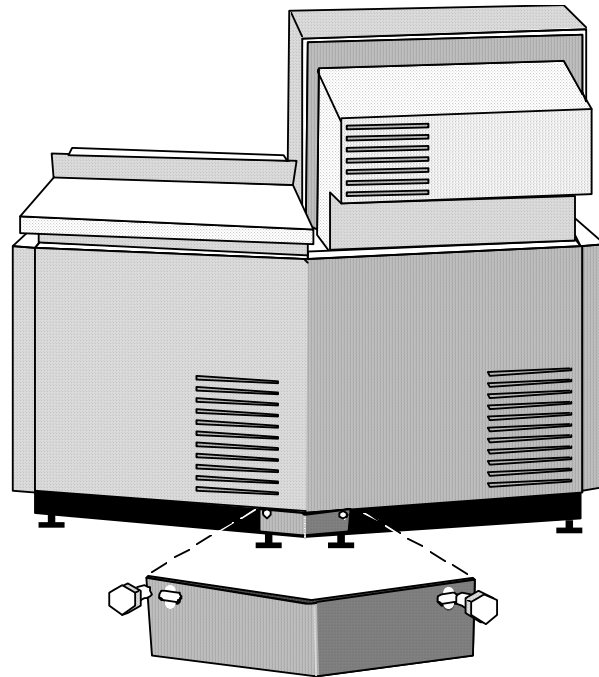


Figure 3-2 Installation of Complexing Plate under Keyboard

An angled strap, 51201261, fastens the back of the two USs together. See Figure 3-3.

Illustration



11352

Figure 3-3 Installation of the Complexing to Lower Rear Corners

3.6 Console Complexing—Z-Furniture

Scope

Z-Console furniture can be complexed in two forms.

- A semicircular arc, where each station is arced 10° from the adjacent station, is formed by bolting the stations together.
- An inline console where wedges are added in between each station to close the gaps left between the front surfaces.

3.7 Complexing Stations for an Arced Configuration

Procedure

The following steps are required for the joining of two stations together. The complexing tie plates (2) are attached to one station, the second station is set in place next to the first, the flexible spacers are inserted, and then the second station is bolted to the two tie plates.

Table 3-1 Complexing Stations for an Arced Configuration

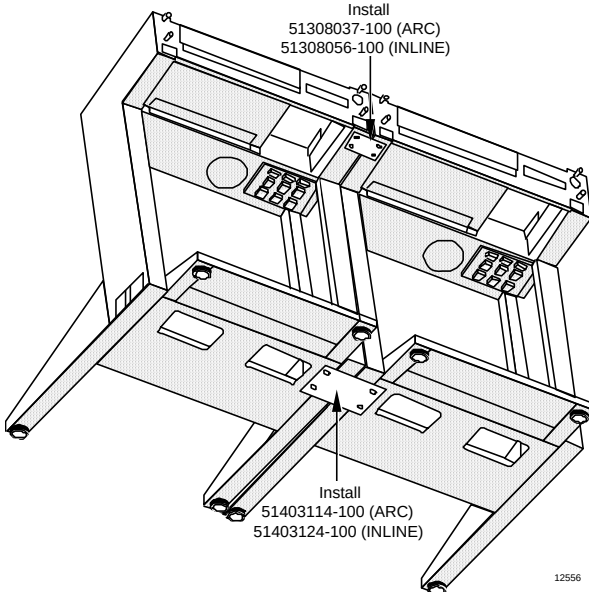
Step	Action
1	Install a Wedge Tie Plate Arc, 51403114-100, to the front of the bottom corner of one of the two stations bases. 
2	Install a Tie Plate Arc, 51308037-100, under the keyboard of the same station.
3	Place the second station next to the first station.
4	Insert a Flexible Spacer, Bottom Arc, 51308033-100, between the two lower monitors as shown in Figure 3-5. The spacer fits in the groove behind the bezel, on both stations, with the top even with the middle of the CRT Mount (wedge.)

Table 3-1 Complexing Stations for an Arced Configuration

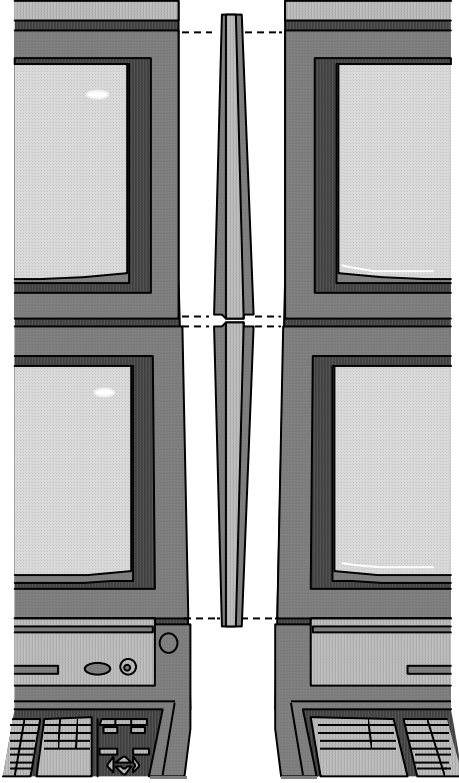
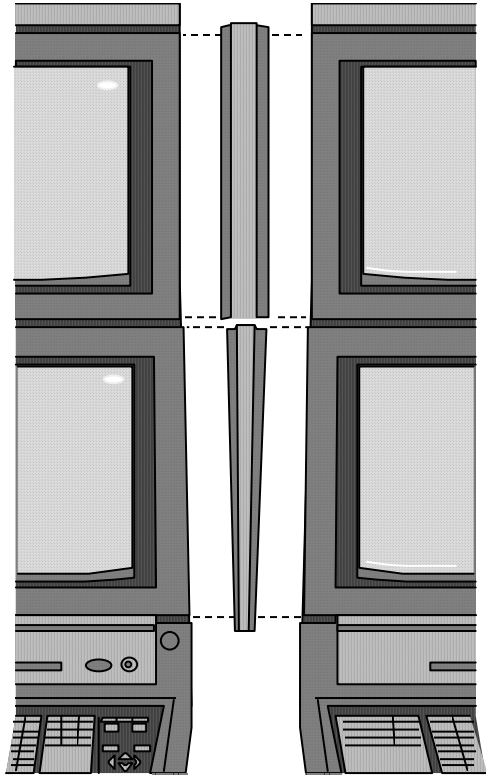
Step	Action
5	Upper Monitors: If there is an upper monitor in both stations, and they are tilted forward, insert a Flexible Spacer, Bottom Arc, 51308033-100, between the two upper monitors. The spacer fits in the groove behind the bezel.  12557 The top insert must be tight to the insert below it. Trim the excess upper end of the Flexible Spacer.

Table 3-1 Complexing Stations for an Arced Configuration

Step	Action
6	If there is an upper monitor in both stations, and they are not tilted forward, insert a Flexible Spacer, Top Arc, 51308036-100, between the two upper monitors. The spacer fits in the groove behind the bezel.  The top insert must be tight to the insert below it. Trim the excess upper end of the Flexible Spacer.
7	Bolt the second station to both complexing tie plates.

3.8 Complexing Stations for an Inline Configuration

Procedure

The following steps are required for the joining of two stations together. The complexing tie plates (2) are attached to one station, the second station is set in place next to the first, the flexible spacers are inserted, and then the second station is bolted to the two tie plates.

Table 3-2 Complexing Stations for an Inline Configuration

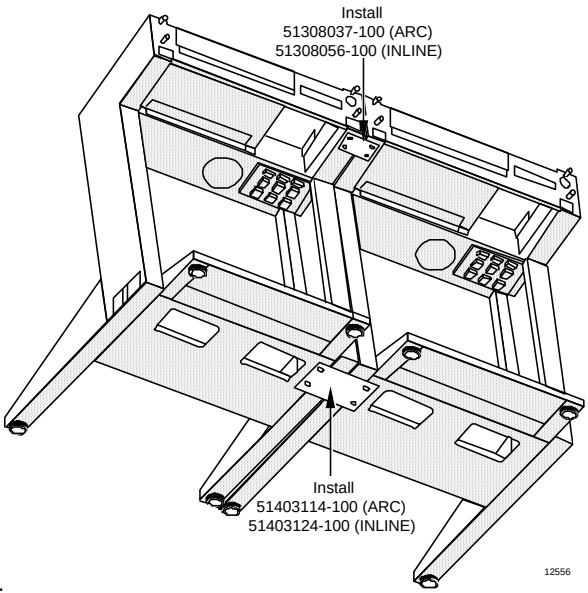
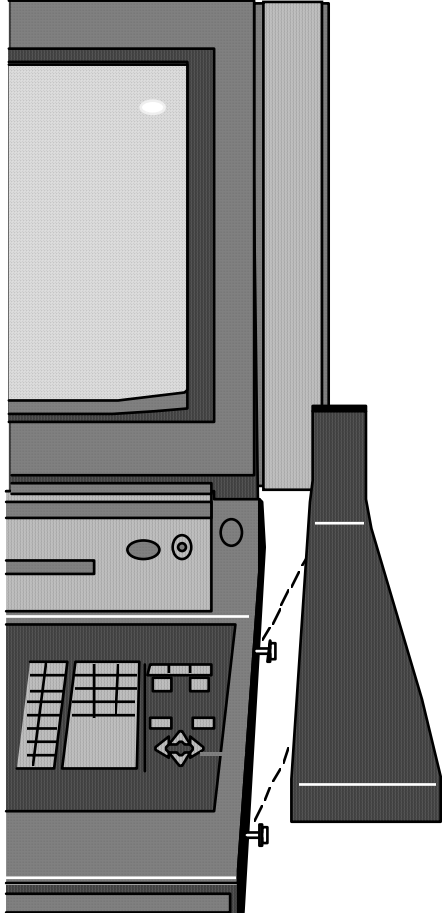
Step	Action
1	Install a Wedge Tie Plate Inline, 51403124-100, to the front of the bottom corner of one of the two stations  bases.
2	Install a Tie Plate Inline, 51308056-100, under the keyboard of the same station. See the above.
3	Place the second station next to the first station.
4	Insert the two long bolts to the keyboards of each station. Start a nut on each bolt.
5	Insert the Inline Wedge Assembly, 51202034-100, between the keyboards and tighten the nuts.

Table 3-2 Complexing Stations for an Inline Configuration

Step	Action
6	Insert a Flexible Spacer, Bottom Inline, 51308038-100, between the two lower monitors.  12559 The spacer fits in the groove behind the bezel, on both stations, with the top even with the middle of the CRT Mount (wedge.)
7	If there is an upper monitor in both stations, and they are tilted forward, insert a Flexible Spacer, Bottom Inline, 51308038-100, between the two upper monitors.
8	Bolt the second station to both complexing tie plates.

3.9 Installing the Cable Access Port Covers

Installation

Install a Cable Cover, 51202027, in the outside leg of the far right and far left station. Insert the Cable Cover plate from the inside of the leg, over the mounting stud, and retain with a nut. The protruding part of the plate will fill the hole, such that the outside of the cabinet leg is smooth.

Illustration

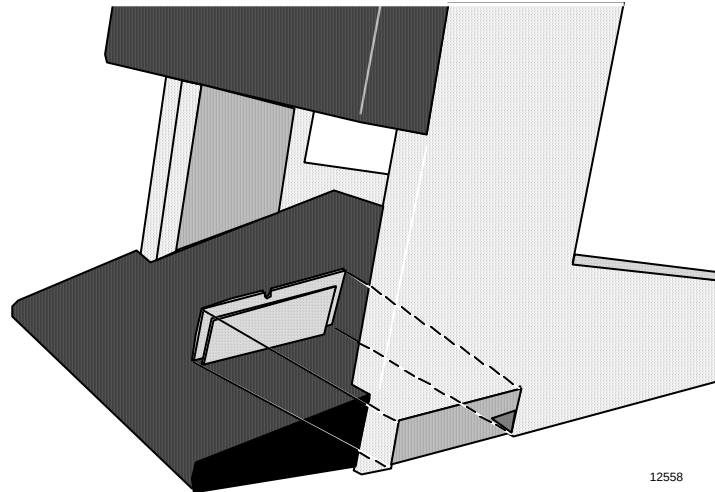


Figure 3-4 Location of Cable Access Ports on Z-Furniture

3.10 LCN Cable Installation

Scope

As shown in Figure 2-2, LCN cabinets have the intra-cabinet cables installed in the factory. The system is then tested in a system configuration and the cables inside the cabinet are left in place to be shipped with the equipment. The remaining installation task is for the cabinets and the consoles to be cabled together as shown in Figure 3-9, Figure 3-10, and Figure 3-11.

Illustration

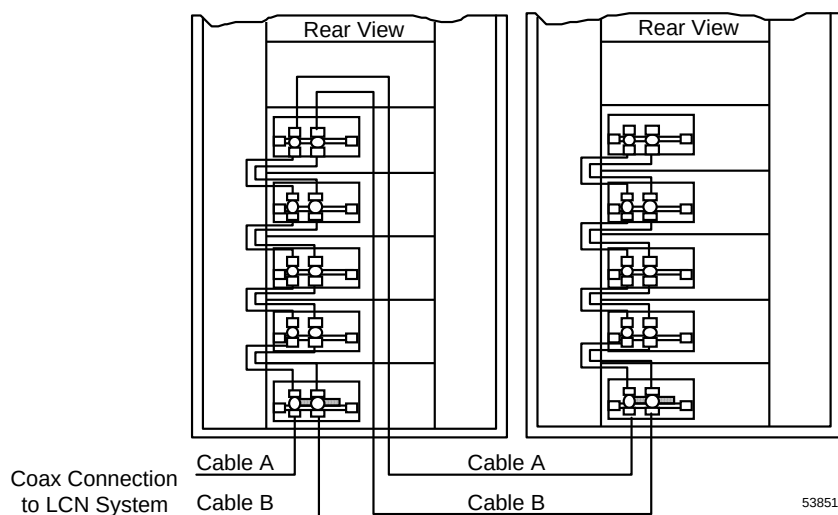


Figure 3-5 Intra-cabinet Coax Cabling Between LCN Cabinets

Cable connections to classic furniture

After the Universal Stations have been complexed (bolted) together, coax cable can be connected as shown in Figure 3-10 or Figure 3-11. Cables can exit and enter each US through the bottom or through the side openings.

Illustration

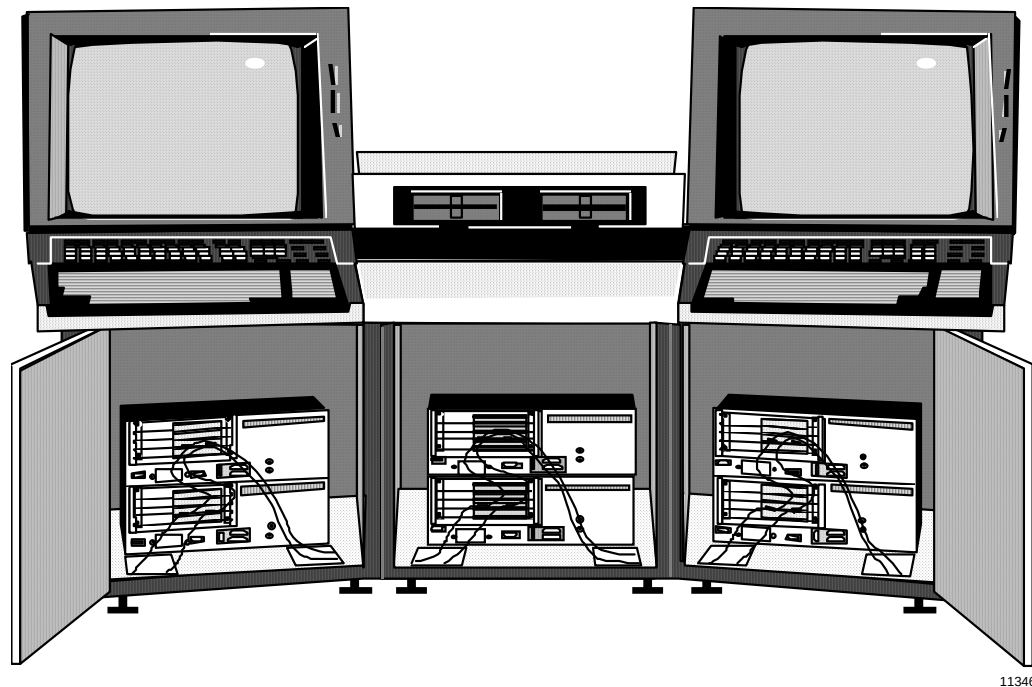


Figure 3-6 Console Cabling between Classic Furniture Stations

Cables connections to Z-furniture

Z- furniture cable must be done in the left leg of the station. The right leg is reserved for ac wiring—this applies to wiring that goes all the way to the top monitor position in the station. Excess signal cables are allowed to loop into the left leg.

Illustration

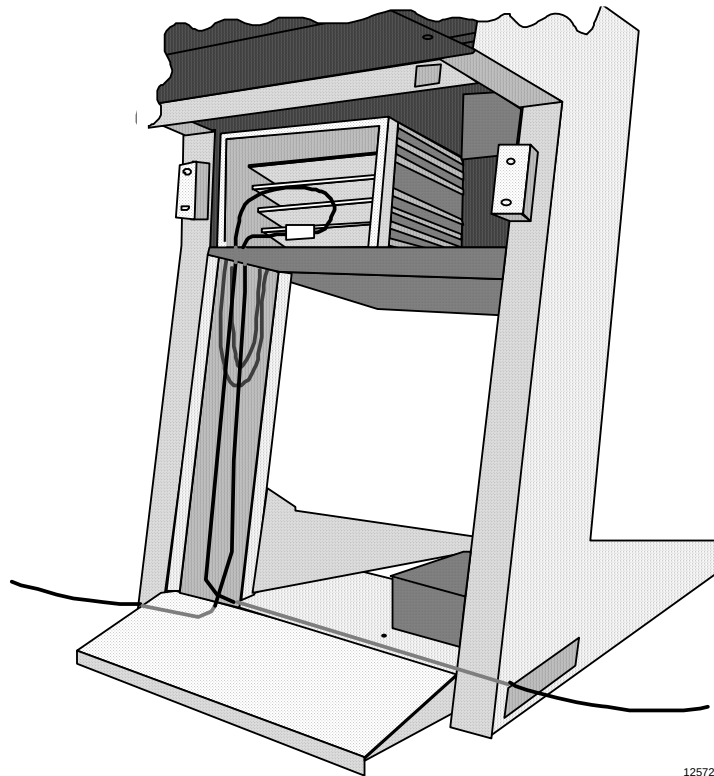


Figure 3-7 Z-Furniture Cabling Routing

3.11 LCN Cabling Rules

Specifications

- a maximum coax cable length that is a total of 300 meters (984 feet). This includes cable segments.
- a minimum cable length of 2 meters (6 feet) between any two nodes on the network.

The RS-485 interface on the 4 of DNPKG modules is not a part of the restriction.

WARNING

TDC 3000^X LCN coaxial cables and associated components have been designed and tested by Honeywell IAC to meet strict standards of performance. Other coaxial cables and components have not been tested and qualified for use in the LCN system; therefore, no substitutions are approved by Honeywell IAC.

Various lengths of coaxial cable are available as standard products and must be ordered from Honeywell IAC.

Components of the LCN

Table 3-3 LCN Coax Cable Component Specifications

Component	Specification																		
Maximum cable length	300 meters (984 feet)																		
Coaxial cable type	System Cables (LCN coaxial-cable set) are available in standard and variable lengths. Each set consists of two cables, marked A & B, in the following lengths: <table> <tr> <th>Honeywell Part Number</th><th>Length (meters)</th></tr> <tr> <td>51308112-003</td><td>3 meters</td></tr> <tr> <td>008</td><td>8 meters</td></tr> <tr> <td>015</td><td>15 meters</td></tr> <tr> <td>030</td><td>30 meters</td></tr> <tr> <td>050</td><td>50 meters</td></tr> <tr> <td>100</td><td>100 meters</td></tr> <tr> <td>200</td><td>200 meters</td></tr> <tr> <td>XXX</td><td>Variable length Max = 300 meters</td></tr> </table>	Honeywell Part Number	Length (meters)	51308112-003	3 meters	008	8 meters	015	15 meters	030	30 meters	050	50 meters	100	100 meters	200	200 meters	XXX	Variable length Max = 300 meters
Honeywell Part Number	Length (meters)																		
51308112-003	3 meters																		
008	8 meters																		
015	15 meters																		
030	30 meters																		
050	50 meters																		
100	100 meters																		
200	200 meters																		
XXX	Variable length Max = 300 meters																		
Minimum cable length between nodes	2 meters (78 inches). Use 51308111-002, which consists of two cables, one marked Cable A, and one marked Cable B. All LCN cables must be marked. It is essential that Cable A is Cable A to all nodes. Make certain that cables are not crossed. Do not substitute with unauthorized cables.																		
BNC T-connectors	51190728-105. Note that these are not the same as those used on the Data Hiway.																		
Termination resistors	30732052-001 resistors are required at each end of both coaxial cables.																		
For fiber optic cable ordering information, see <i>LCN Guidelines</i> in the <i>LCN Installation</i> binder.																			
LCN cabling <u>must not</u> be routed outside the control room building. Consider enhanced LCN Extenders (XLCNE2) and Fiber Optic cables for outside conditions.																			
See Section 2 of <i>LCN Guidelines</i> in the <i>LCN Installation</i> binder for specific LCN cabling rules.																			

Color Coding

Attention

Use only Honeywell supplied LCN cabling, and always connect cables marked "A" to the A connector of a node, and cables marked "B" to B connector of a node (never interchange "A" and "B" cables). It is strongly recommended that the color-coding system of marking coax cable, T-connectors, terminators, and connectors (board) be used. These materials are available from your Honeywell representative.

LCN Coax Cable Color Code Cable A = Yellow Cable B = Green

3.12 Typical LCN Cable Connections

Scope

The LCN coax cable system consists of a dual run of coax cable called the “A Cable” and the “B Cable.” Each cable connects to the coax connector on the module interface board (CLCN A/B, CLCN A, or CLCN B) with a T-connector. Refer to Figure 3-12 and Figure 3-13.

Five-Slot and Ten-Slot Modules have one set of coax connections for each module. These connections are located on the CLCN A/B card in slot 2 at the rear of the module. A coax cable (minimum length of 2 meters (6 feet)) is used to connect one module to another.

DNPKG Modules have one set of connections for each DNPKG module. The Cable A connection is located on the CLCN A board in the bottom slot (1) of the top node in a DNPKG module. The Cable B connection is located on the CLCN B board in the bottom slot (1) of the bottom node in a DNPKG module. A coax cable (minimum length of 2 meters (6 feet)) is used to connect one module to another.

Illustration

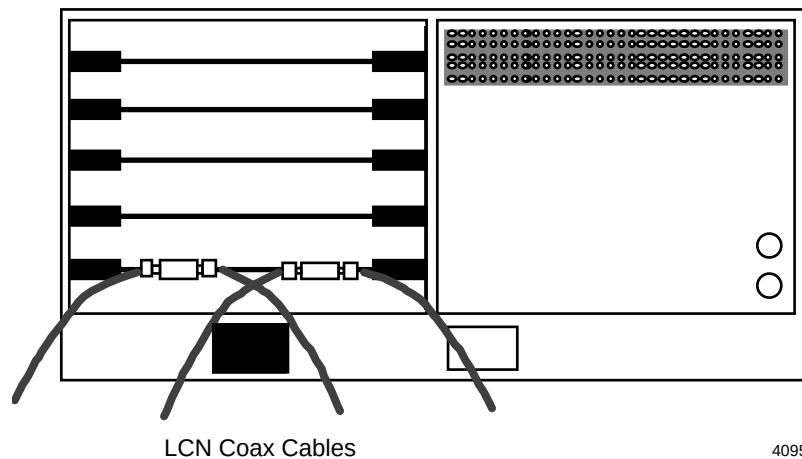


Figure 3-8 LCN I/O Board Coax Connections

Illustration

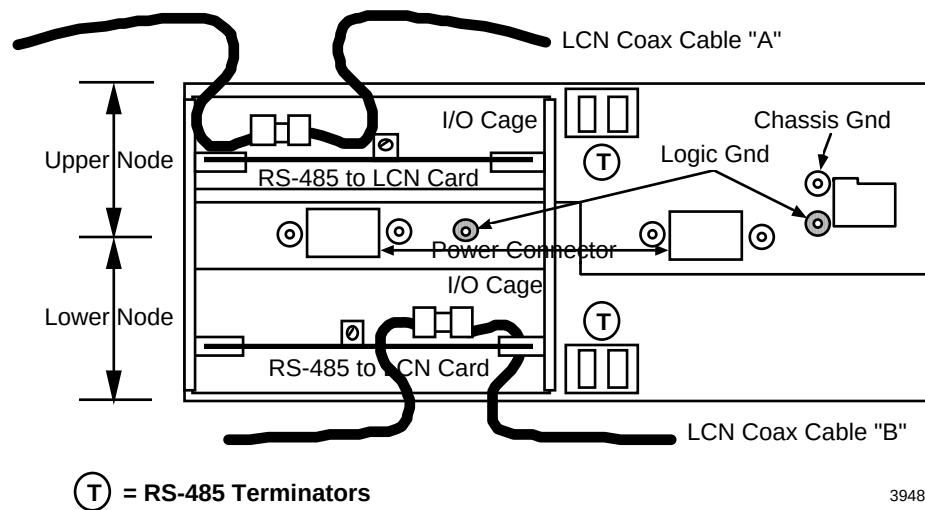


Figure 3-9 KLCNA/KLCNB Board Coax Connections

3.13 Enhanced LCN Extender (XLCNE2 Cabling)

Scope

Enhanced LCN Extenders (XLCNE2s) and fiber optic cables, up to eight km (26,246 ft.) in length, can be used to interconnect LCN coaxial bus segments. For information about the hardware and connections required to implement this, refer to Section 3 of *LCN Guidelines*.

3.14 Installation of Engineer's Keyboard (EC-PC101)

Install an Engineer's Keyboard on a Universal Station (US) that is designated to use the Engineer's Personality or do engineering functions using the Universal Personality. Connect the keyboard cable to the US by plugging it into the connector located on the right-hand side of the gooseneck under the Operator's Keyboard.

Attention

In an Arc configuration, keyboard trays cannot be extended on adjacent Z-Consoles

- Extend the keyboard tray by grasping the tray by the hole, located under the tray in the center, and pulling forward.
- Keyboard cable connection is located behind the Removable Media door.
- Set the keyboard on the extended tray with the keyboard legs in the elevated position.

Illustration

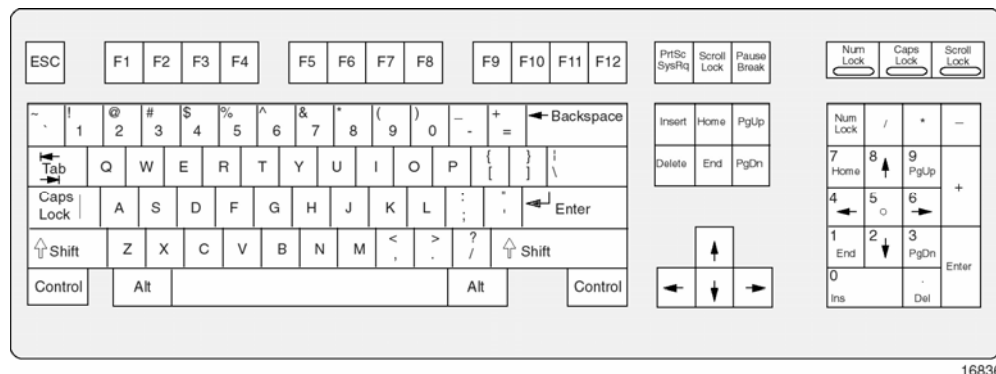


Figure 3-10 EC-PC101 Engineer's Keyboard

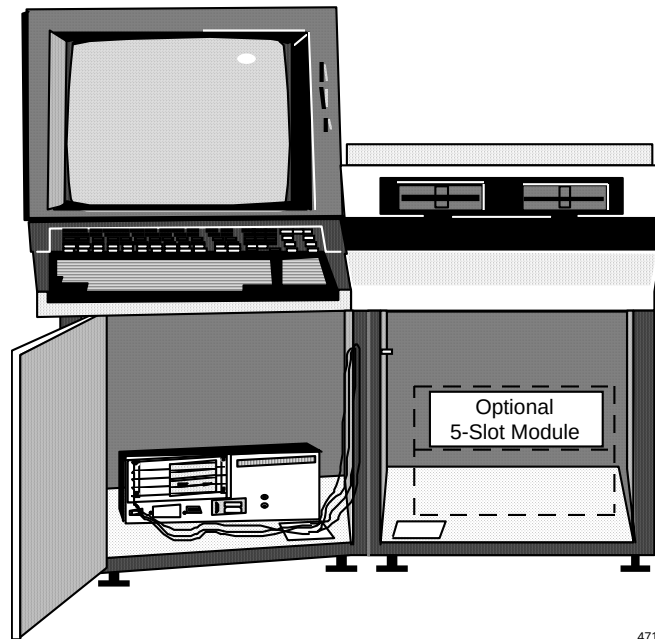
3.15 Interconnection of Peripherals Located in Separate Single Bay US Furniture

To interconnect the cartridge or floppy drive(s) that are located in single bay furniture, do the following:

Table 3-4 Interconnect of Peripherals

Step	Action
1	Place the Universal Station (US) and the single bay furniture next to each other such that the single bay furniture has the folded cables located next to the US. The cartridge/floppy interface cable is folded and clamped to the side of the cabinet, just inside the front door as shown in Figure 3-11.
2	Unfasten the cable clamps that hold the cartridge/floppy cable, but allow the cable to still pass through the cable clamps. Refasten the cable clamps.
3	Route the cable through the hole in the side of the cabinet (or in the bottom of the cabinet and up through the hole in the US) as shown in Figure 3-11.
4	The peripheral cables are folded and retained by cable clamps inside the work surface cabinet. <u>For Classic furniture</u> • Unfasten the cable, still passing them through the cable clamp. • Pass the cable through the cable pass-through hole in the cabinet side, into the Universal Station (or Universal Station^X) cabinet. • Connect the peripheral interface cable as follows: – Cartridge cable, 51192054-102 to the EPDGC or TPDGC I/O board SCSI connector J2. <u>For Ergonomic furniture</u> • The cables are stored in the foot of the work surface furniture. Free the cable and pass the cable through the cable pass-through hole in the cabinet side, into the Universal Station (or Universal Station^X) cabinet. • Run the interface cable up the left side of the console leg to the I/O card file. • Connect the peripheral interface cable(s) as follows: – Cartridge cable, 51192054-102 to the EPDGC or TPDGC I/O board SCSI connector J2.
5	If there is any excess cable hanging below the cabinet, pull it into the single bay cabinet and fasten it in the cable clamps on the side of the cabinet.

Illustration



4714

Figure 3-11 Interconnection of Peripheral Cables in Separate Stations

4. System Grounding

4.1 Ground Wiring Overview



SHOCK HAZARD

Troubleshooting, opening this equipment and removing any panels or covers will expose the **user** to the risk of a shock hazard. There are no **user** serviceable parts inside this equipment. Refer all servicing only to **qualified service personnel**.

Scope

System-grounding requirements are specified in the *LCN Site Planning* manual. Both Universal Stations and cabinets are pre-wired with the logic grounds and chassis grounds already connected to the LG, HG, and Safety Ground connection points. At installation time, the connection points must be wired to building steel or the appropriate ground stake. In summary, system grounding consists of two steps:

- Step 1 is to ensure that the cabinet/station metal frame is grounded to ac-power safety ground to protect personnel and to establish a good noise ground connection. See the next subsection.

Connect the safety ground lug in each cabinet to the safety ground stake or intermediate star plate. Use a separate wire from each cabinet to the connection point. Only one star plate is allowed between any one cabinet and the safety ground stake collection star plate, located just inside the building.

- Step 2 is to single-point ground Hiway Gateway (HG) cable shields to MRG. This is only done when an HG is used without an HTD. **This must be done if the HTD is removed from the system.** See Figure 13-5.
- Step 3 is to ground the Local Control Network (LCN) cables in one of two different schemes.

4.2 Grounding LCN Cabinets and Stations

Scope

Each LCN cabinet and station is equipped with a Safety Ground Terminal. It is recommended by Honeywell and required by many codes, for personnel protection, that each cabinet and station be connected to building steel or other designated safety ground before connecting ac power wiring. Make sure that the safety wire (green or yellow green), if insulated, from the cabinet and station frame ground post is connected to the safety ground terminal block in the control room power panel, dedicated branch circuit power panel, or building steel.

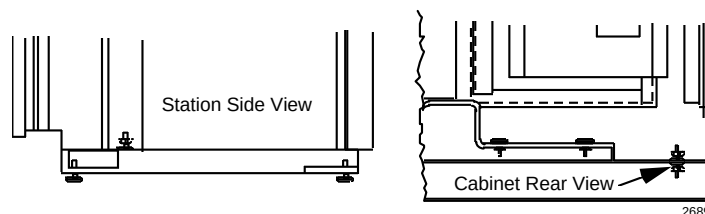


Figure 4-1 Safety Ground Terminal in Classic Stations and Cabinets

The individual modules are chassis grounded to the cabinet or station frame by the mounting hardware. Most module have the logic ground connected to the chassis ground on the rear of the module.

MRG grounding

Those modules that require separate grounding of the logic ground from the module to Master Reference Ground (MRG) are wired to HG terminal in the station and then to the MRG star plate:

Five-Slot Modules

Hiway Gateway modules that connect to Data Hiways that do not have HTDs must have a ground wire installed from the DHI I/O board to the MRG.

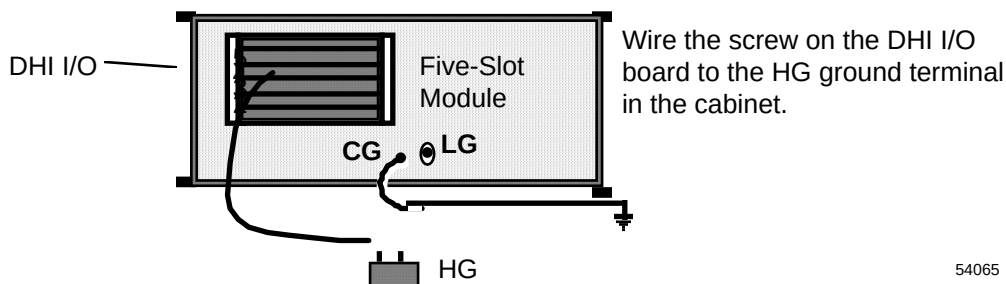


Figure 4-2 Logic Ground (LG) Connection to Safety ground Terminal

Dual Node Modules

If an HG is used without an HTD, the DHI I/O board ground screw must be connected to the HG connection post (shown in the figures of this section) and the HG connection post must be wired with 4-gauge wire to the Master Reference Ground (MRG), as shown in the *Data Hiway Subsystem Site Planning* manual.

If the HG, used without an HTD, is located in a cabinet, the LG/HG service post must be grounded to an MRG Star Plate with a 5.3 m² (#4 AWG) wire. Daisy chaining of MRG and HG grounds is not allowed.

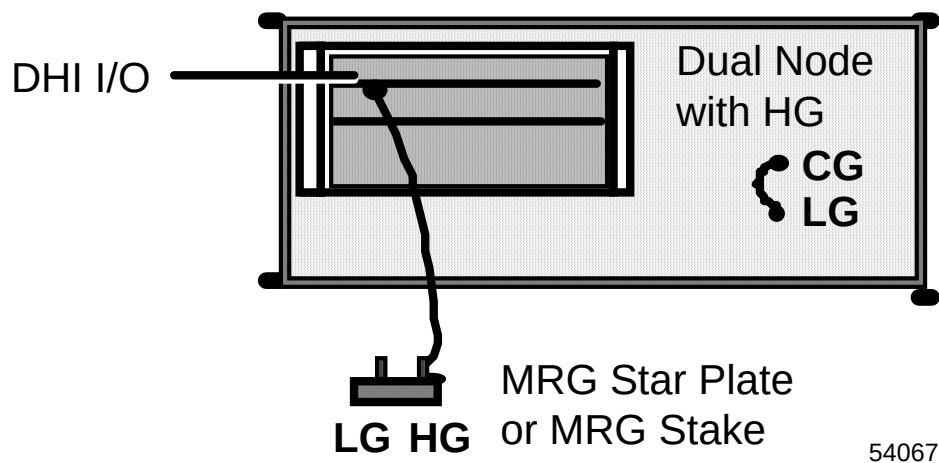


Figure 4-3 DHI I/O Ground Connection in a Dual Node Module

4.3 Cabinet Logic Ground

Scope

If required, the utility cabinet has a Logic Ground Terminal board mounted in the rear for the purpose of collecting multiple logic ground wires.

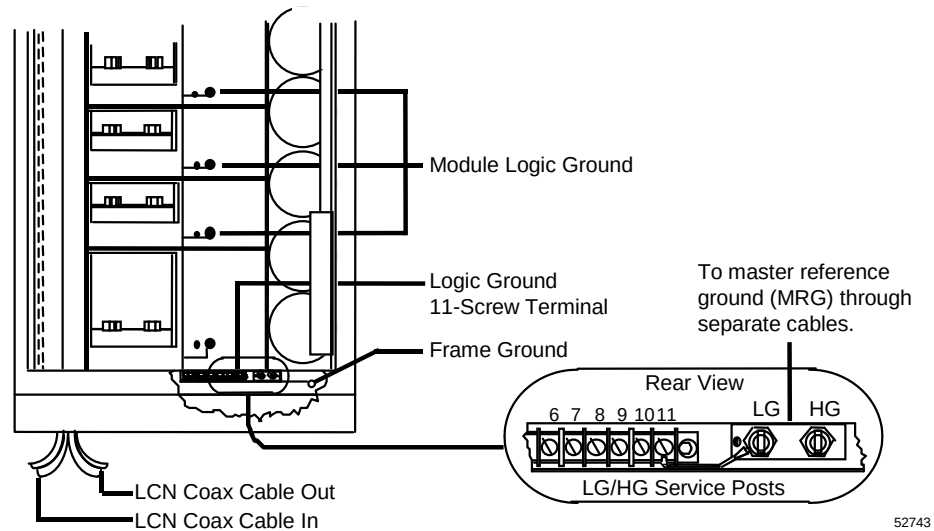


Figure 4-4 Cabinet Logic ground Terminal Board

4.4 Grounding LCN Cables

Scope

There are two schemes for grounding LCN Cables:

- Systems with the 12.5 KHz system clock
- Systems with the 5 MHz digital system clock

12.5 KHz system clock

- Systems with the 12.5 KHz system clock are self-grounding, by way of the cable shield, when the cable is connected to the BNC receptacle on the CS/R paddleboard in the clock source node.

5 MHz system clock

- In systems with the 5 MHz bit digital system clock (a segment with only K2LCN boards in the modules), the coax ground connection is made once for each cable (A and B) by a wire from the KLCNA or KLCNB board to logic ground. This wiring is to be done in one module for each cable, preferably in the Module designated as clock source modules.
- In a segment that has had the processor board (EMPU, HMPU or HPK2) replaced with K2LCN boards, the CS/R board must be retained but placed in an empty slot in a module (the CS/R PWA is now used for grounding only).

If the system includes Hiway Gateway (HG) module(s) and a Hiway Traffic Director (HTD), the Data Hiway cable shield grounding rules specified in the Data Hiway Subsystem Site Planning manual must be followed. An overview of HG preferred-access cabling is presented in

Figure 13-4 or Figure 13-5.

If the system does not include the HTD, the Data Hiway cable shield is grounded at the primary HG as follows: connect an AWG #14 wire from terminals 11 and 13 on terminal TB1 of the primary HG, DHI I/O board to the HG service post at the bottom of the cabinet or station. See Figure 4-3. **The HG service post must be connected to Master Reference Ground (MRG) through AWG #4 wire.**

The LCN cable connectors, like the Data Hiway cable connectors, must not be allowed to touch the cabinet frame. All LCN cable connectors that enter or exit the cabinet or station must be covered with heat-shrink tubing insulation.

Note that all LCN cable runs must be terminated the same way as the Data Hiway cables. The 75-ohm terminators are interchangeable; however, the T-adapters are not.

Attention

For LCN cable, use BNC T-adapter P/N 51190728-105 (Kings No. KC-99-52 or Amphenol 31-208-1051). For Data Hiway cable, use BNC T-adapter P/N 30732055-001.

In an all K2LCN segment of an LCN coax system, a ground must be provided for the coax shield by placing a wire between the grounding screw of the KLCN CA/KLCN CB boards and the logic ground terminal on the back of a Dual Node module. This should be done at only one point on each cable in a coax segment. Ground Cable A and Cable B in different modules.

Verify the integrity of the system ground wiring after the system power has been connected (see Section 7). Any grounding problems must be identified and immediately corrected to eliminate the possibility of any of the following conditions:

- Shock hazard to personnel
- System operating problems
- Damage to equipment

4.5 Grounding Stations in Z-Console Furniture

Ground lugs

The new US furniture has two collection point ground lugs for in-station grounds (one for chassis ground and one for MRG). There is also a ground lug for collecting the following:

- Logic Ground—in an HG on a Data Hiway that does not have an HTD (see Figures 4-2 and 4-3).
- The HG terminal must be connected to the MRG ground stake with an insulated 5 m² (#4 AWG) wire.
- Third post (Safety Ground)—this post is connected to the station sheet metal and can be wired to the safety ground in the power panel or to building steel with an insulated 5 m² (#4 AWG) wire.

Attention

The power cord has a safety ground connection. This second Safety Ground connection is required by some codes.

Illustration

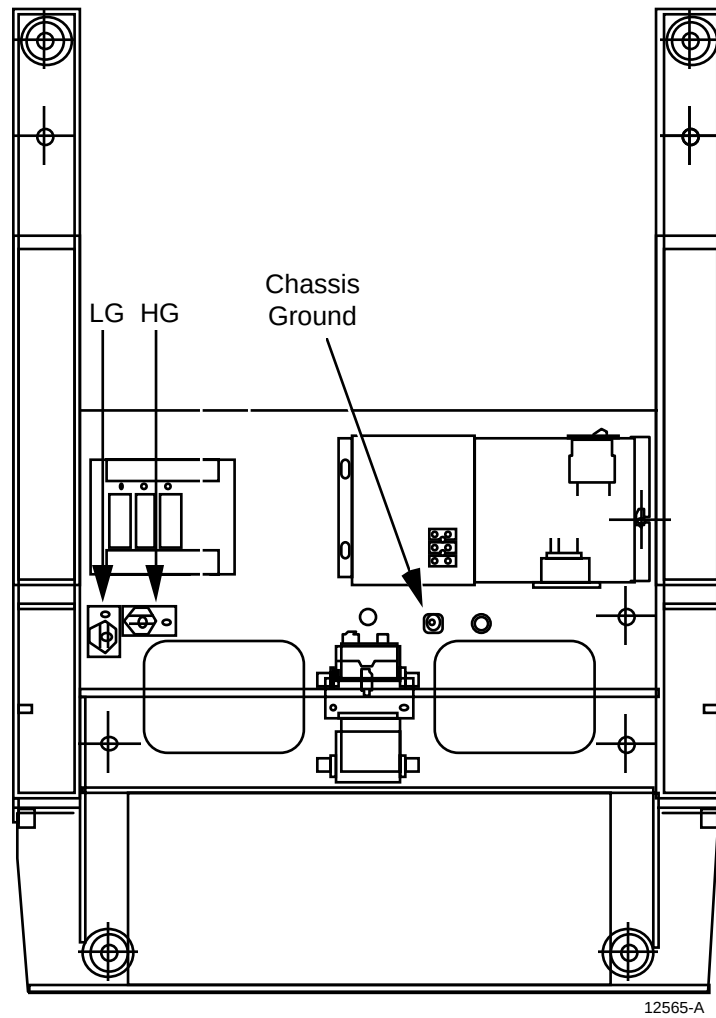


Figure 4-5 Safety Ground Terminal in Z-Furniture

5. Power Entry Wiring

5.1 Power Entry Wiring for Classic Furniture and Cabinets

Scope

Both the Universal Station (each implemented bay) and the equipment cabinets include a power entry box, located at the bottom of the cabinet or console bay. The power entry-box circuit breakers, filter, and receptacles are wired at the factory. System power wiring to the terminal board, TB1, in each power entry box is furnished and installed by the user. Note that TB1 accepts solid or stranded wire up to AWG #12/2.5 mm².

Table 5-1 Connecting AC Power to Classic Stations

Step	Action
1	 SHOCK HAZARD Troubleshooting, opening this equipment and removing any panels or covers will expose the user to the risk of a shock hazard. There are no user serviceable parts inside this equipment. Refer all servicing only to qualified service personnel. Trip the circuit breakers on the power entry box and at the control room ac power-supply panel or dedicated branch circuit.
2	Remove the access cover from the power entry box and wire TB1 as shown in Figure 5-1 or Figure 5-2.
3	Strip approximately 7.5 mm (0.3") of insulation, and then insert wire into tubular breakers in the OFF position.
4	Replace the access cover on the power entry box.

5.2 Wiring Recommendation—Classic Furniture

Scope

Wiring entry to classic furniture can be made from the bottom of the circuit breaker box shown in Figure 5-1.

- Wiring should be AWG 14-3.

<u>SERVICE</u>	<u>BREAKER SIZE</u>
120 Vac	15 Amp
240 Vac	15 Amp

Classic station ac wiring

Wiring the power in a Universal Station located in Classic furniture is shown in Figure 5-1.

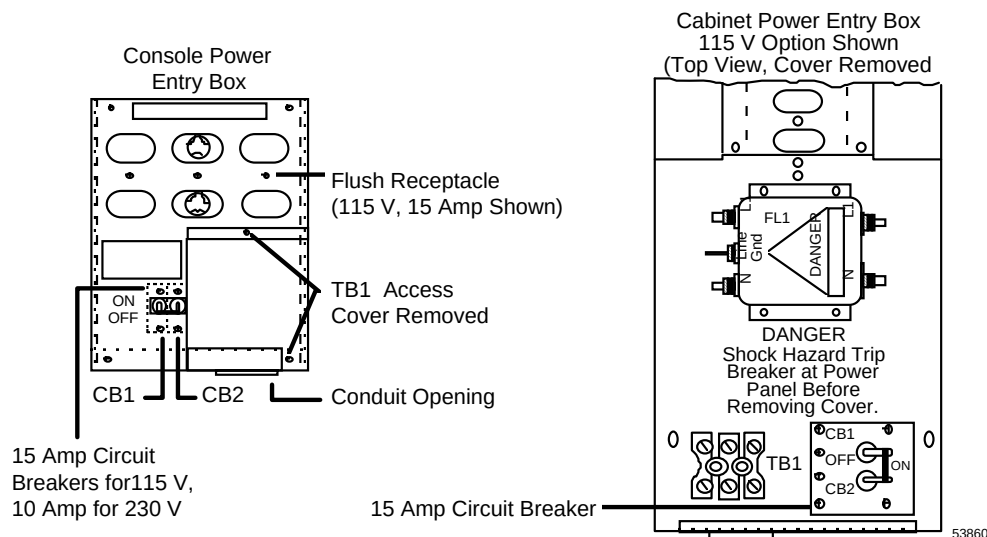


Figure 5-1 Classic Station Furniture Power Entry Box

LCN Cabinet ac wiring

Wiring the power in a LCN cabinet is shown in Figure 5-2.

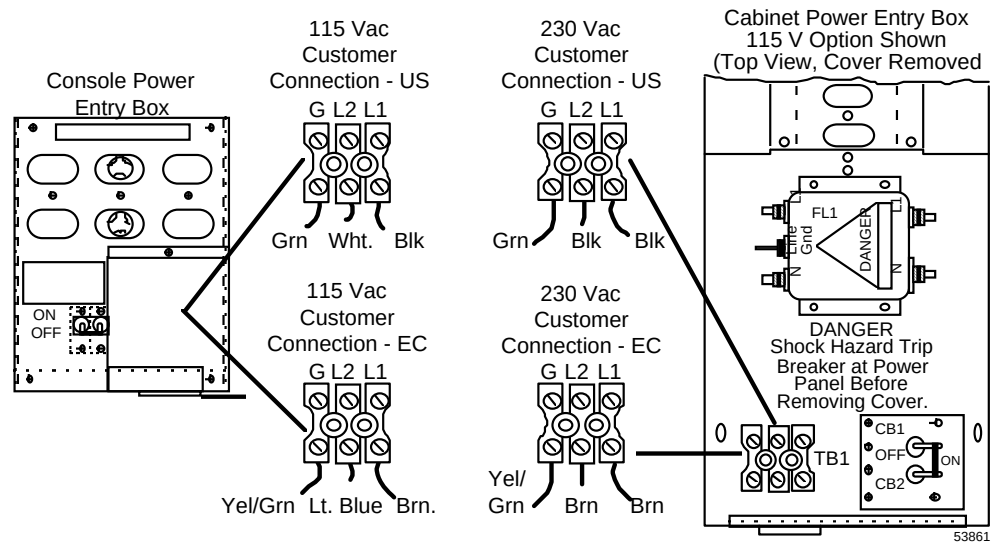


Figure 5-2 LCN Cabinet Power Entry Box

LCN cabinets can optionally have a second power entry box and distribution strips mounted beside the primary box. This is to provide power from a separate power source for redundant modules. This second power entry should be wired to a different power distribution system. See *TPS System Site Planning—Power Distribution*.

5.3 WIRING RECOMMENDATION—Z-Furniture Stations

The Z-Furniture Station is designed to accommodate conduit access from the rear of the station or from beneath the floor.

- Figure 5-3 shows access from the rear of the station.
- Figure 5-4 shows access from beneath the station.
- Figure 5-5 show the interior of the connection box.

Rear power entry

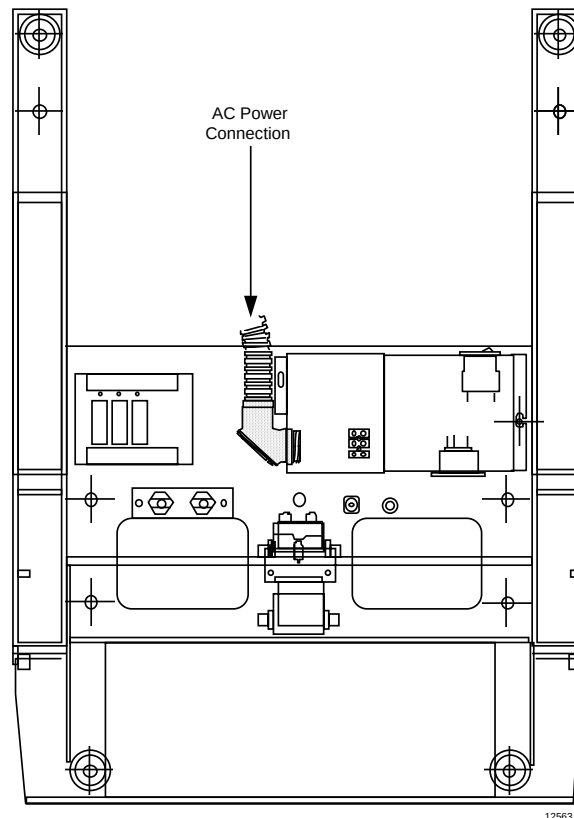


Figure 5-3 Power Entry to Z-Furniture Stations (Rear Entry)

Bottom power entry

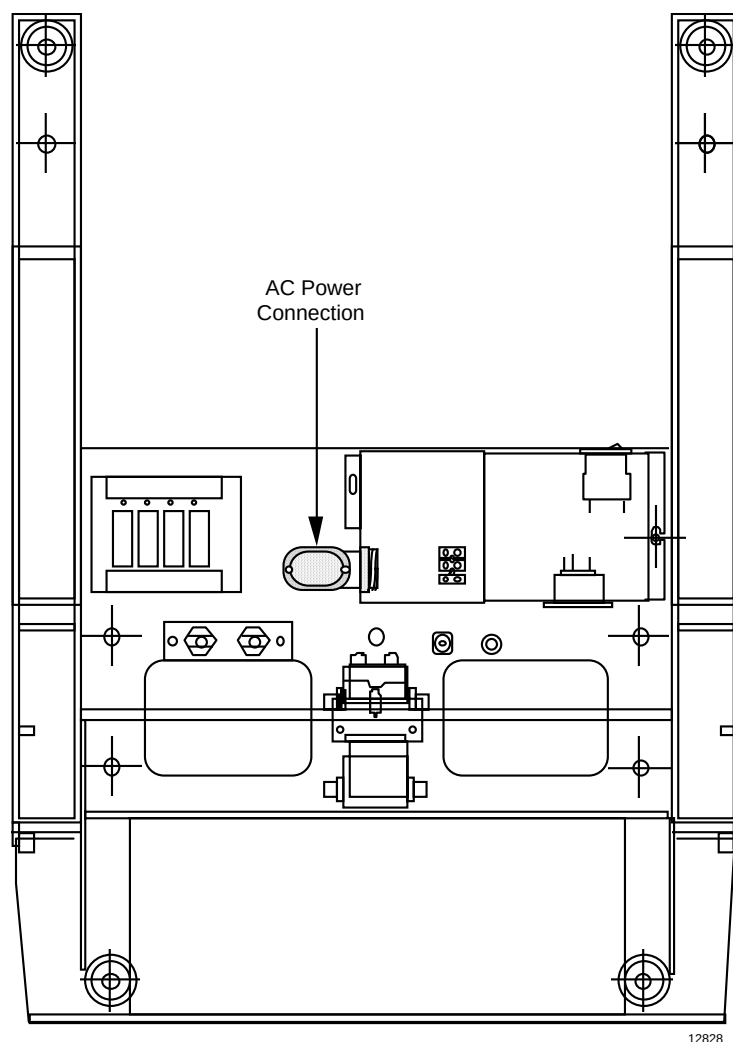


Figure 5-4 Power Entry to Z-Furniture Stations (Bottom Entry)

Materials

Wiring entry to a Z-furniture Station can be made from the rear or through the 1 1/2" hole in the bottom of the cabinet. The following conduit parts were used in the design of the access.

Table 5-2 Conduit Parts for Installation of an Z-Furniture Station

Function	Source	Part Number
3/4" 90° Pulling Elbow Male to Female	Appleton	MFL-75
	Bridgeport	72-DC
	Midwest/Cooper	811
	O-Z/Gadney	MF-075
	RACO	2663
3/4" Zinc Die Cast Screw-in Connector with Steel Locknut	Appleton	SGC-75DC
	Bridgeport	521-DC
	Midwest/Cooper	772
	O-Z/Gadney	
	RACO	
3/4" Clamp Connector for Onmetallic Sheathed and Flexible Wire	Honeywell	51190714

- Wiring should be AWG 14-3/2.5 mm² or larger for greater cable lengths. See *TPS System Site Planning*.

<u>SERVICE</u>	<u>BREAKER SIZE</u>
120 Vac	15 Amp
240 Vac	15 Amp

Power connection

Wire the AC service to the terminals as indicated by the symbols in Figure 5-5.

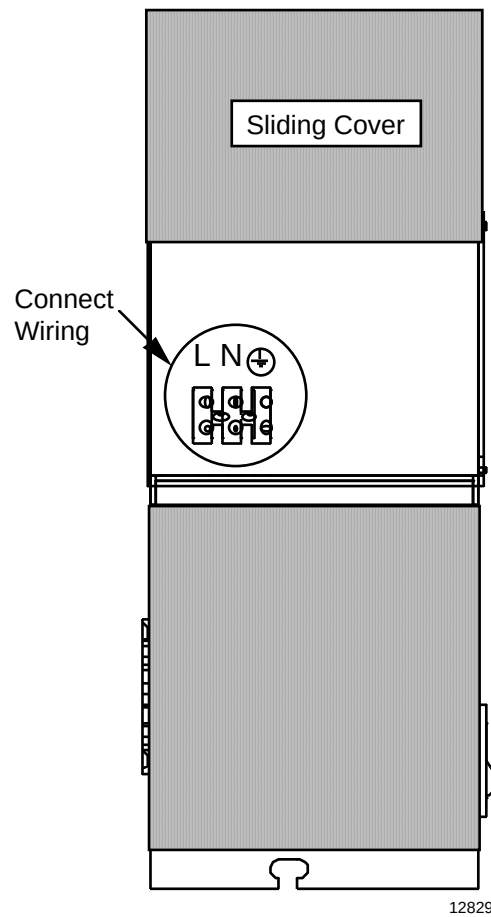


Figure 5-5 Z-Furniture Station Power Connections

6. Equipment Care and Handling

6.1 Electrostatic Discharge (ESD)

Scope

Electrostatic Discharge (ESD) and its destructive affect on system electronics components must be understood and kept under control, especially during installation and servicing of all TPS products.

What ESD is and how it can be controlled is summarized by Technical Bulletin, TB NO. ESD-1. How to handle system components is presented in Appendix A of the *LCN System Checkout Manual*. Follow these guidelines:

6.2 Clean Up

Establish practices

With the Equipment Cabinet(s) and Universal Station(s) physically installed and ready for a power-up check, please take time for a general cleanup of the area. From this point on, the habit of maintaining site cleanliness should be developed. The following steps are recommended:

- Make a final check of all shipping containers and wrappings for small parts or smaller containers.
- Arrange for the storage of special containers in which the peripheral devices are shipped. These may be required if such devices are shipped to a new location or returned for repair.
- Discard all unneeded packing materials.
- Have the room cleaned and arrange for it to be kept clean, but avoid raising dust. Damp-mop tile floors or vacuum carpeted floors frequently.

6.3 Touch-Up Colors

For all touch-up requirements, refer to the factory or to your Honeywell project engineer.

7. Power-Up Check

7.1 Overview

Scope

This section tells you how to initially power up the equipment; carefully follow the instructions.

7.2 Power-Up Procedure

Check power wiring

To ensure that the customer service panel or control room power-distribution panel is correctly wired, and to power up the LCN equipment cabinet(s) and Universal Station(s), proceed as follows:

Attention

The following procedures apply to each Power Entry Box, one at a time.

1. Each station (US, UXS, GUS) has a rocker switch located on the rear side of the base, between the legs, that controls ac power to the station. Turn the switch to the OFF position.
2. Locate the power entry box near the bottom of each implemented cabinet and console bay and open the two circuit breakers (see Figure 5-1). Remove TB1 access cover.
3. Locate each LCN module front panel and turn the power switch to the OFF position. Note that a Dual Node Module will have two power switches if both nodes are installed.
4. Close the breakers on the control room power-distribution panel and check for proper voltage at the point of power connection (TB1) (see Figure 5-1 for US and cabinets and Figure 5-2 for Z-Consoles). If okay, open the distribution panel breakers and then replace TB1 access cover(s).

NOTE

The following procedures apply to each cabinet, OIS, and Universal Station, one at a time.

5. Close the breakers on the control room power panel, close the two circuit breakers on the power entry boxes, and turn the OIS rocker switches to the ON position. Find an unused duplex receptacle, or unplug one module, and check for fault voltage between the ground pin on the duplex and the cabinet frame. If voltage is found, correct the problem before proceeding.
6. Make sure that each LCN module power cord is plugged in and that the margin-test jumper is in the normal (center) position.
7. Turn the power switch ON each module, one at a time, and check that the green POWER STATUS LED is on. Remember that a Dual Node Module may have two power switches. Correct any problems before proceeding to the next module. Refer to the Five/Ten-Slot Service manual or Dual Node Module Service manual if service is required.
8. A Deskside Global User Station has a power-on LED on the front of the electronics tower for deskside models, and an LED on the front of the console module located at the rear of the station.

8. System Configuration

8.1 Overview

Scope

All LCN modules, Hiway boxes, and peripheral devices require a unique system device address.

8.2 System Pinning

Scope

There are no LCN module adjustments; however, the LCN I/O boards have an address pack that must be characterized for the particular node it occupies on the LCN. This normally is done for each system during factory test.

Method

The address of the nodes in an LCN must start with "1" and be assigned in increasing numerical order. In planning for future additions to the system, reserve a number or set of numbers for that. All other addresses are assigned in the order of module placement. Nodes can be added later during system expansion, and can be done so without changing original assignments if gaps in the node numbering are placed to allow more node numbers to be added later.

Attention

At this time, a topology map should be initiated and maintained.

If the LCN I/O or K2LCN board is replaced on site, or if an additional module is added, the new board or module (including the Universal Work Station) must be set up for the node it will occupy; for example, Figures 8-1 and 8-2 show the location of the address pack on the LCN I/O board and the K2LCN board, pinned for an address of three (3).

Attention

The address pinning is logically ORed together if an LCN I/O board is used with a K2LCN board. **The address must be pinned only on the LCN I/O board and all jumpers removed from the K2LCN board.**

Universal Station^X

A Universal Station^X node contains a WSI I/O board that is to be pinned as shown in Figure 8-3. The WSI board has a Normal Shutdown switch that should be in the NORMAL position. The SHUTDOWN position disables the Work Station Processor and returns the node to a UNP personality.

Application Module^X

An Application Module^X node contains a WSI2 board that is to be pinned for used with an HMPU processor or K2LCN processor. The WSI board has a Normal Shutdown switch that should be in the NORMAL (to the right) position. The SHUTDOWN position disables the Work Station Processor.

Global User Station

The Global User Station node number is configurable by software. Refer to *Configuration User's Guide, GU02-100*.

Data Hiway

For Data Hiway device addressing and response priorities, refer to *Data Hiway Subsystem Site Planning* manual. For the Process Interface Units (PIUs), refer to HLPIU-M, LLPIU-M, or LEPIU-M Service manuals.

Peripheral pinning

For peripheral device pinning or function/selection, refer to the *service manual for Universal Station, Universal StationX, TPS Node, or the appropriate service manual for the Global User Station platform as appropriate*.

UWS keyboard pinning

For keyboard pinning options on the Universal Work Station, refer to Appendix A, paragraph A.7.

K2LCN

If a node does not have an LCN I/O (or CLCNA/B) board, the LCN node address must be configured on the K2LCN or K4LCN board.

Figure 8-3 illustrates LCN node address selection on the K2LCN board, assembly 51401551-x00. The pinning block, TS2, is located at coordinate 9C on the board. The absence of a jumper is a binary “1.” Parity must be an odd number.

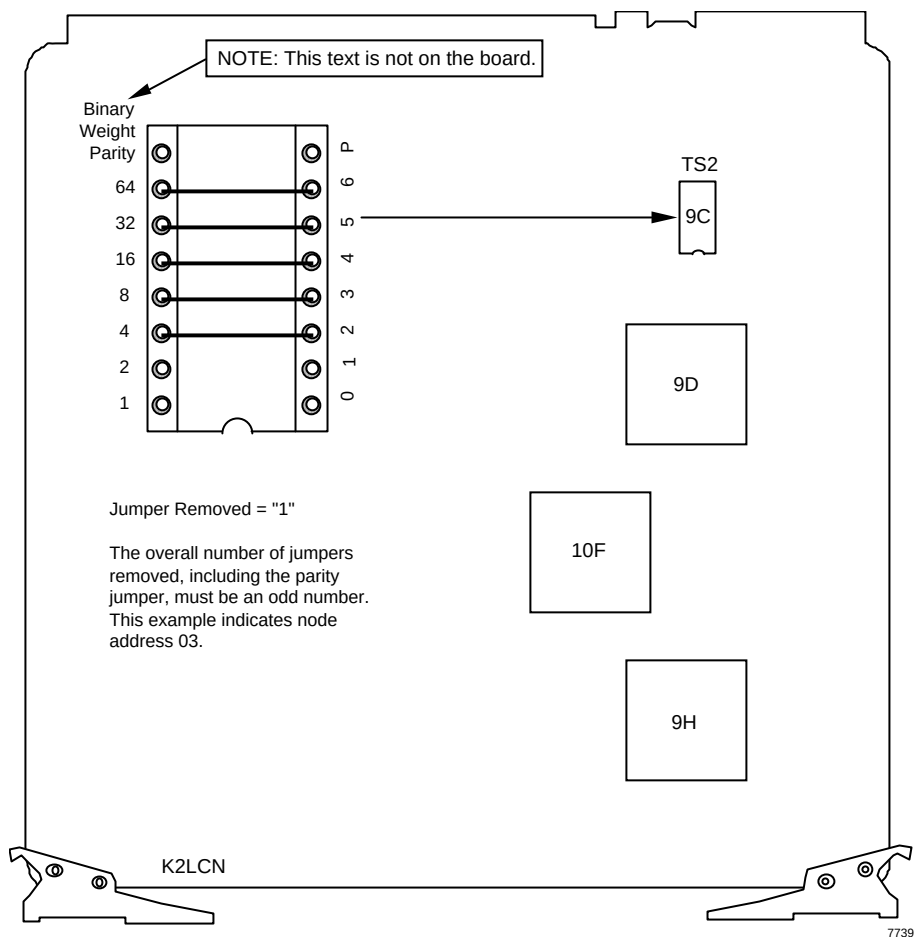


Figure 8-3 K2LCN Board (51401551-x00)

Figure 8-4 illustrates LCN node address selection on the K2LCN board, assembly 51402615-x00. The pinning block, TS1, is located at coordinate 7A on the board. The absence of a jumper is a binary “1.” Parity must be an odd number.

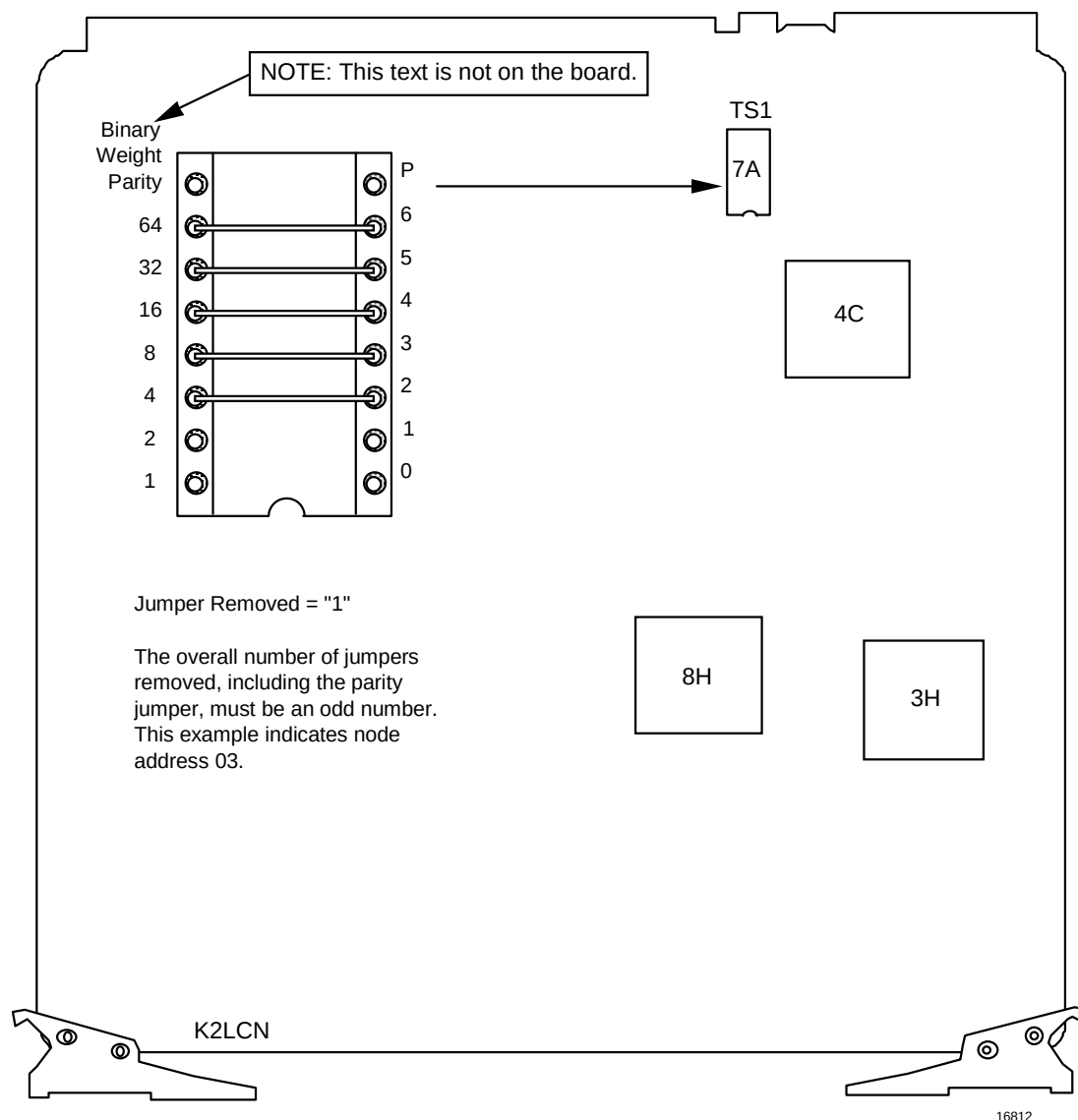


Figure 8-4 K2LCN Board (51402615-x00) LCN Node Address Pinning

Figure 8-6 illustrates LCN node address selection on the K4LCN board, assembly 51402755-100. The pinning block, TS1, is located at coordinate 7A on the board. The absence of a jumper is a binary “1.” Parity must be an odd number.

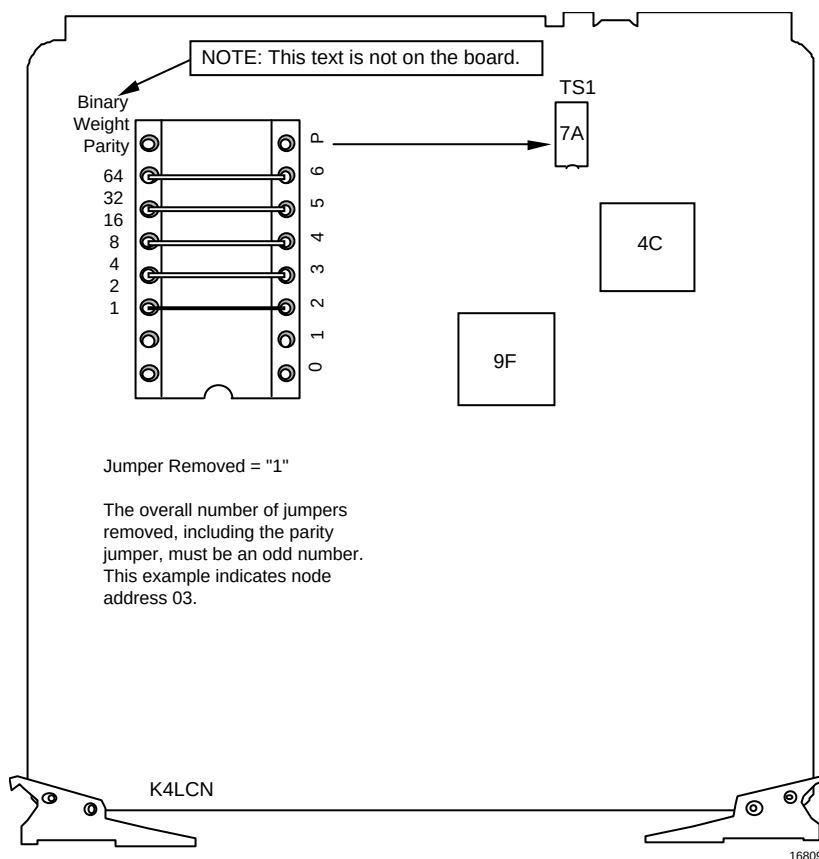


Figure 8-6 K4LCN Board (51402755-100) LCN Node Address Pinning

WSI I/O Board Pinning

Figure 8-7 illustrates the pinning on a WSI I/O board.

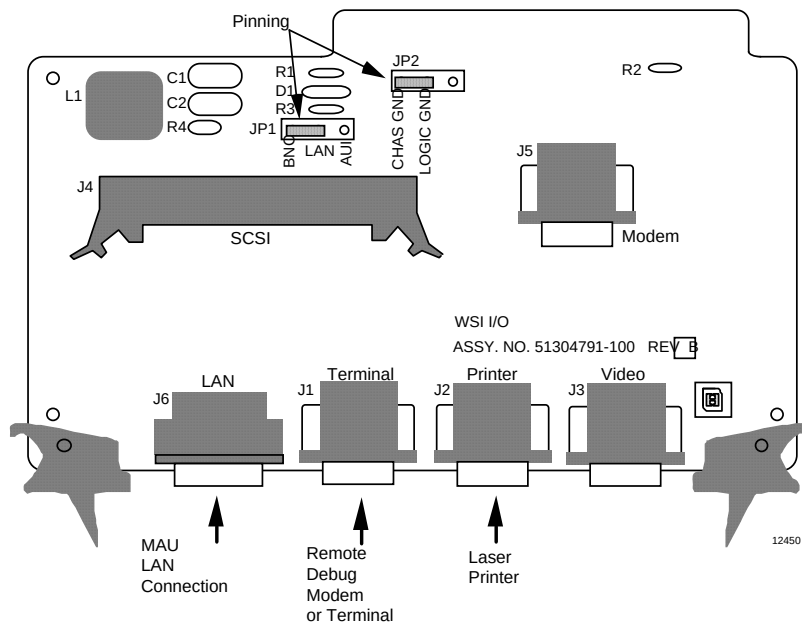


Figure 8-7 WSI I/O Board Pinning

WSI2 I/O

Figure 8-8 illustrates the pinning on a CE Compliant WSI2 I/O board.

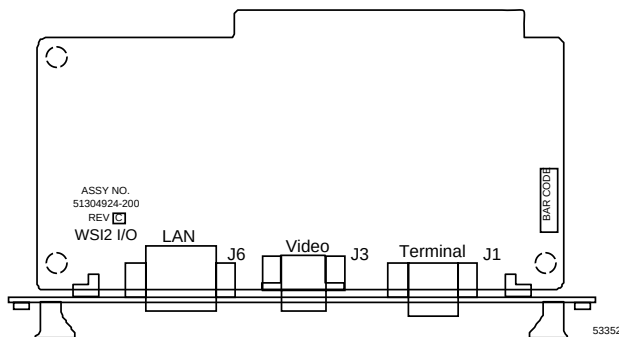


Figure 8-8 WSI2 I/O Board Pinning

TPDG I/O Board Pinning

Figure 8-9 illustrates the pinning on a non-CE Compliant TPDG I/O board.

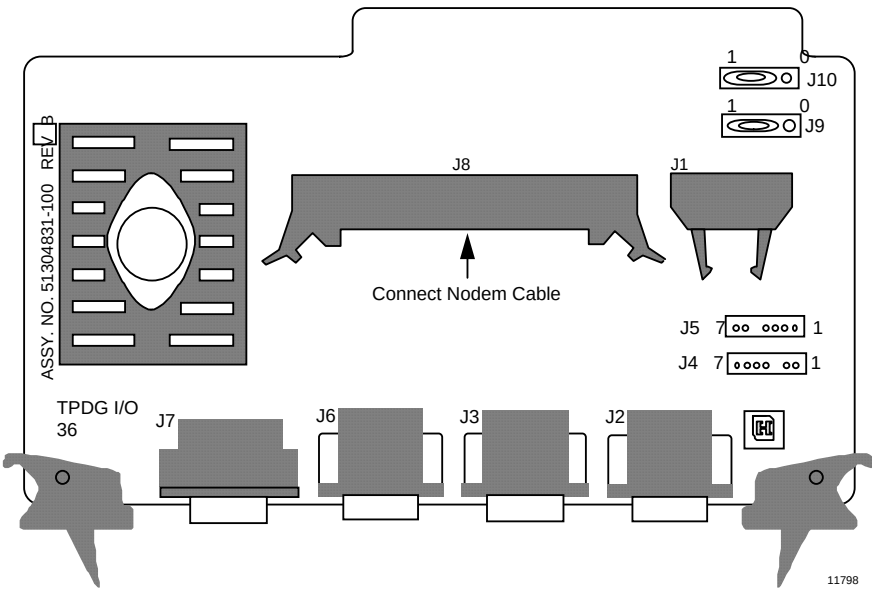


Figure 8-9 TPDG I/O Board Pinning

TPDGC I/O

Figure 8-10 illustrates the pinning on a CE Compliant TPDGC I/O board.

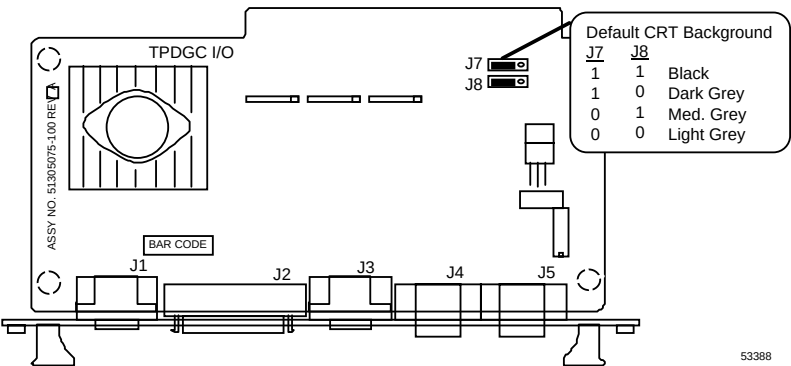


Figure 8-10 TPDGC I/O Board Pinning

9. UCN Tap Installation

9.1 Installation In LCN Cabinets



SHOCK HAZARD

Troubleshooting, opening this equipment and removing any panels or covers will expose the **user** to the risk of a shock hazard. There are no **user** serviceable parts inside this equipment. Refer all servicing only to **qualified service personnel**.

Scope

Install two taps in the LCN cabinet that contains the NIM module(s) for this UCN. See Figure 9-1. Mount each tap bracket to the two bolts located at the top of the rear side of the cabinet.

Perform this procedure to mount taps in an LCN cabinet.

Table 9-1 Installing a UCN Tap in an LCN Cabinet

Step	Action
1	Locate and remove the four bolts at the top, rear side of the cabinet. There are eight bolts total. Remove the four located nearest the center of the cabinet.
2	Mount a tap mounting bracket inside the left-hand and right-hand side of the cabinet and replace the four bolts, tightening them until the brackets are held firmly in place.
3	Mount the tap to each mounting bracket. - Left side of the cabinet—Isolated port marker must be located on right side of the mounting bracket. - Right side of the cabinet—Isolated port marker must be located on the left side of the mounting bracket. - Tighten the mounting bolt to only 25 inch pounds—just enough to hold the tap fixed in place. Do not crush the body of the tap.
4	Connect a 14 AWG/2.5 mm² ground wire from each tap directly to the nearest safety ground. Do not daisy chain the connection.
5	Connect a drop cable, 51195153-XXX (XXX is the length in meters), to each drop cable connection 1. It is very important to tighten the drop cable connectors to 25 inch pounds to ensure an error-free UCN system.
6	Continue to connect drop cables to the tap until all are connected.

Table 9-1 Installing a UCN Tap in an LCN Cabinet

Step	Action
7	Go to instruction for connecting trunk cable in <i>UCN Guidelines</i> (UN12-510), subsection A.3.

Illustration

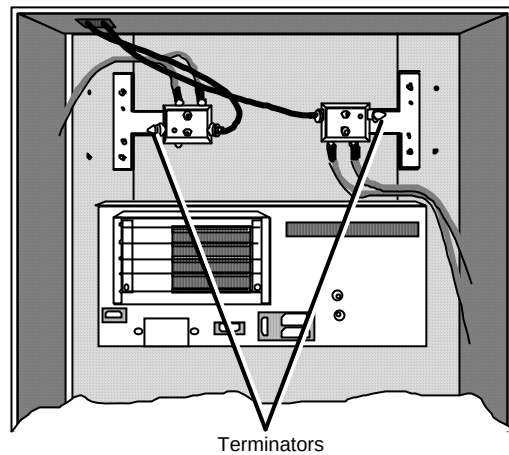


Figure 9-1 Taps Mounted in an LCN Cabinet

Attention

For FTE switch installation, refer to section 16. *FTE Switch Installation*. ENIM does not require taps as it connects to the EUCN using Honeywell qualified industrial Ethernet switches.

9.2 Installation in a Universal Station

Scope

A NIM module can be mounted in a Universal Station only if there is an empty console furniture cabinet located adjacent to it. This is because of the bend-radius requirements of UCN Trunk cables. To do this, two mounting brackets must first be mounted inside the lower cavity of the station. Two sets of holes will have to be drilled in the cabinet.

9.3 Installation in a Z-console Universal Station

Scope

If a NIM is to be installed in a Universal Station, the taps are mounted to the mounting brackets as shown in Figure 9-2.

Procedure

Table 9-2 Installing UCN Taps in a Z-Console

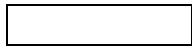
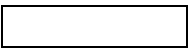
Step	Action
1	Mount a tap to each mounting bracket. - Isolated port marker must be located on right side of the mounting bracket. - Mount the tap with a bolt through the hole in the tap.  Do not mount with the grounding stud. - Tighten the mounting bolt to only 25 inch pounds—just enough to hold the tap fixed in place.  Do not crush the body of the tap.
2	Connect a 14 AWG/2.5 mm² ground wire from each tap directly to the nearest safety ground. Do not daisy chain the connection.
3	Connect a drop cable, 51195153-XXX (XXX is the length in meters), to each drop cable connection 1. It is very important to tighten the drop cable connectors to 25 inch pounds to ensure an error-free UCN system.
4	Connect a drop cable, 51195153-XXX (XXX is the length in meters), to each drop cable connection 1. It is very important to tighten the drop cable connectors to 25 inch pounds to ensure an error-free UCN system.

Table 9-2 Installing UCN Taps in a Z-Console

Step	Action
5	Install the rest of the UCN Cable System as shown in the publication UN20-100 that is found in the bag of taps. The same instructions are found in Appendix A of <i>UCN Guidelines</i> .

Illustration

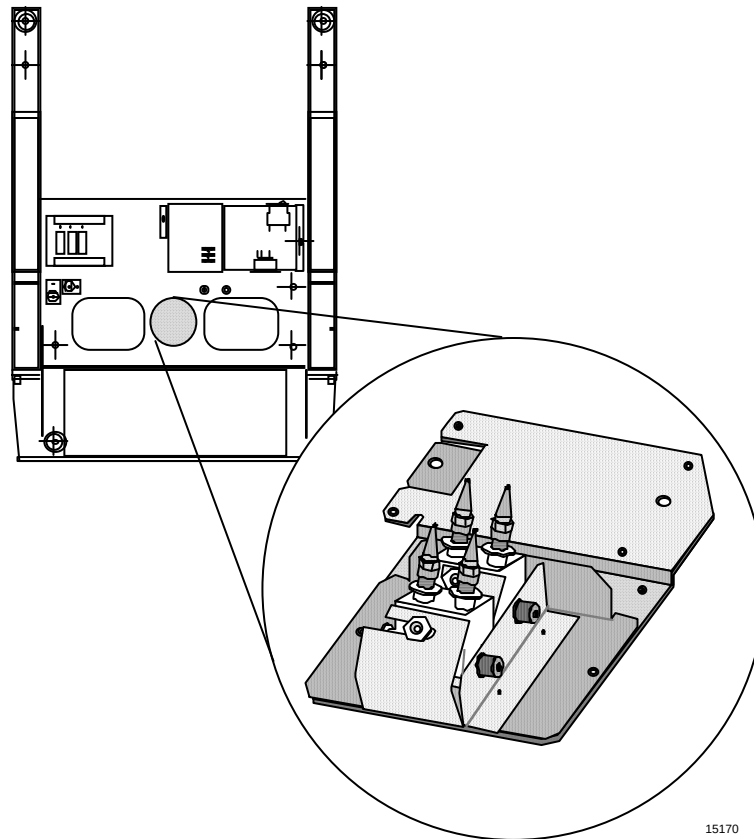


Figure 9-2 Taps Mounted in a Z-Console Universal Station

10. RULA Installation

10.1 Installation Instructions

Scope

Provides for the installation of a Remote User Link Adapter (RULA) in a Universal Station, Operator Interface Station (OIS), Micro TDC 3000, and Universal Work Station.

In general, the Dayna SCSI/Link—3 unit is installed on the support tray in a half height upper pod next to a Cartridge Drive. **There must be an existing cartridge drive installed.**

- The old power cable for the cartridge drives is disconnected and a new cable that has an addition connector to provide power to the Dayna SCSI/Link—3 module, is connected to the old cable. The old cable connection is then re-established.
- The old SCSI Interface Cable for the cartridge drives is disconnected and a new cable that has an additional connector to provide connection to the Dayna SCSI/Link—3 module, is connected to the old cable. The old cable connection is then re-established.
- The Dayna SCSI/Link—3 adapter is mounted to a support bracket with hook and loop fastener strips.
- The Ethernet connection is now ready to be made.
Detailed instructions for the four applications are found in two places:
- Upgrade Kit Index UK20-100
- Individual copies are included in the shipment of the hardware.
- The four kits are:
 - MP-RLKT11 Universal Station applications
 - MP-RLKT12 Micro TDC 3000 applications
 - MP-RLKT13 Operator Interface Station applications
 - MP-RLKT14 Universal Work Station applications

11. Scanner Application Module Installation

11.1 Planning the Installation

Placement

The Scanner Application Module (SAM) can be placed where any other Five-Slot Module can be placed.

Grounding

The Scanner Application Module is grounded the same as an Application Module in a Five-Slot Module with the logic ground connected to the safety ground.

AC Power

Connect the ac power cords attached to the module to the ac power strip in an LCN cabinet or to the ac distribution box in a Universal Station or Universal Station^X.

LCN Cable Connection

The LCN A and LCN B cables are connected to the LCN I/O board as any other node in the system—see Section 3.

Communication Connection to MCO

There are two types of connections between the SAM and MCO. If the RS232 interface is used, the distance between the SAM and MCO cabinet is limited to <20 meters (66 ft). If the RS422 interface is used for longer distances, adapters that convert the RS232 signal to RS422 level is used.

For RS232 interface, a shielded cable (Part No. 51196082-115) with three wires is sufficient for the transmission. The cable is connected at the main cable strip of the cabinet.

For RS422 interface, a cable with five wires is sufficient for the transmission. RS422 transmits differential signals that are not sensitive to disturbances.

Current loop interfaces can also be used.

Illustration

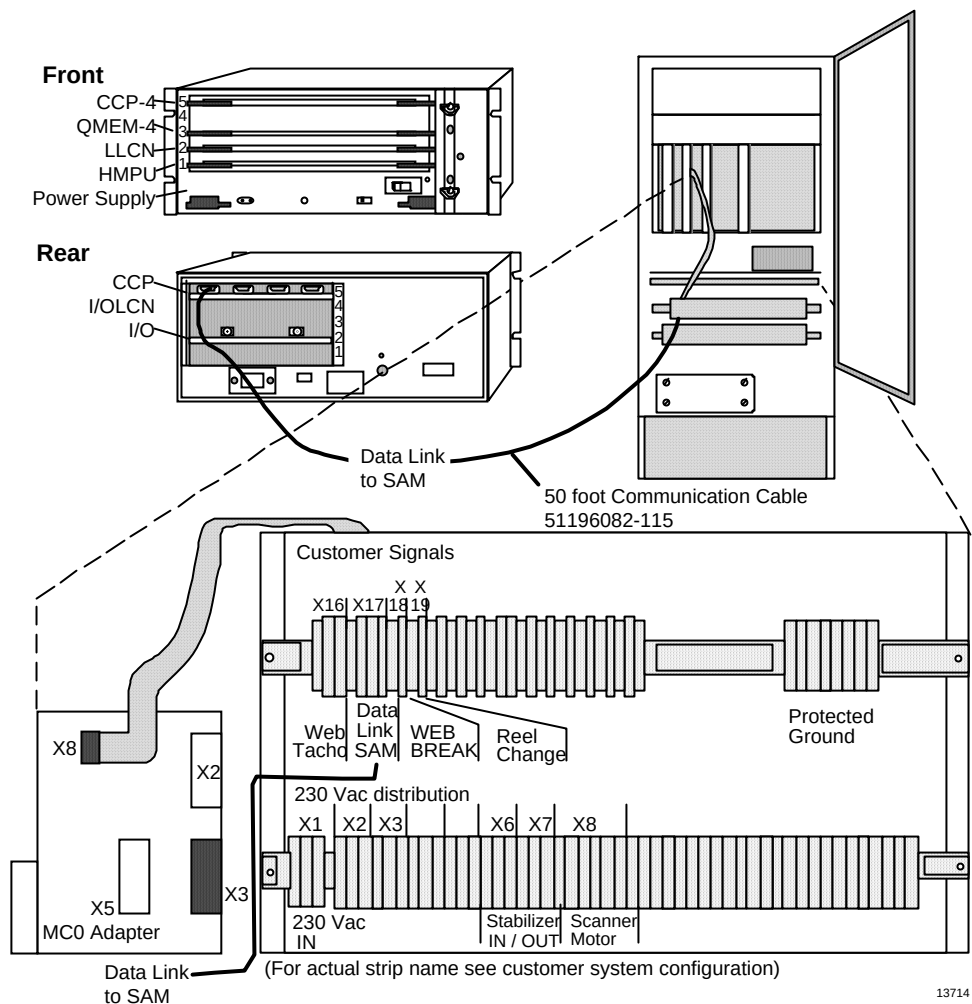


Figure 11-1 Configuration and Cabling of Scanner Application Module (SAM)

12. Installation of PC Based TPN-Connected Nodes

12.1 Referral

Scope

The installation of PC based TPN-Connected Nodes is supported by the "Planning, Installation and Service for TPN-Connected Nodes (PC manufacturer and model number)" publication for the PC that you have.

The PC Based TPN-Connected Nodes include, but are not limited to, the following:

- Global User Station (GUS)
- Application Processing Platform (APP)
- Experion Station TPS (ES-T)
- Experion Server TPS (ESVT)

Publications for the nodes are found on the documentation CD-ROM disk and are also available in hard copy.

13. Options Installation for Classic Universal Stations

13.1 Classic Furniture Universal Station Complexing



SHOCK HAZARD

Troubleshooting, opening this equipment and removing any panels or covers will expose the **user** to the risk of a shock hazard. There are no **user** serviceable parts inside this equipment. Refer all servicing only to **qualified service personnel**.

Scope

A complete complexing kit, P-ZCKY1, is available from Honeywell for on-site expansion. The kit includes hardware for complexing the keyboard and work-surface support trays, for complexing the console bases, and the trim kit for the ends of each console complex. See Section 3 for installation instructions. When viewed from the LCN, each station is seen as a separate unit.

13.2 Module Installation

Scope

The LCN electronics modules are shipped complete with power cord, front cover, attachment hardware, and LCN inter-cabinet cables, including tees and terminators. Refer to installation drawings for connection details. See Section 8 for system pinning (node addressing).

13.3 Peripheral Installation

Scope

The standard built-in peripherals, cartridge disk drives, floppy disk drives, and trend pen recorders, are installed and cabled into the Universal Station^X at the factory. Removal and replacement procedures are provided in the applicable service manual and installation drawings. The matrix printer is packaged separately for shipping.

Peripherals installation procedure

Table 13-1 Matrix Printer Installation

Step	Action
1	Unpack the printer, position printer on a work surface or tabletop.
2	Route the power cord and data cables down the back of the console and up through the cable cutouts.
3	Connect the power cord into any unused receptacle on the power entry box.
4	Connect the data cable connections as shown in Figure 13-1 and. Figure 13-2. See subsection A.7 for cabling information for the Universal Work Station.

Illustration

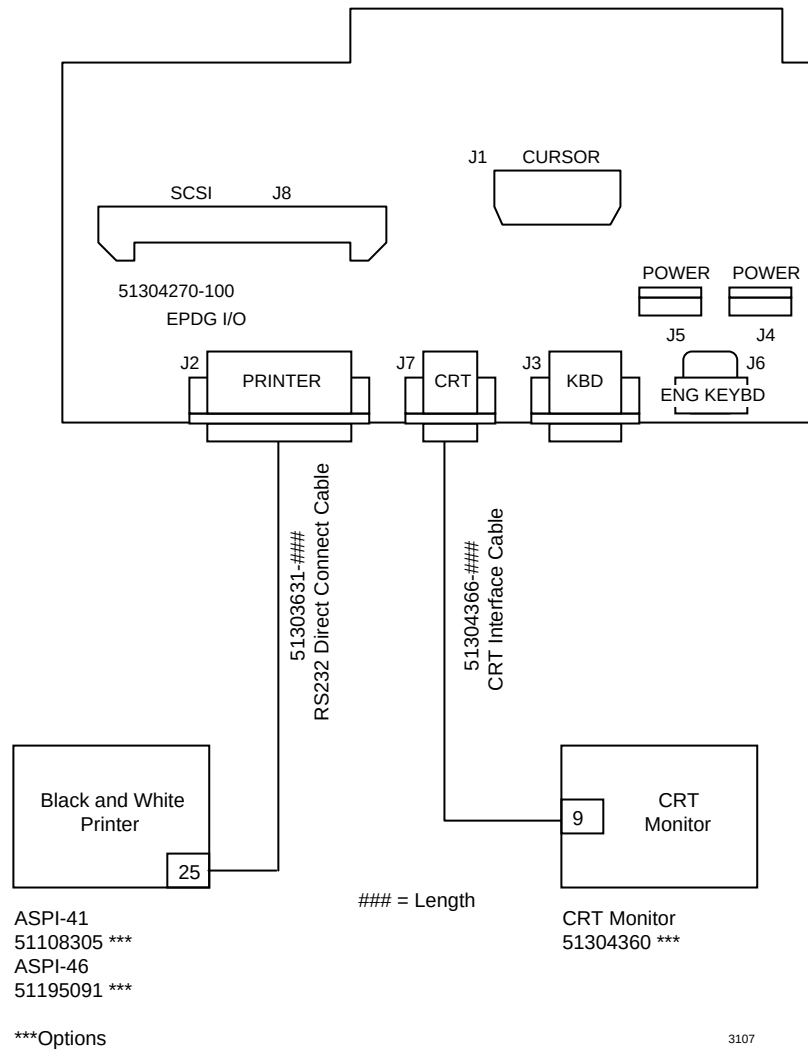


Figure 13-1 Printer and Monitor Cabling (EPDG Board)

Illustration

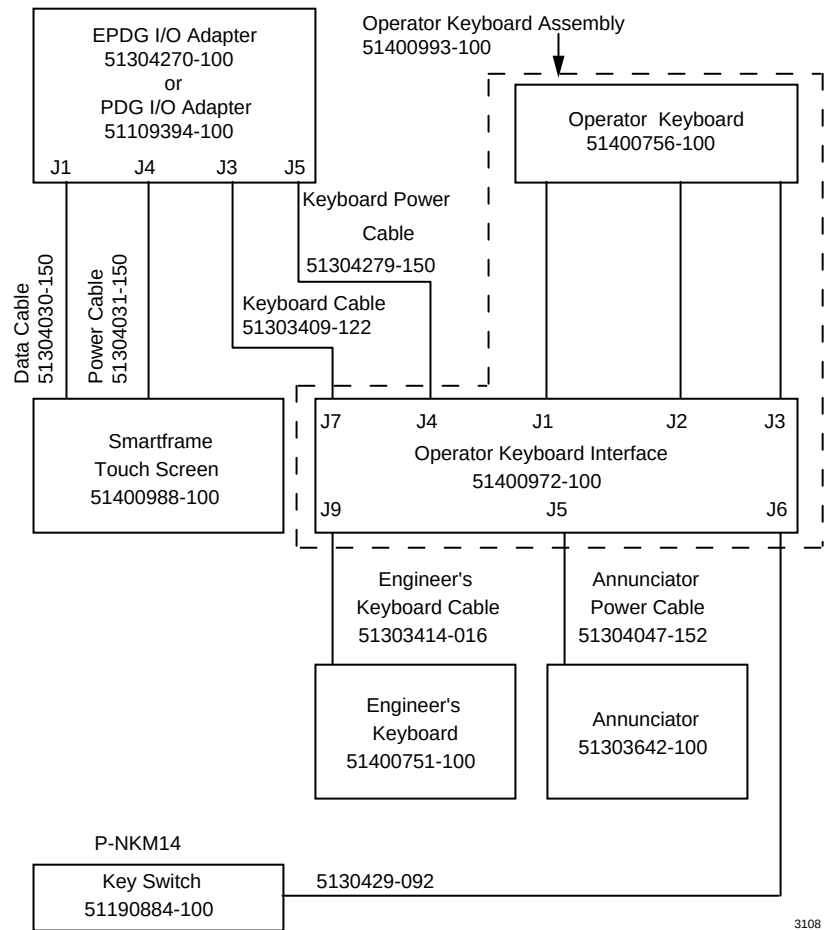


Figure 13-2 Keyboard Cabling

13.4 Trend Pen Assembly Removal and Assembly

Scope

The trend pen recorders, up to three, and the power/interface chassis are mounted in a separate assembly, above the CRT shroud. Mounting brackets and M5 hex-head screws are used to secure it in place. To remove and replace the recorders, refer to the *Universal Station Service* in the *LCN Service-1* binder or *Universal Station^X Service* in the *Universal Station^X* binder. The standard trend-pen cable connections are shown in Figure 13-3.

Trend Pen cabling

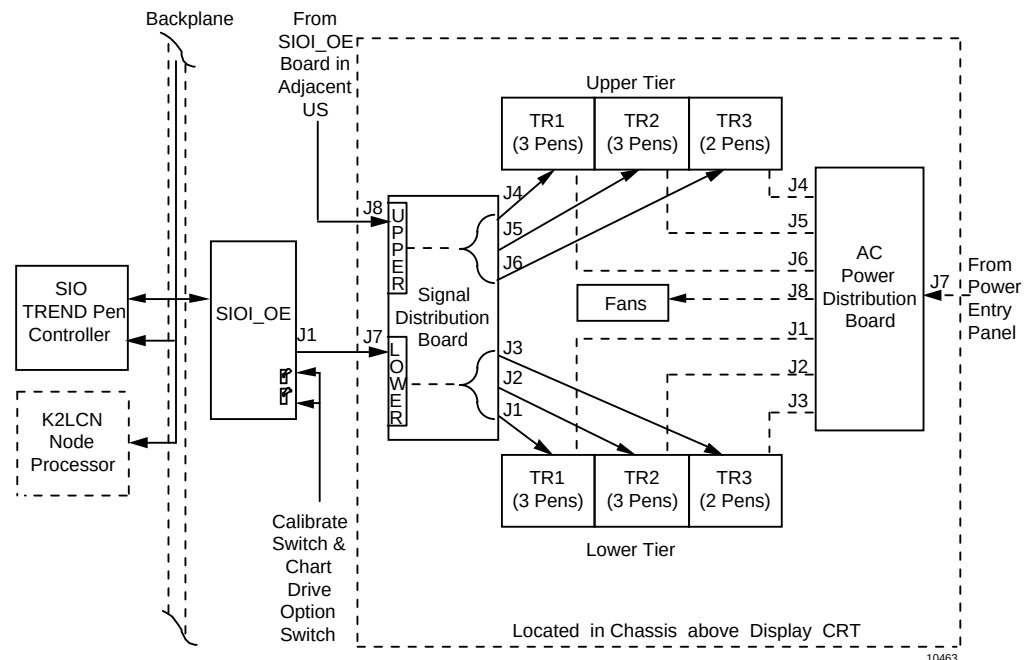


Figure 13-3 Trend Pen Cabling

13.5 Hiway Gateway Cabling

Scope

For an overview of Hiway Gateway preferred access cabling with or without a Hiway Traffic Director (HTD), see Figure 13-4 or Figure 13-5. For Data Hiway cable specifications and address priority assignments, refer to the *Data Hiway Subsystem Site Planning* manual in the *Basic System Summary* binder.

Overview of Preferred Access Cabling - With HTD

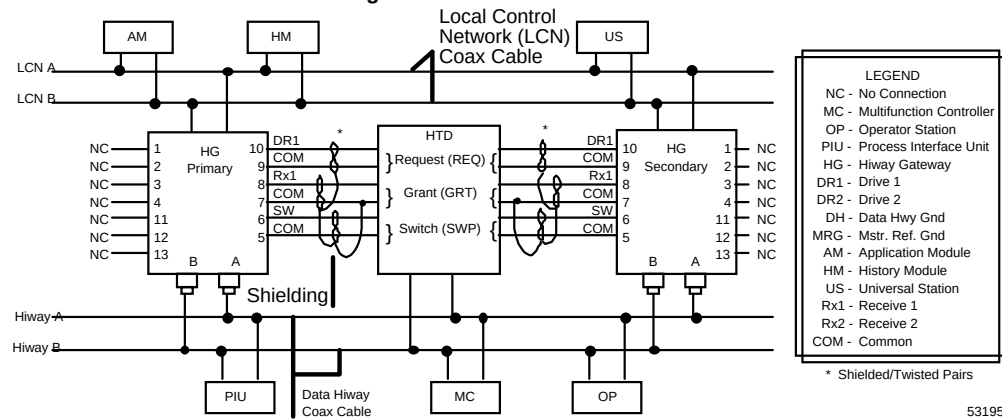


Figure 13-4 Data Hiway Wiring with the HTD

Overview of Preferred Access Cabling - Without HTD

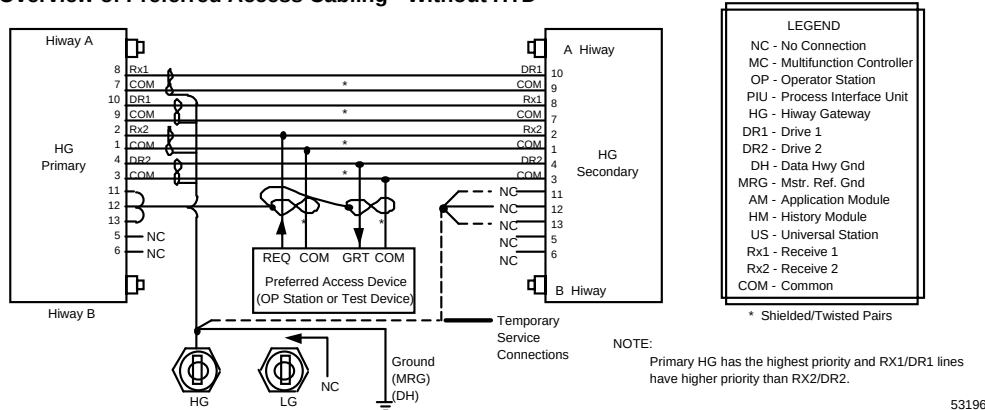


Figure 13-5 Data Hiway Wiring without the HTD

13.6 Enhanced Universal Work Station EUWS) Installation

Scope

The EUWS consists of a 5-slot LCN module with optional media storage devices (either dual floppy-disk drives or dual high-capacity cartridge drives), all of which are mounted in a freestanding enclosure. Peripherals supplied are a 14" CRT color monitor, either an engineering keyboard or a supervisory keyboard, or both, a mouse for cursor control and an optional printer.

The power cable, peripheral cables, and LCN cables enter and exit through the rear panel. Access to the 5-slot module cards is through the front cover and access to the I/O cards is through the rear. Some details of I/O cabling are shown in Figures A-5, A-6, and A-7. See Section 8 for address pinning.

For complete installation and operating details, see the *Enhanced Universal Work Station Installation, Operation, and Service* manual packed with the equipment.

Cable Routing

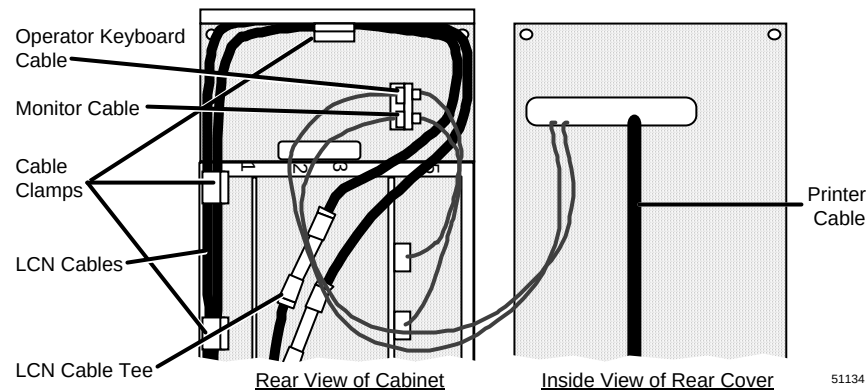


Figure 13-6 Cable Routing at Rear of EUWS

EPDGP Pinning

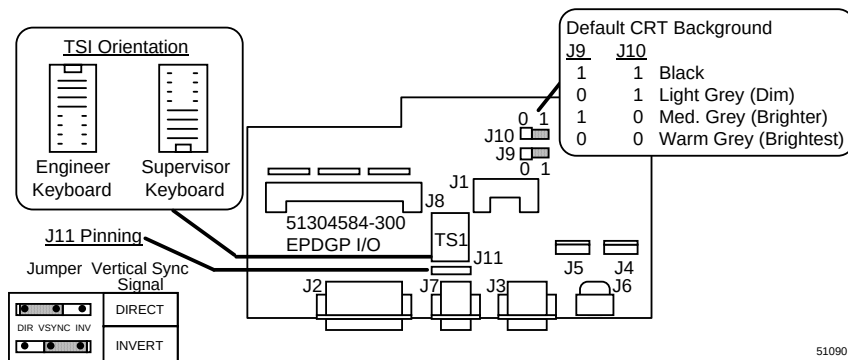


Figure 13-7 EPDGP I/O board Pinning

EPDGP I/O Cabling for EUWS

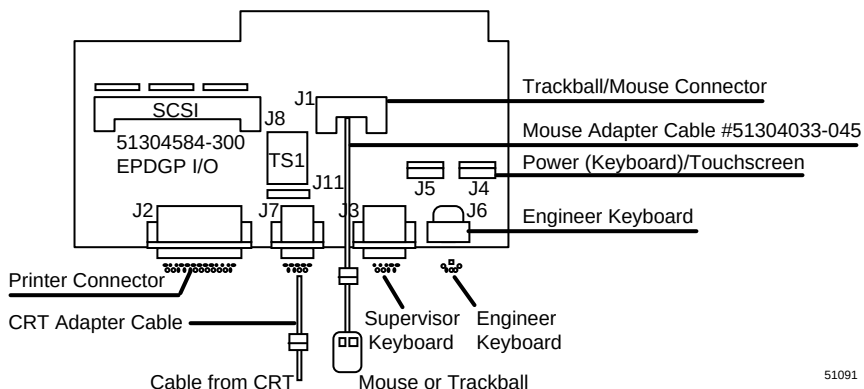


Figure 13-8 EPDGP I/O Cabling for EUWS

13.7 Redundant Application Module Cabling

Board to board cabling (HMPU/AMR)

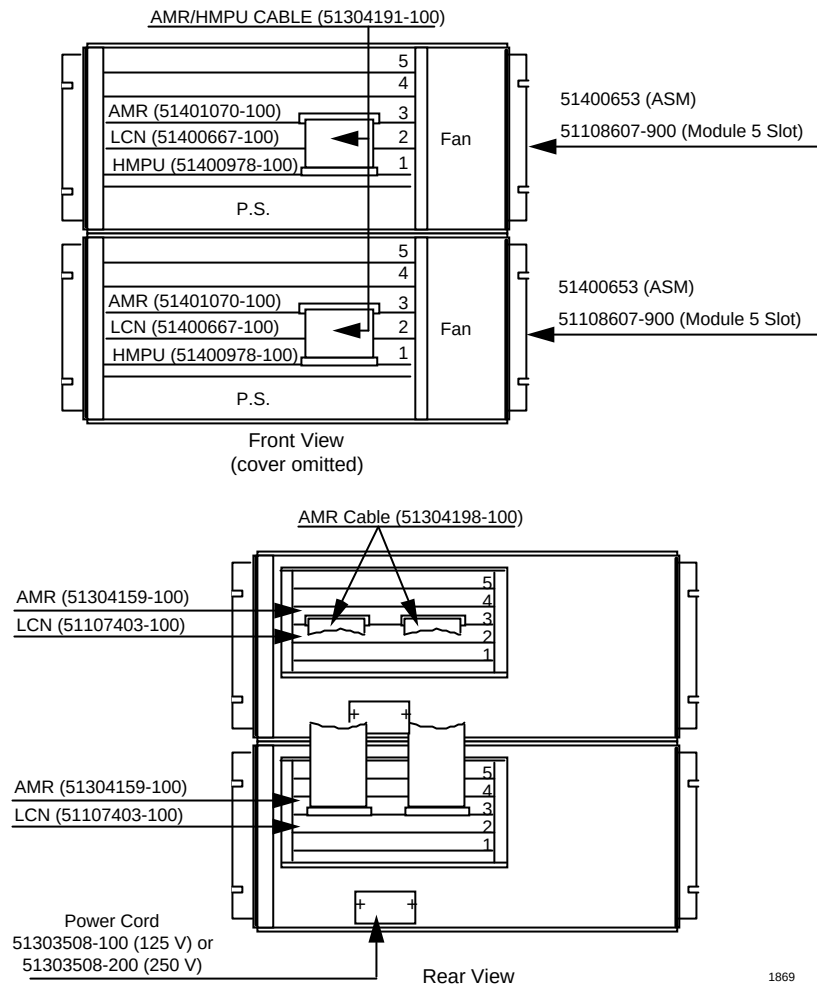


Figure 13-9 Redundant Application Module Cabling with HMPU/AMR Boards

Board to board cabling (K4LCN/EAMR)

This is an example of cabling for Enhanced Redundant Application Modules using K4LCN/EAMR boards as shown in Figures 13-10 through 13-17. The boards can be installed in either node position of the Dual Node Module. There are four combinations of upper/lower node positions.

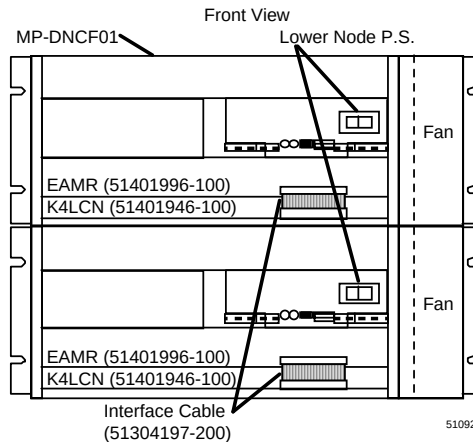


Figure 13-10 Redundant Application Module Cabling (K4LCN/EAMR)

Figures 13-10 and 13-11 show the front and back views of the same application.

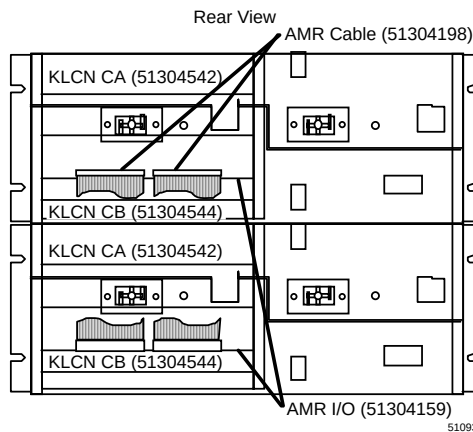


Figure 13-11 Redundant Application Module Cabling (K4LCN/EAMR)

Figure 13-12 and 13-13 show the front and back views of the same application.

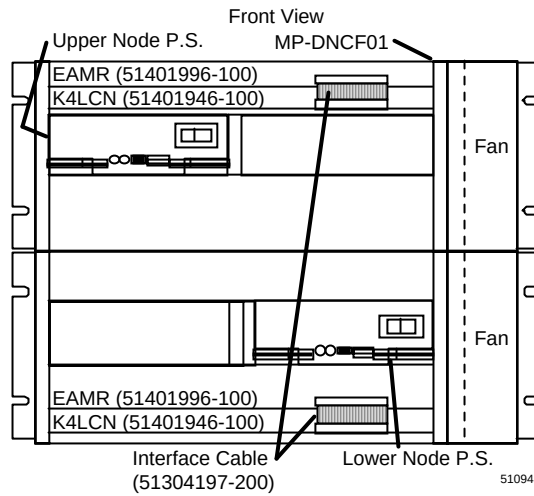


Figure 13-12 Redundant Application Module Cabling (K4LCN/EAMR)

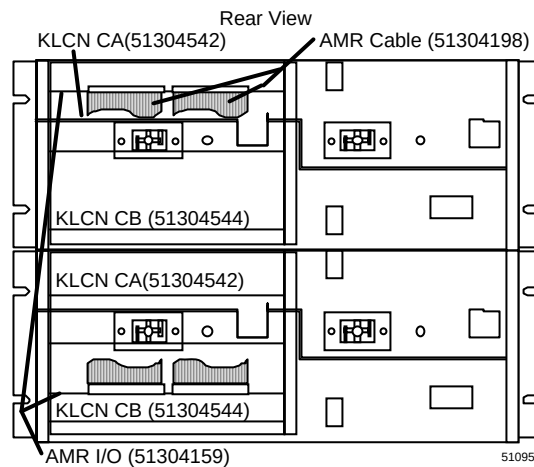


Figure 13-13 Redundant Application Module Cabling (K4LCN/EAMR)

Figure 13-14 and 13-15 show the front and back views of the same application.

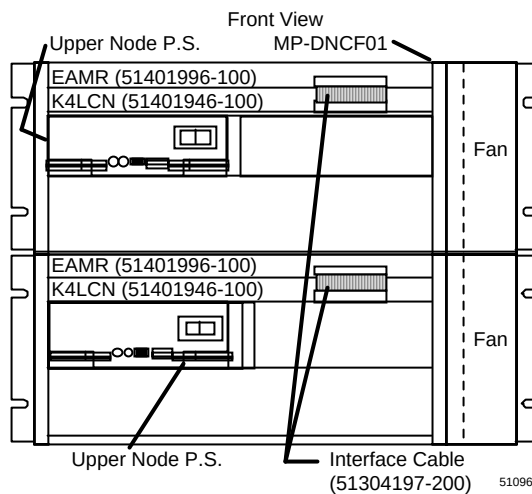


Figure 13-14 Redundant Application Module Cabling (K4LCN/EAMR)

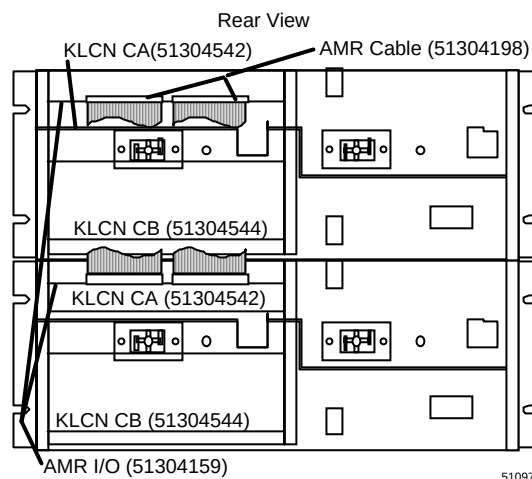


Figure 13-15 Redundant Application Module Cabling (K4LCN/EAMR)

Figure 13-16 and 13-17 show the front and back views of the same application.

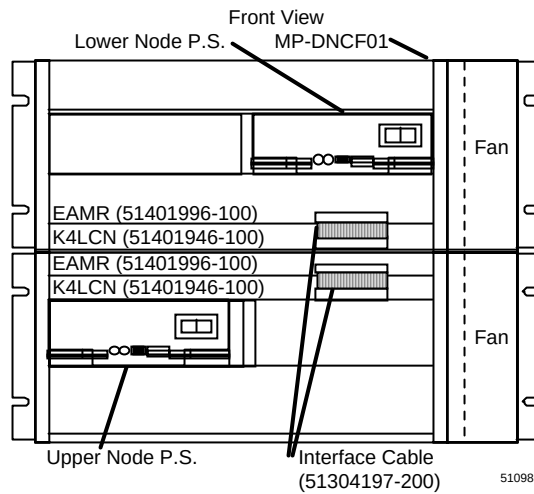


Figure 13-16 Redundant Application Module Cabling (K4LCN/EAMR)

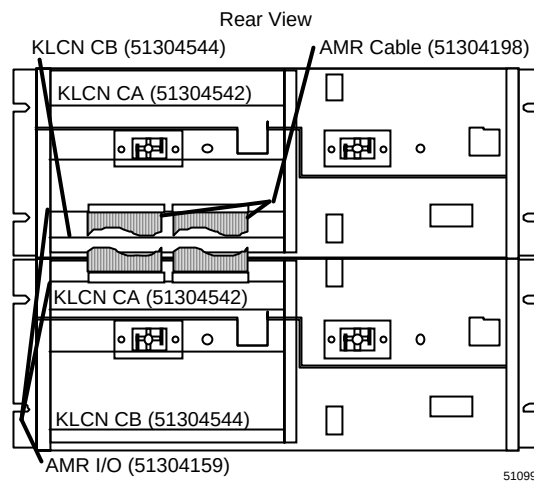


Figure 13-17 Redundant Application Module Cabling (K4LCN/EAMR)

13.8 Universal Station Cabling (Classic Furniture)

Cabling

The Universal Station cabling shown in Figures A-18, A-19, and A-20, when all combined, shows the complete cabling for a Universal Station^X.

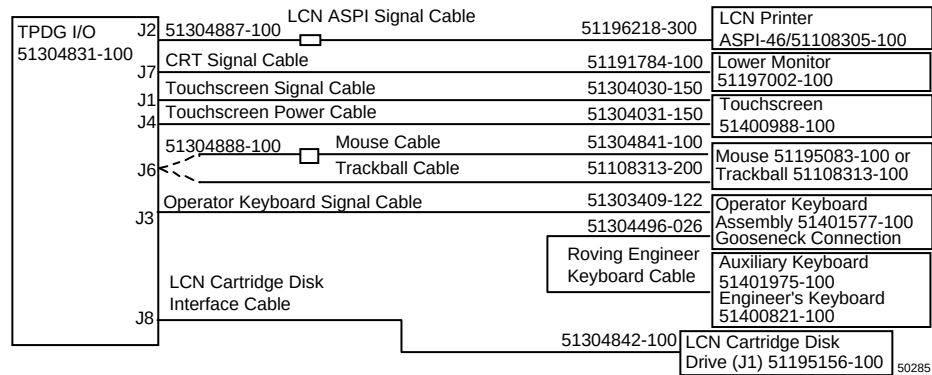


Figure 13-18 Universal Station Internal cabling—Left Half

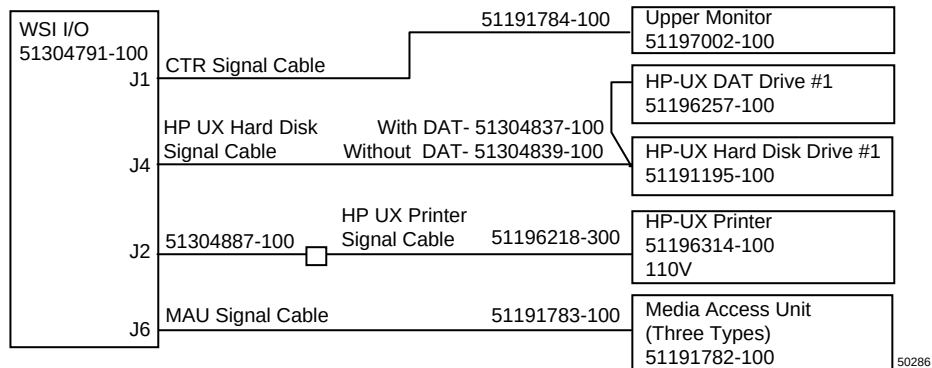


Figure 13-19 Universal Station Internal cabling—Right Half

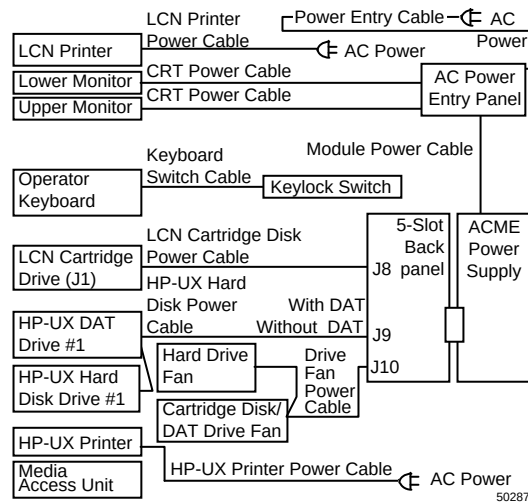


Figure 13-20 Universal Station Cabling

13.9 Universal Station Cabling (Z-Console)

Cabling

The Universal Station (Ergonomic) internal cabling is shown in Figure A-20. Figure 13-21 is supported by Table 13-2, which gives the part numbers for the items referred to by the number in the bubble.

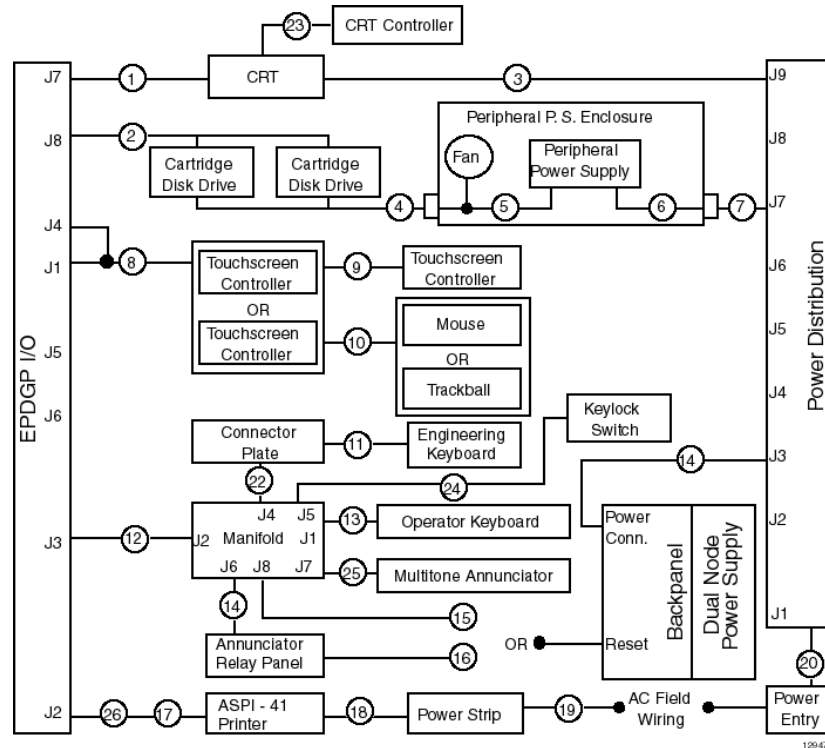


Figure 13-21 Universal Station (Z-Console) Internal Cabling

Table 13-2 Universal Station Internal Cabling key List (Z-Console)		
Item	Part Number	Identification
1	51191784-100	CRT Signal Cable
2	51304842-200	Cartridge Disk Interface Cable (Single Drive)
	51304199-300	Cartridge Disk Interface Cable (Dual Drive)
3	51308020-100	Monitor/P.S. AC Power Cable
4	51308055-100	Drive Cable, DC Power
5	51308051-100	P.S. Enclosure DC Cable (51197004-100) P.S.)
	51308051-200	P.S. Enclosure DC Cable (51197004-200) P.S.)
6	51308052-100	P.S. Enclosure AC Cable (51197004-100) P.S.)
	51308052-200	P.S. Enclosure AC Cable (51197004-200) P.S.)
7	51308020-200	Monitor/Ps AC Power Cable
8	51308047-100	Touchscreen Interface Cable
	51308082-100	Mouse I/C Cable (EPDG)
	51308073-100	Trackball I/C Cable (EPDG)
9	—	Touchscreen I/C Cable (part of Touchscreen)
10	—	Mouse /Trackball Cable (part of subassembly)
11	—	Keyboard Cable (part of subassembly)
12	51308059-100	Manifold/Keyboard Interconnect Cable
13	51308058-100	Keyboard I/F Cable
14	51308019-100	Module AC Power Cable (120 Vac) Right or Close Power Supply
	51308019-100	Module AC Power Cable (120 Vac) Left or Far Power Supply
	51308066-100	Module AC Power Cable (240 Vac) Right or Close Power Supply
	51308066-200	Module AC Power Cable (240 Vac) Left or Far Power Supply
15	51303420-081	Module Status Cable
16	51303420-086	Module Status Cable
17	51303631-003	ASPI-41 Printer Signal Cable (3 Meter)
	51303631-008	ASPI-41 Printer Signal Cable (8 Meter)
	51303631-015	ASPI-41 Printer Signal Cable (15 Meter)

Table 13-2 Universal Station Internal Cabling key List (Z-Console)

Item	Part Number	Identification
18	51191839-100	Cable, Power, Shielded, PVC Jacket
19	51191840-203	Cable, Power, Shielded, PVC Jacket
20	51308054-100	AC Power Entry Cable (120 Vac)
	51308067-100	AC Power Entry Cable (240 Vac)
21	51304047-152	Annunciator Power Cable
22	51308060-100	Eng. Keyboard I/F Cable
23	—	Monitor Remote Harness (supplied with monitor)
24	—	Keylock Switch Cable (part of subassembly)
25	—	Multitone Annunciator cable (part of subassembly)
26	51304506-300	Printer Adapter Cable

13.10 Universal StationX Cabling (Z-Console)

Cabling

The Universal Station^X (Ergonomic) internal cabling is shown in Figures A-22 and A-23. Figures A-22 and A-23 is supported by Table A-3, which gives the part numbers for the items referred to by the number in the bubble.

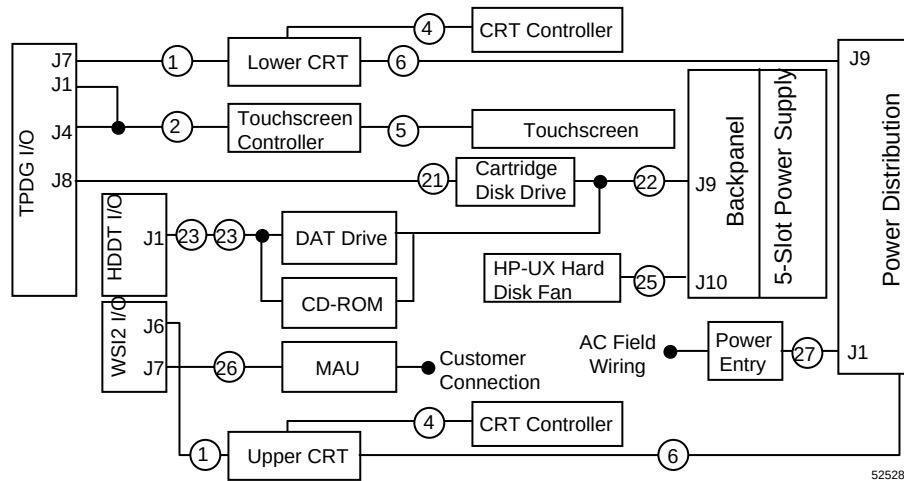


Figure 13-22 Universal Station^X Internal Cabling (Z-Console)

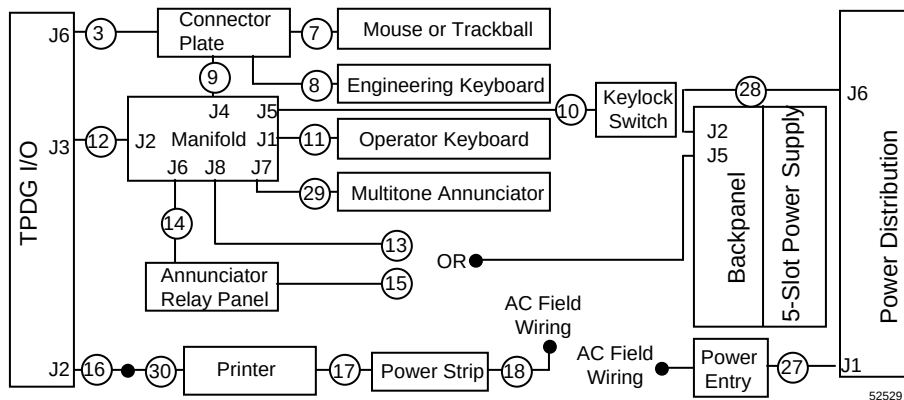


Figure 13-23 Universal Station^X Internal Cabling (Z-Console)

Table 13-3 Universal StationX Internal Cabling key List (Z-Console)

Item	Part Number	Identification
1	51191784-100	CRT Signal Cable
2	51308047-100	Touchscreen Interface Cable
3	51308057-100	Mouse/Trackball Interconnect Cable
4	—	Monitor Remote Harness (supplied with monitor)
5	—	Touchscreen I/C Cable (part of touchscreen)
6	51308020-100	Monitor/P.S. AC Power Cable
7	—	Mouse Trackball Cable (part of subassembly)
8	—	Keyboard Cable (part of subassembly)
9	51308060-100	Eng. Keyboard I/F Cable
10	—	Keylock Switch Cable (part of subassembly)
11	51308058-100	Keyboard I/F Cable
12	51308059-100	Manifold/Keyboard I/C Cable
13	51303420-081	Module Status Cable
14	51304047-152	Annunciator Power Cable
15	51303420-086	Module Status Cable
16	51196218-300	Printer Signal Cable (3 meter)
17	51191839-100	Cable, Power, Shielded, PVC Jacket
18	51191840-203	Cable, Power, Shielded, PVC Jacket
19	51196218-300	Printer Signal Cable (3 Meter)
20	51191839-100	Cable, Power, Shielded, PVC Jacket
21	51304842-200	Cartridge Disk Interface Cable (Single Drive)
	51304199-300	Cartridge Disk Interface Cable (Dual Drive)
22	51308080-200	Cartridge/DAT Drive Cable DC Power
23	51308083-200	Hard Disk Signal Cable w/o DAT (drive)
	51308084-200	Hard Disk Signal Cable with DAT (drive)
24	51304838-100	WDD Power Cable
25	51304585-200	DC Fan Power Cable
26	51191783-100	MAU Signal Cable
27	51308054-100	AC Power Entry Cable (120 Vac)
	51308067-100	AC Power Entry Cable (240 Vac)
28	51308019-200	Module AC Power Cable (120 Vac)

Table 13-3 Universal StationX Internal Cabling key List (Z-Console)

Item	Part Number	Identification
	51308066-200	Module AC Power Cable (240 Vac)
29	—	Multitone Annunciator Cable (part of assembly)
30	51304880-100	Printer Ferrite Cable

14. Options Installation for Universal Stations in Z-Console

14.1 Overview

Scope

This section provides brief installation procedures for adding an option or feature to an existing Universal StationX node in the new ergonomic furniture. These procedures, together with the information contained in the various service manuals, provide the necessary information to modify or expand your system.

14.2 Console Complexing (Physical)

Reference

Refer to Section 3 for instructions about complexing Z-furniture stations into physical consoles.

14.3 Modular Installation

Reference

Refer to the Section 2 in Universal Station (Ergonomic) Service for detailed instructions for adding an upper monitor and adding an additional card file.

14.4 Peripheral Installation and Cabling

Scope

The standard built-in peripherals, cartridge disk drives, and trend-pen recorders, are installed and cabled into the Universal Station at the factory. Removal and replacement procedures are provided in the applicable service manual and installation drawings. Upgrade kits contain individual upgrade kit installation instructions that provide step-by-step instructions for adding or expanding options.

The matrix printer is packaged separately for shipping.

After unpacking, position the printer on the tabletop. Route the power cord and data cables down the back of the console and up through the cable cutouts. Connect the power cord into any unused receptacle on the power entry box. The data cable connections, as applicable to the Universal Station, are shown in Figures B-1, Figure B-2, and Figure B-3.

Printer installation

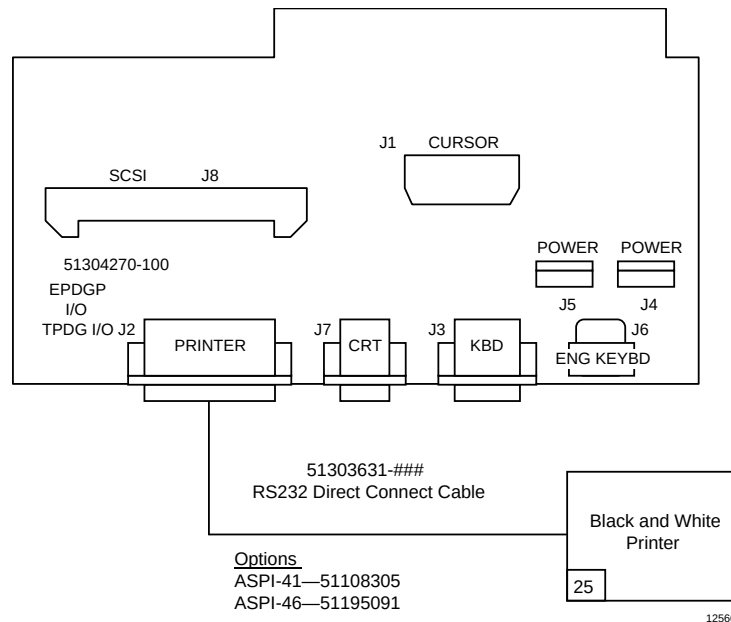


Figure 14-1 Peripheral Cabling (EPDG I/O or TPDG I/O Board)

Manifold cabling

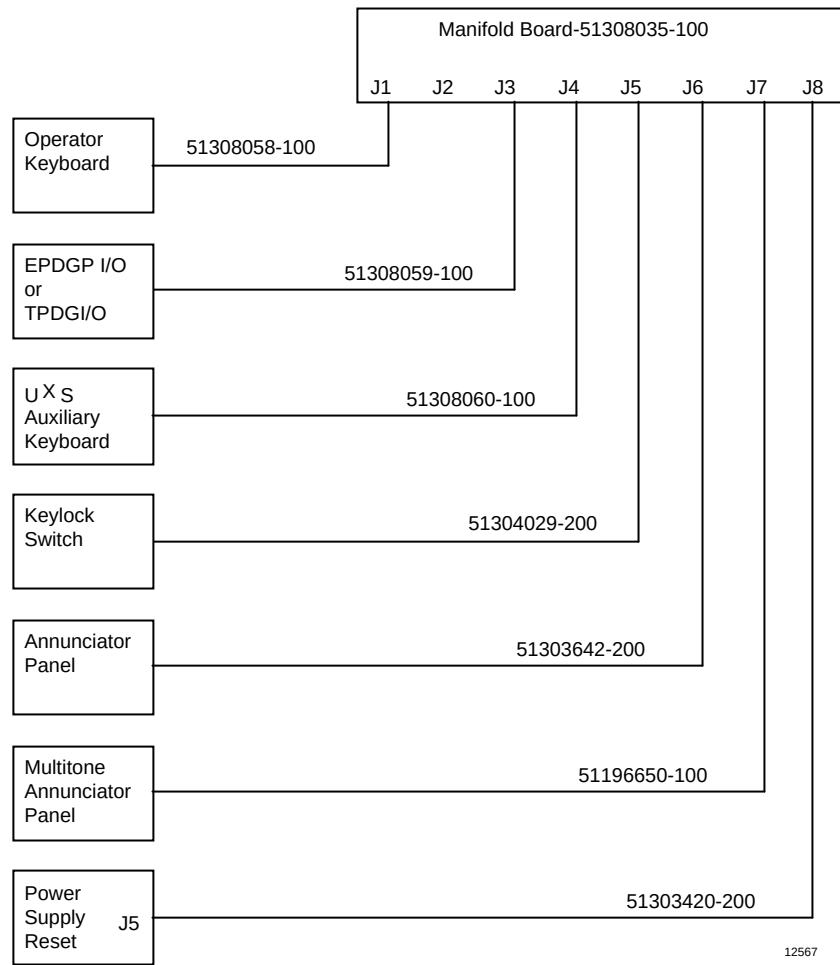
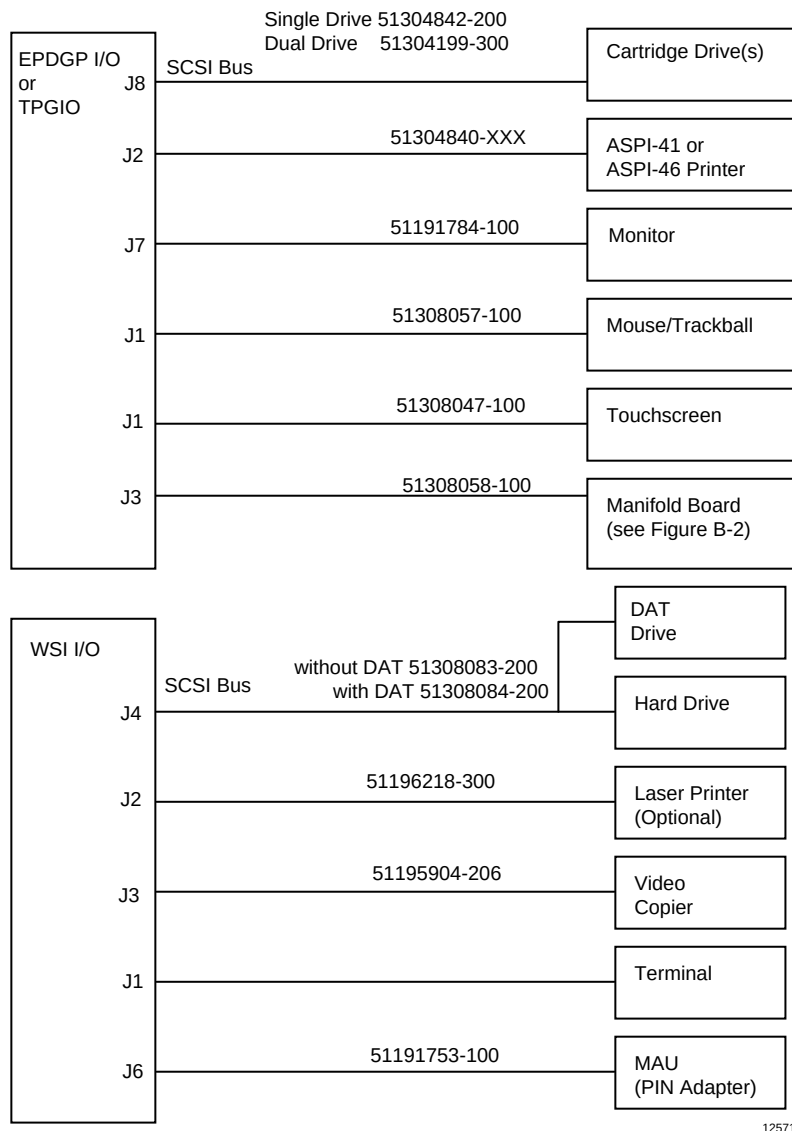


Figure 14-2 Manifold Cabling

U^XS cablingFigure 14-3 Universal Station^X Interface Cabling

US Node Cabling

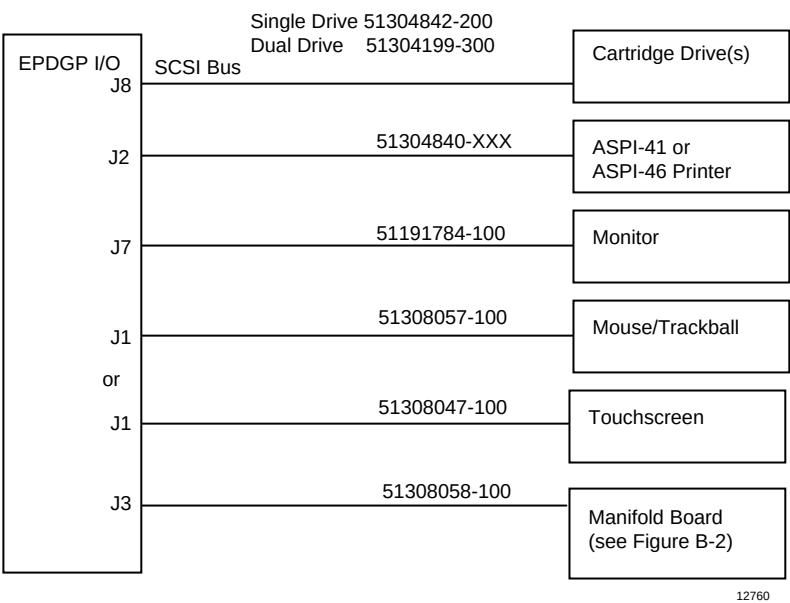


Figure 14-4 US Node Cabling

15. Installations of CE Compliant Options

15.1 Overview

Scope

CE Compliant hardware consists of a special set of I/O boards, for each control board type, that have a metal faceplate mounted to the edge of the board that provides grounding for the shielded cables connected to the I/O board. These boards are connected to the module chassis by a retaining screw, located on the right and left ends of the faceplate.

Figure 15-1 shows an EPDGC I/O board as an example of an I/O board and the grounding screw locations.

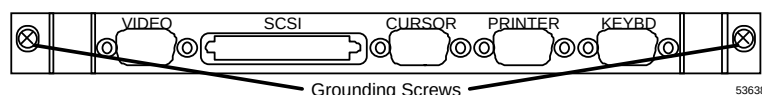


Figure 15-1 EPDGC I/O Faceplate

Attention

These screws must always be firmly into the I/O card file to provide the required grounding.

Cables

Classic and ergonomic Universal Stations, Universal Station^Xs, Micro TDC 3000s, and Universal Work Stations have a complete new set of cables that connect the components in each application.

Cabling diagrams and tables

Each cabling application is shown in this appendix with each different cable numbered for identification. Each number is listed in the table following the diagram. Each entry in the table lists the part number and part identification title. The part identification title is assigned for the first application of the cable. A second usage of the same part may have a title not associated with the application.

15.2 US Cabling—CE Compliant Classic Furniture

Scope

Cabling for Universal Station in classic furniture is shown in Figure 15-2 through Figure 15-12.

US cabling—CE Classic furniture with touchscreen

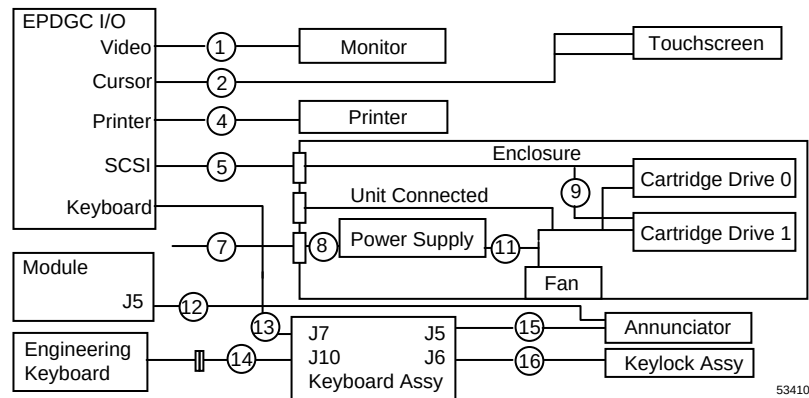


Figure 15-2 US Cabling—CE Classic Furniture with Touchscreen

Cable key list

Table 15-1 US Cabling—CE Classic Furniture with Touchscreen		
Ref. #	Part Number	Application Notes
1	51308120-100	Classic Console EC Video Cable
2	51308123-200	EC Smartscreen Cable
3		Not Applicable
4	51308103-003	Printer Cable – 3 meters/9.8 ft.
	51308103-008	Printer Cable – 8 meters/26.3 ft.
	51308103-015	Printer Cable — 15 meters/49.2 ft.
5	51192054-102	Cable, SCSI Shielded, 50 position
6		Not Applicable
7	51308094-600	Classic Module AC Power Cable
8	51308052-200	P.S. Enclosure AC Cable
9	51305070-100	Cable, SCSI (1 Drive)
	51305070-200	Cable, SCSI (2 Drives)
10	51308055-100	Drive Cable DC Power
11	51308051-200	DC Power Cable, Fan
12	51303420-586	Module Status Cable
13	51308126-100	Keyboard Cable
14	51304496-026	Engineer's Keyboard Cable
15	51304047-652	Annunciator Power Cable
16	51304029-500	Keylock Switch Cable

US cabling—CE Classic with mouse and trackball

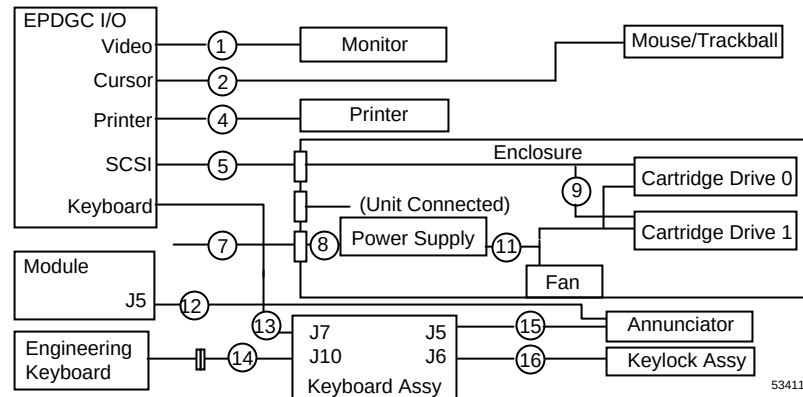


Figure 15-3 US Cabling—CE Classic Furniture with Mouse and Trackball

Cable key list

Table 15-2 US Cabling—CE Classic Furniture with Mouse/T.S.		
Ref. #	Part Number	Application Notes
1	51308120-100	Classic Console EC Video Cable
2	51308099-100	EPDGC Mouse/Trackball I/F Cable
3		Not Applicable
4	51308103-003	Printer Cable – 3 meters/9.8 ft.
	51308103-008	Printer Cable – 8 meters/26.3 ft.
	51308103-015	Printer Cable – 15 meters/49.2 ft.
5	51192054-102	Cable, SCSI Shielded, 50 position
6	51305070-100	Cable, SCSI (1 Drive)
	51305070-200	Cable, SCSI (2 Drives)
7	51308129-100	Skynet-Cart Drive Power Cable
8	51308125-100	Cartridge Drive Encl Power Harness
9	51308051-200	DC Power Cable, Fan
10	51308055-100	Drive Cable, DC Power
11	51308051-200	DC Power cable, Fan
12	51303420-586	Module Status Cable
13	51308126-100	Keyboard Cable
14	51304496-026	Engineer's Keyboard Cable
15	51304047-652	Annunciator Power Cable
16	51304029-500	Keylock Switch Cable

15.3 US Cabling—CE Compliant Z-Console Furniture

Scope

Cabling for the Universal Station in ergonomic furniture is shown in the following figure and table. The number in the circles refers to the line item in the table below.

US cabling— Z-Console with touchscreen

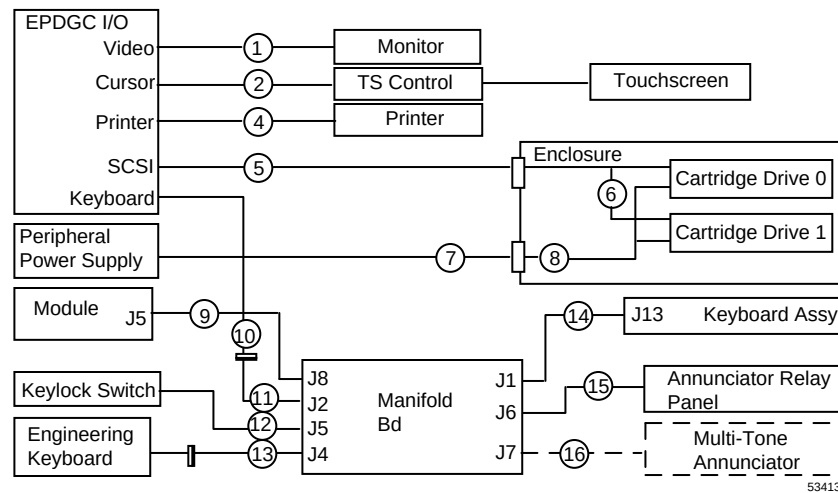


Figure 15-4 US Cabling—CE Z-Furniture with Touchscreen

Cable key list

Table 15-3 US Cabling—CE Z-Furniture with Touchscreen		
Ref. #	Part Number	Application Notes
1	51301784-200	Cable, CRT Signal
2	51308099-100	Mouse/Trackball Interface Cable
3		Not Applicable
4	51308103-003	Printer Cable – 3 meters/9.8 ft.
	51308103-008	Printer Cable – 8 meters/26.3 ft.
	51308103-015	Printer Cable – 15 meters/49.2 ft.
5	51192054-101	Cable, SCSI Shielded, 50 position
6	51305070-100	Cable, SCSI (Single Drive)
	51305070-100	Cable, SCSI (Dual Drive)
7	51308129-100	Skynet-Cart Drive Power Cable
8	51308125-100	Cartridge Drive Enclosure Power Harness
9	51308051-200	Module Status Cable
10	51308105-100	EPDGC/TPDGC Keyboard I/C Cable
11	51303420-622	Annunciator Cable
12	51304029-200	Keylock Switch Cable
13	51308060-100	Eng Keyboard I/F Cable
14	51308058-100	Keyboard I/F Cable
15	51304047-552	Annunciator Power Cable
16	51196650-100	Keyboard Alarm Assy (Multitone Annunciator)

15.4 Universal StationX Cabling—CE Classic Furniture

Scope

Cabling for Universal Station^X in classic furniture is shown in the following figure and table.

U^XS cabling—CE classic furniture

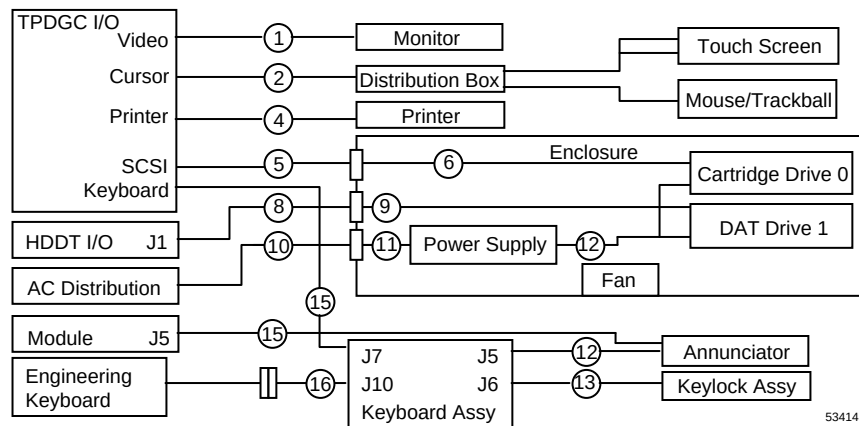


Figure 15-5 U^XS Cabling—CE Classic Furniture

Cable key list

Ref. #	Part Number	Application Notes
1	51191784-100	Cable, CRT Signal
2	51308119-100	TPDGC I/O Distribution Box Cable
3	51308123-100	EC Smartscreen Cable
4	51308103-003	Printer Cable – 3 meters/9.8 ft.
	51308103-008	Printer Cable – 8 meters/26.3 ft.
	51308103-015	Printer Cable – 15 meters/49.2 ft.
5	51192054-102	Cable, SCSI Shielded, 50 position
6	51305070-100	Cable, SCSI
7		Not Applicable
8	51192054-102	Cable, SCSI, Shielded, 50 Pos
9	51305070-200	Cable, SCSI
10	51308094-600	Classic Module AC Power Cable
11	51308051-200	P.S. Enclosure DC Cable
12	51308052-200	P.S. Enclosure AC Cable
13	51308055-100	Drive Cable DC Power
14	51303420-586	Module Status Cable
15	51308126-100	Keyboard Cable
16	51304496--026	Engineering Keyboard Interface Cable
17	51304047-652	Annunciator Power Cable
18	51304029-500	Keylock Switch Cable

15.5 Universal StationX Cabling—Z-Furniture

Scope

Cabling for Universal Station^X in ergonomic furniture is shown in the following figure and table. The number in the circles refers to the line item in the table below.

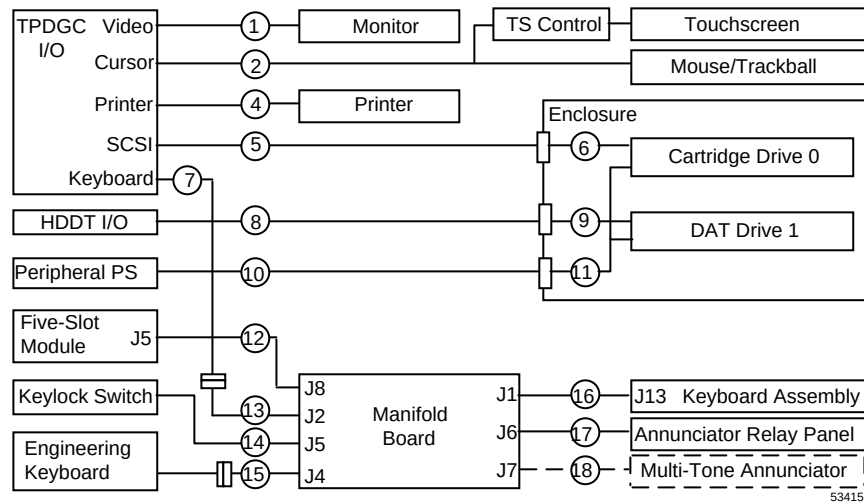


Figure 15-6 U^XS Cabling—CE Z-Furniture with DAT Drive

Cable key list

Table 15-4 U^XS Cabling—CE Z-Furniture with DAT Drive		
Ref. #	Part Number	Application Notes
1	51191784-200	Monitor Interface Cable
2	513408102-100	Touchscreen/Mouse/Trackball Interface Cable
3		Not Applicable
4	51308103-003	Printer Cable – 3 meters/9.8 ft.
	51308103-008	Printer Cable – 8 meters/26.3 ft.
	51308103-015	Printer Cable – 15 meters/49.2 ft.
5	51192054-101	Cable, SCSI Shielded, 50 position
6	51308128-200	EC SCSI Cable
7	5130105-100	EPDGC/TPDGC Keyboard I/C Cable
8	51192054-101	Cable, SCSI Shielded, 50 position
9	51308128-100	EC SCSI Cable
10	51308130-100	Cartridge Drive Power Cable
11	51308125-100	Cartridge Drive Encl Power Harness
12	51198051-200	Module Status Cable
13	51303420-622	Annunciator Cable
14	51304029-200	Keylock Switch Cable
15	51308060-100	Eng Keyboard I/F Cable
16	51308058-100	Keyboard I/F Cable
17	51304047-552	Annunciator Power Cable
18	51196650-100	Keyboard Alarm Assy (Multitone Annunciator)

15.6 MicroTDC3000 Cabling

Scope

Cabling for a Micro TDC 3000 is shown in the following figure and table. The number in the circles refers to the line item in the table below.

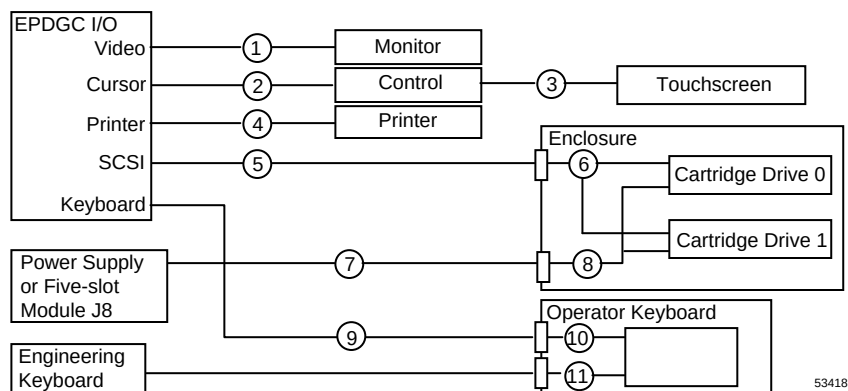


Table 15-5 MicroTDC300 Cabling—Standard

Cable key list

Table 15-6 MicroTDC3000 Cabling—Standard		
Ref. #	Part Number	Application Notes
1	51192126-200	Video Cable
2	51308099-100	Mouse/Trackball Interface Cable
3		Not Applicable
4	51308103-003	Printer Cable – 3 meters/9.8 ft.
	51308103-008	Printer Cable – 8 meters/26.3 ft.
	51308103-015	Printer Cable – 15 meters/49.2 ft.
5	51192054-101	Cable, SCSI Shielded, 50 position
6	51305070-200	Cable, SCSI
7	51308110-100	Cartridge DC Power Cable
8	51308109-100	Cartridge DC Power Cable (in enclosure)
9	51308105-200	Keyboard Interface Cable
10	51308104-100	Keyboard Interface Cable (in keyboard tray)
11	51304496-046	Engineer's Keyboard Cable (in keyboard tray)

Touchscreen option

Cabling for a Micro TDC 3000 is shown in the following figure and table. If the touchscreen special has been installed, the following figure and table represents the cabling. The number in the circles refers to the line item in the table below.

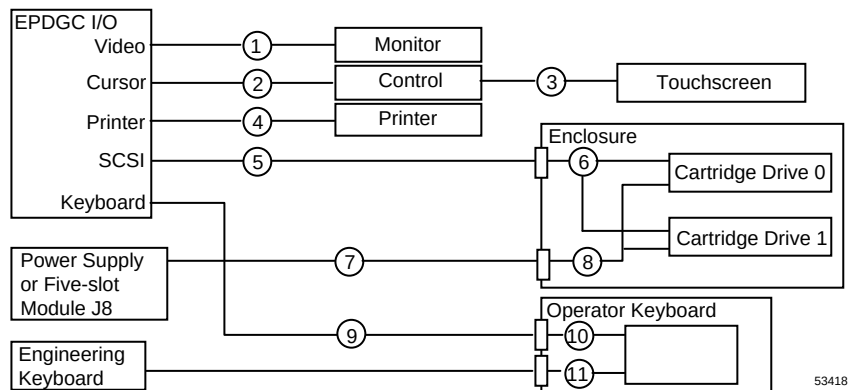


Figure 15-7 MicroTDC3000 Cabling—Touchscreen

Cabling key list

Table 15-7 MicroTDC3000 cabling—Touchscreen		
Ref. #	Part Number	Application Notes
1	51192126-200	Video Cable
2	51308097-100	Touchscreen Interface Cable
3	51308114-100	Touchscreen Interface Cable
4	51308103-003	Printer Cable – 3 meters/9.8 ft.
	51308103-008	Printer Cable – 8 meters/26.3 ft.
	51308103-015	Printer Cable – 15 meters/49.2 ft.
5	51192054-101	Cable, SCSI Shielded, 50 position
6	51305070-200	Cable, SCSI
7	51308110-100	Cartridge Drive Power Cable
8	51308109-100	Cartridge Drive Power Cable (in enclosure)
9	51308105-200	Keyboard Interconnecting Cable
10	51308104-100	Keyboard Interface Cable (in keyboard tray)
11	51304496-046	Engineer's Keyboard Cable (in keyboard tray)

15.7 Application ModuleX (AXM) Cabling

Cabling

Cabling for Applications Module^X (A^XM) mounted in classic furniture, is shown in the following figure and table. The number in the circles refers to a line item in the table below.

A^XM in Classic Furniture

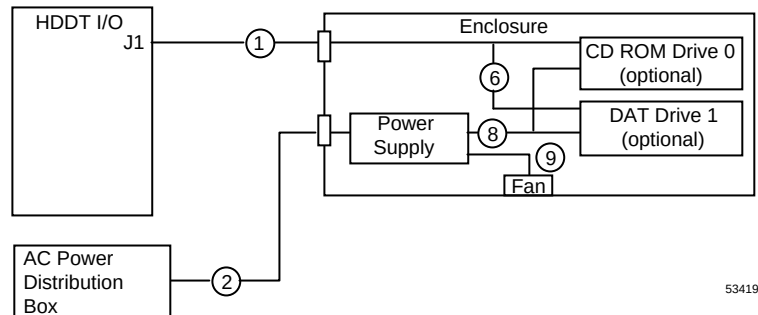


Figure 15-8 A^XM Cabling in Classic Furniture

Cable key list

Table 15-8 A ^X M Cabling in Classic Furniture		
Ref. #	Part Number	Application Notes
1	51192054-102	Cable, SCSI Shielded, 50 position
2	51305070-100	Cable, SCSI
3	51308094-600	Classic Module AC Power Cable
4	51308052-200	DC Power Cable, Power Supply End
	51308055-100	DC Power Cable, Cartridge End
5	51308051-200	DC Power Cable, Fan

HDDT/DAT drive cabling

The cable connections for an Applications Module^X (A^XM) in both classic and ergonomic furniture, are shown below.

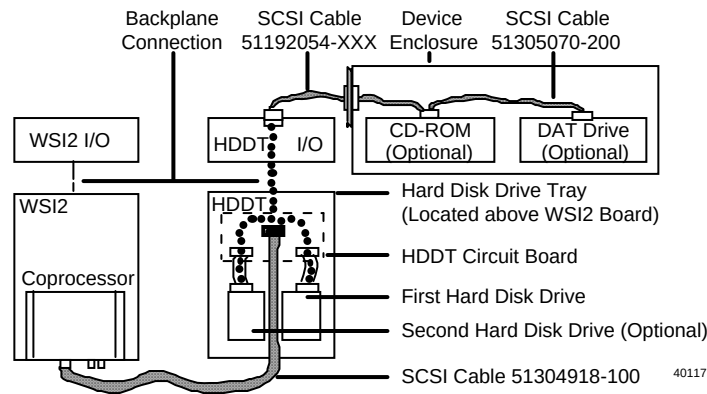


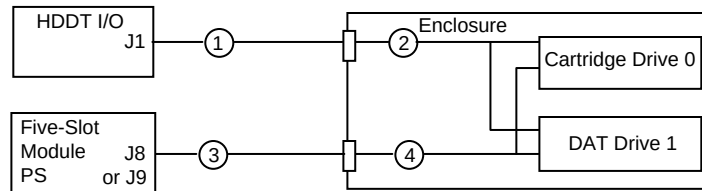
Figure 15-9 HDDT/DAT Drive Cabling

Drive location

When an Application ModuleX is located in a station.

A^XM Cabling in Z-Furniture

Cabling for Applications Module^X (A^XM) mounted in classic furniture, is shown in the following figure and table. The number in the circles refers to a line item in the table below.



53420

Figure 15-10 A^XM Cabling in Z Furniture

Cable key list

Table 15-9 A ^X M Cabling in Z-Furniture		
Ref. #	Part Number	Application Notes
1	51192054-101	Cable, SCSI Shielded, 50 position
2	51308128-100	HDDT-DAT SCSI Interface Cable—Enclosure
3	51308130-100	DC Power Cable
4	51308125-100	DC Power Cable—Enclosure

16. FTE Switch Installation

16.1 Overview

Scope

With TPN R684, you can replace the traditional coaxial UCN cable system with Fault Tolerant Ethernet (FTE) based communication network. This network utilizes Ethernet cables rather than coaxial cables. The Enhanced Network Interface Module (ENIM), which is FTE-based, provides a gateway function between the LCN and an Enhanced Universal Control Network (EUCN).

The EPNI2 in conjunction with the EPNI2 I/O enables communication between the Module Bus and the FTE network. The EPNI2 functions as a 10/100 Mbit/sec 802.3 Network Interface.

16.2 Installation of Ethernet switches in LCN cabinet

Ethernet switches connect the ENIM and the EHPM over the EUCN network. The number of switches required depends on the distance between the ENIM and the EHPM and the number of EHPM added to the EUCN network.

There are three Ethernet switch configurations offered for mounting on the LCN cabinet. You can select a configuration based on your requirement:

Table 16-1 Level 2 LCN Cabinet Switch Configuration

LCN Cabinet Ethernet Switch Configuration #	Criteria for Selecting Switch Configuration	Ethernet Switch Upgrade Configuration Description	Model Number	Mounting Space Consumed
1	The number of EHPM clusters is less than or equal to nine. One redundant Level 2 dual-purpose uplink port (see note) and up to eight redundant Level 2 copper ports are available to communicate with EHPM clusters having a routed cable distance less than or equal to 100 meters of the level 2 switch.	• 8 redundant Level 1 copper ports • 8 redundant Level copper ports • 2 redundant Level 2 dual-purpose ports • 2 AC power adapters	NE-ZFTEB2	28 inches
2	The number of EHPM clusters is less than or equal to nine. One redundant Level 2 dual-purpose uplink port (refer to the note) and up to eight redundant Level 2 fiber ports are available to communicate with EHPM clusters..	• 8 redundant Level 1 copper ports • 8 redundant Level 2 fiber ports • 2 redundant Level 2 dual-purpose ports • 2 AC power adapters	NE-ZFTEB3	28 inches
3	The number of EHPM clusters is less than or equal to seventeen. One redundant Level 2 dual-purpose uplink port (refer to the note), up to eight redundant Level 2 copper ports and up to eight redundant Level 2 fiber ports are available to communicate with EHPM clusters.	• 8 redundant Level 1 copper ports • 8 redundant Level 2 copper ports • 8 redundant Level 2 fiber ports • 2 redundant Level 2 dual-purpose ports • 2 AC power adapters	NE-ZFTEB4	28 inches

NOTE

If the redundant Level 2 dual-purpose port is configured to use optical fiber, it can serve as an uplink port for an EHPM at a routed cable distance of 2 kilometers or less. If the redundant Level 2 dual-purpose port is configured as a copper port, it can serve as an uplink port for an EHPM at a routed cable distance of 100 meters or less.

For more information about EUCN topology and planning, refer to the *Universal Control Network Planning* and *LCN Planning* manuals.

Attention

Select a cabinet with 14 inches or 28 inches of rack space at the top of the cabinet or reconfigure the cabinet to position the switch bracket at the top of the cabinet. Refer to “Table 16 1 Level 2 LCN Cabinet Switch Configuration” for selecting a configuration for the Ethernet switch.

Procedure

Perform the following procedures to mount the Ethernet switches in a LCN cabinet.

Table 16-2 Connecting the Power adapter to the Ethernet switches

Step	Action
1	Note: Perform this procedure on a worktable. On the left side of the Ethernet switch, pry open the side panel using a flat-blade screwdriver.
2	The power adapter has upper and lower latches to secure it to the switch. On the power adapter, pull the upper latches up and the lower latches down.
3	Align the connectors on the switch and the power adapter, and slide the power adapter and the switch together.
4	Push in the upper and lower latches to secure the power adapter to the switch. If you are assembling NE-ZFTEBC kit, skip steps 5 through 8.
Adding Expansion Modules	
5	On the right side of the Ethernet switch, pry open the side panel using a flat-blade screwdriver. Repeat this step on the expansion module to remove the cover from the left side.
6	On the Ethernet switch and the expansion module, pull the upper latches up and the lower latches down.
7	Align the connectors on the switch and the expansion module, and slide the switch and the expansion module together.
8	Push in the upper and lower latches to secure the expansion module to the switch. Repeat steps 5 through 8 to add another expansion module.
9	1) Remove the two-position connector plug from the receptacle on the power adapter. 2) Plug in the DC power clip connecting the power adapter and the Ethernet switch. 3) Then, tighten the screws on the DC power clip.

Step	Action								
10	Connect the power supply cable to the input terminal block connector. 1) Remove the cover and then the screws on the input terminal block connector. 2) Place the appropriate cable connectors under each screw and tighten it. 3) Replace the safety cover on the input terminal block connector. <table border="1"> <thead> <tr> <th>Wire color</th><th>Connection</th></tr> </thead> <tbody> <tr> <td>Green</td><td>Ground</td></tr> <tr> <td>Blue</td><td>Neutral</td></tr> <tr> <td>Brown</td><td>Line</td></tr> </tbody> </table>	Wire color	Connection	Green	Ground	Blue	Neutral	Brown	Line
Wire color	Connection								
Green	Ground								
Blue	Neutral								
Brown	Line								
11	1) On the lower right hand corner of the switch, remove the screw. 2) Place the ground wire, replace the screw, and tighten it. The other end of the ground wire must be connected to the chassis ground under the head of one of bracket mounting screws.								

Illustration

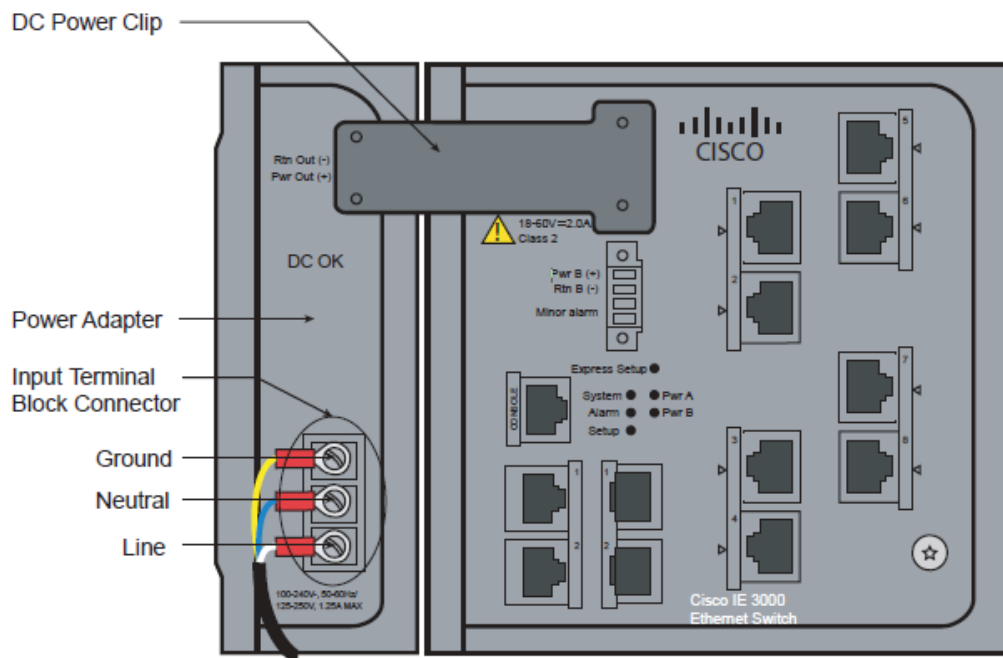


Figure 16-1 Power adapter connected to an Ethernet Switch

Table 16-3 Installing the assembled switches on the mounting bracket

Step	Action
1	Note: Perform this procedure on a worktable. Place the mounting bracket on a worktable.
2	Place the assembled switch on the mounting bracket and secure it using the available hex head machine screw.
3	Route the plug end of the AC power cord, which is connected to the input terminal block connector of the power adapter, through the grommet hole. Ensure that the power cord remains behind the mounting bracket.

Illustration

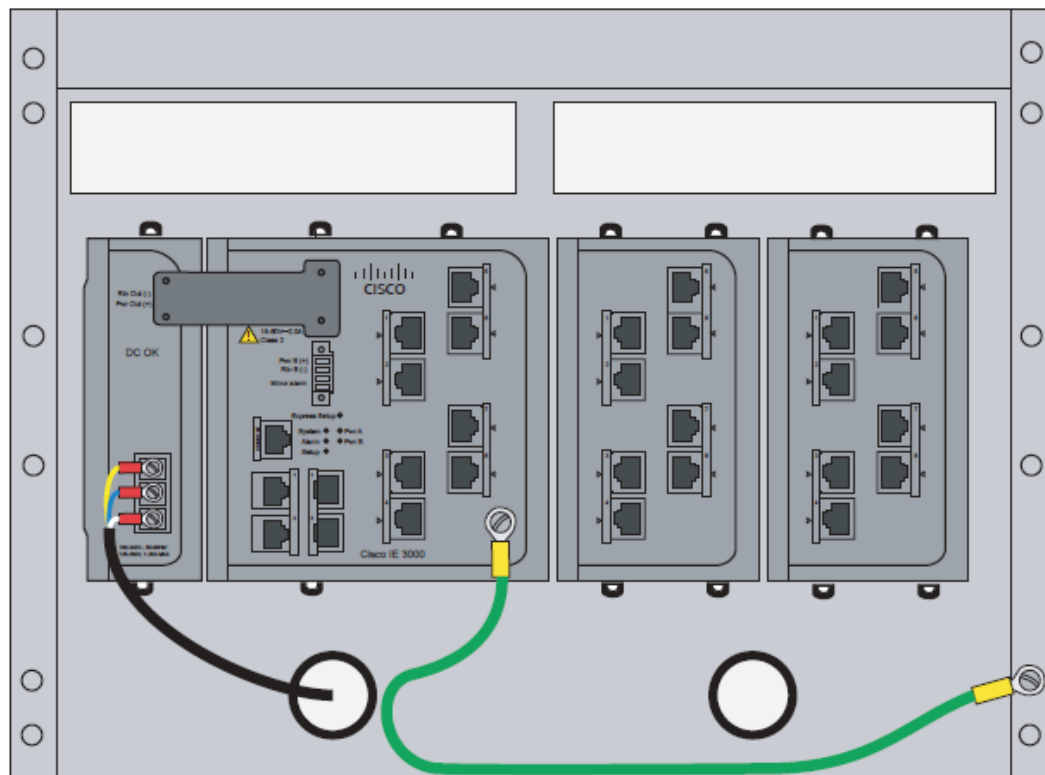


Figure 16-2 Mounting Bracket with Ethernet switches installed

Table 16-4 Installing Ethernet switches in the LCN Cabinet

Step	Action
1	Note: Select a LCN cabinet that has the required rack space at the top based on the selected switch configuration mentioned in "Table 16 1, Level 2 LCN Cabinet Switch Configuration." If not, reconfigure the cabinet and create the required space to position the switch bracket in the top. Lift and place the assembled mounting bracket with the Ethernet switches into position in the LCN cabinet.
2	Secure the assembled mounting bracket with four hex head screws.
3	Connect the ground wires. Place the free end of the ground wire from the Ethernet switch under the head of the screw used to secure the assembled mounting bracket. Repeat this step for the redundant Ethernet switch.
4	From the rear of the mounting bracket, route all the FTE A cables through one of the grommet holes and all the FTE B cables through the other.
5	Plug the labeled FTE A and B cables into appropriate ports on the switches. Ensure that the connectors are plugged in straight and there is not strain on the cable. For information about specification and installation of copper and fiber optic cabling to support FTE systems, refer to <i>FTE Cabling Best Practices (WP-07-01-ENG)</i>.
6	Insert the plug end of the AC power cord in to an available receptacle in the cabinet power entry. Ensure that the power cord and all the FTE cables remain behind the mounting bracket. Note: If the LCN cabinet has a power entry that supports 120/240V Twist Lock Plugs, then use the AC power cables (P/N 51305409-100 for 120V / 51305410-100 for 240V) in addition to the AC power cord (51307101-200) provided with the upgrade kit to connect to the Cisco power adapter.
7	Check the FTE network connection.

17. FTE Switch Configuration

17.1 Honeywell's Switch Configuration Files

Honeywell provides switch configuration files that, when implemented, configure the FTE switch ports for different node types according to defined requirements. Table 17-3 lists a few switch configuration files.

Location of switch configuration files

Switch configuration files are available from the following locations.

- SSOL or Network Services – this location contains the newest switch files
- For TPS users: on the Utilities and Load Modules media

Procedure

Perform the following procedure to obtain the latest switch configuration files.

Table 17-1 Obtaining the latest switch configuration files

Step	Action
1	In your web browser application, type the following URL: https://www.honeywellprocess.com/en-US/support/Pages/default.aspx
2	Click Login to My Account . Note: If you are a new user, register for access first.
3	Type your user name and password, and then click Login .
4	Under SUPPORT , on the Search bar, type " CISCO Configuration Files ". Search results are displayed.
5	In the list, click the " CISCO Configuration Files ".
6	Save the .zip file to your local machine.
7	Double-click the downloaded .zip file to open it.
8	Read the Important Notice.txt file for recent update information.
9	Archive all existing switch configuration files: • Navigate to Program Files\Honeywell\FTEDriver\SwitchConfigurationFiles\Cisco Catlyst <Model of switch>. • Rename the Switch Configuration folder to Old Switch Configuration. • Create a new Switch Configuration folder.
10	Run the switch configuration executable: • Double-click the Cisco_Configuration_MMMYY.exe file, where MMMYY is the latest month and year of the file. • Click Yes if you receive a Win-Zip caution.

Table 17-1 Obtaining the latest switch configuration files

Step	Action
11	From the Win-Zip self extractor dialog, accept C:\ drive as the default or select the drive where Honeywell software is installed. Results: All the switch configuration files will be unzipped to the following location: Program Files\Honeywell\FTEDriver\SwitchConfigurationFiles

Configuration order for switch ports

The specific configuration file you choose defines your switch options and how each switch port is configured. Uplink ports are configured first, FTE port on EUCN switches are configured second. The following table summarizes the switch port configuration settings.

Table 17-2 Switch port configuration settings					
Configuration order	Port Type	Spanning Tree	Status	Duplex	Speed
1 st	Uplink ports	Enabled	Enable	Full	100 Mbps
2 nd	FTE	Portfast/BPDUGuard	Enable	Full	Auto

Switch configuration examples

This section contains examples that illustrate how switch ports are configured using the switch configuration files.

Table 17-3 provides details on the specific parameters implemented in each switch configuration file. The Level column indicates the network level for which the switch configuration file must be used.

Note: All the switch configuration files with the prefix “eucn_” in the paths `\FTEDriver\Switch_Configuration_Files\cisco ie3000\200 or less node l1 and split configurations` and `FTEDriver\Switch_Configuration_Files\cisco ie3000\greater than 200 node l1 and split configuration files` are for the IE3000 switch on the EUCN network.

Table 17-3 Cisco IE3000 switch configuration files						
File name	Level	Switch type	No. of uplinks	No. of 10 MB FTEBs	No. of FD 100 MBs	VLAN101 configured
eucn_L1_base_ie3000_4	1	4 port Base Unit	Gigabit only	4	0	No
eucn_L1_base_ie3000_16	1	Two 8 port Expansions	0	16	0	No

8base_8e xpansion_ ie3000_s plit*	1 and 2	8 port Base and 8 port Expansion	Gigabit only	7	7	No
* These are split configurations. These are preferred for level1/2 combinations. 2 interfaces are lost to the L1/L2 crossover in the switch.						
IE3000 switches are loaded with an image with the “base” characteristic in the file name. Additional ports are desired expansion modules which can be added to the base switch. After loading a “base” configuration a configuration with the “expansion” characteristic in the file name can also be loaded. The two split configurations are an exception. They are configured to have an 8 port base module and an 8 port copper expansion module. The “expansion” configuration files cannot be applied to the split switch configurations without modification.						

17.2 Configuring Cisco Switches

Use the procedures in this section to install the switch configuration files to the node, and configure the Cisco switches for FTE using the switch’s command line interface and the correct switch startup configuration file.

Before you begin

Before beginning the procedures in this section, verify the following:

Table 17-4 Prerequisites for Cisco switch configuration	
✓	Task
	You have an RS-232 cable configured, as required by the switch vendor, to connect the computer’s serial port to the switch’s CONSOLE port.
	Ensure that Telnet is enabled on the operating system used as the interface to the switch.
	You have reviewed the specific vendor’s switch user guide, if necessary.
	HyperTerminal must be installed.

Tasks for configuring a Cisco switch

The following table lists the tasks for configuring the Cisco switches in an FTE network.

Table 17-5 Cisco switch configuration	
✓	Task
	Connect to the switch. See Section <i>Connecting locally to the switch</i> .
	Verify the switches have the IOS version qualified by Honeywell as listed in the SCN for your release. Note: If the version is not the same as that listed in the SCN, contact Honeywell Network Services for the procedure to upgrade the IOS.
	Configure the switch. See Section <i>Configuring switch interface options</i> .

	Load the switch configuration file. See Section <i>Loading the switch configuration file</i> .
--	--

Connecting locally to the switch

Use the following procedure to connect to the switch and start HyperTerminal.

Attention

Do not power up the switch until instructed to do so.

Table 17-6 Connecting locally to the switch

Step	Action
1	Connect the RS-232 cable to the Switch's CONSOLE port and the computer's serial port.
2	Click Start > All Programs > Honeywell > HyperTerminal .
3	From the Connection Description dialog box, type a name that describes the connection in the Name box and then click OK .
4	In the Icon box, click the icon for the connection, and then click OK .
5	From the Connect To dialog box, select the serial port being used by the computer in Connect Using box and then click OK .
6	From the Properties page configure the following Port Settings: • Bits per second: 9600 • Data Bits: 8 • Parity: NONE • Stop bits: 1 • Flow control: None
7	Click OK .
8	Power up the switch and go to the next procedure.

Configuring switch interface options

Use the following procedure to enable the configuration dialog and basic management setup in the switch. After configuring the switch options, use the rest of the procedure to setup the switch IP address, enable SNMP traps, and establish the SNMP RO community and the NTP time service. Establishing an IP address allows you to use Telnet and FTP sessions to save and restore configuration options.

Table 17-7 Connecting locally to the switch

Step	Action
TIP	All values to be entered by the user appear in bold . Press ENTER after entering each value.

Table 17-7 Connecting locally to the switch

Step	Action
1	When the following display appears, type all values that appear in bold. Supply your own values for the text that appears in <i>bold italic</i>. <pre> Would you like to enter the initial configuration dialog? [yes/no]: y At any point you may enter a question mark '?' for help. Use ctrl-c to abort configuration dialog at any prompt. Default settings are in square brackets '[]'. Basic management setup configures only enough connectivity for management of the system, extended setup will ask you to configure each interface on the system Would you like to enter basic management setup? [yes/no]: y Configuring global parameters: </pre>
2	The host name is unique for each switch. The following are used as the examples in the switch displays: Cisco_FTE4: example host name Cisco_FTE1: example enable secret FTE4: example virtual terminal password FTE4: example enable password Enter your own host name and password when asked to do so.
3	When the following display appears, type all values that appear in bold. Supply your own values for the text that appears in <i>bold italic</i>. <pre> Enter host name [Switch]: Cisco_FTE4 The enable secret is a password used to protect access to privileged EXEC and configuration modes. This password, after entered, becomes encrypted in the configuration. Enter enable secret: Cisco_FTE1 The enable password is used when you do not specify an enable secret password, with some older software versions, and some boot images. Enter enable password: FTE4 The virtual terminal password is used to protect access to the router over a network interface. Enter virtual terminal password: FTE4 Configure SNMP Network Management? [no]: N </pre>

Table 17-7 Connecting locally to the switch

Step	Action
4	The following is an abridged example of what displays after the configuration. Press the space bar to advance the display when it pauses. <pre> Current interface summary Any interface listed with OK? value "NO" does not have a valid configuration Interface IP-Address OK? Method Status Protocol Vlan1 unassigned NO unset up down FastEthernet0/1 unassigned YES unset down down FastEthernet0/2 unassigned YES unset down down FastEthernet0/3 unassigned YES unset down down FastEthernet0/48 unassigned YES unset down down GigabitEthernet0 unassigned YES unset down down GigabitEthernet0/2 unassigned YES unset down down </pre>
5	After the configuration display is complete, the switch dialog appears. Type all values that appear in bold. <pre> Enter interface name used to connect to the management network from the above interface summary: vlan1 Configuring interface Vlan1: Configure IP on this interface? [yes/no]: N Would you like to enable as a cluster command switch? [yes/no]: N </pre>

Table 17-7 Connecting locally to the switch

Step	Action
6	The following is an abridged example of what displays after the vlan1 configuration. Press the space bar to advance the display when it pauses. <pre> The following configuration command script was created: hostname Cisco_FTE4 enable secret 5 \$1\$gF.3\$3Aikt0lNtdjMLAdknUnht. enable password FTE4 line vty 0 15 password FTE4 no snmp-server ! ! interface Vlan1 shutdown no ip address ! interface FastEthernet0/1 no shutdown no ip address ! interface FastEthernet0/2 ! interface FastEthernet0/3 ! interface FastEthernet0/4 interface FastEthernet0/48 ! interface GigabitEthernet0/1 ! interface GigabitEthernet0/2 ! end </pre>
7	After the configuration display is complete, the following switch dialog appears. Type 2 and press Enter to save the switch configuration. <pre> [0] Go to the IOS command prompt without saving this config. [1] Return back to the setup without saving this config. [2] Save this configuration to nvram and exit. Enter your selection [2]: 2 </pre>

Table 17-7 Connecting locally to the switch

Step	Action
8	The following display appears. This is the end of the switch configuration dialog. Complete the rest of the procedure to setup IP addressing, SNMP traps and the NTP time service for the switch. <pre> Building configuration... [OK] 00:02:36: %LINK-5-CHANGED: Interface Vlan1, changed state to administratively down 00:02:37: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to down 00:02:38: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down Use the enabled mode 'configure' command to modify this configuration. </pre>
9	Use the enable command and the enable secret you previously established: Cisco_FTE1 is used in the following example. Type all values that appear in bold. Supply your own values for the text that appears in bold italic. <pre> Press RETURN to get started!<ENTER> Cisco_FTE4>enable Password:<i>Cisco_FTE1</i> </pre>
10	If VLAN101 is to be used, initialize VLAN 101 by performing these additional steps: • Type vlan101 • Type exit • Type exit Otherwise, go to the next step.
11	To enable Telnet and FTP, use one of the following commands: • To configure vlan1, type int vlan1, or • To configure vlan101, type int vlan101: Type all values that appear in bold. Supply your own values for the text that appears in bold italic. <pre> Cisco_FTE4#config t Enter configuration commands, one per line. End with CNTL/Z. Cisco_FTE4(config)#int vlan1 </pre>

Table 17-7 Connecting locally to the switch

Step	Action
12	The following is used for the IP address and subnet mask in the following switch displays: 10.1.4.253 255.255.255.0 Type all values that appear in bold. Supply your own values for the text that appears in bold italic. <pre>Cisco_FTE4(config-if)#ip address 10.1.4.253 255.255.255.0 Cisco_FTE4(config-if)#no shutdown Cisco_FTE4(config-if)#exit</pre>
Attention Skip steps 13-16, if you do not have an SNMP server.	
13	You need a user name and password for your FTP Server. The following are used as the examples in the switch displays: ps_user - example user name ps_user local – example password FTE4 – example virtual terminal password Type all values that appear in bold. Supply your own values for the text that appears in bold italic. <pre>Cisco_FTE4(config)#ip ftp username ps_user Cisco_FTE4(config)#ip ftp password ps_user local</pre>
Attention One of the nodes on the network must be set up as the FTP server. The FTP server requires a user name and a password that are registered in that machine with rights to allow FTP access. You can then archive and restore configurations using telnet and the FTP server.	
14	The switch generates SNMP traps when the switch reboots or has a link go up or go down. The switch must have a target IP address for the SNMP traps, which will be the IP address of the server that is running the System. Systems with redundant servers need to have both server IP addresses configured in the switches for SNMP. You should also establish a community name for the switch. The following are used as the examples in the switch displays: 10.1.4.15 - Server IP address FTE - Switch Community Name Type all values that appear in bold. Supply your own values for the text that appears in bold italic. <pre>Cisco_FTE4(config)#snmp-server enable traps snmp warmstart linkdown linkup coldstart Cisco_FTE4(config)#snmp-server host 10.1.4.15 FTE snmp</pre>

Table 17-7 Connecting locally to the switch

Step	Action
15	If your system has redundant servers, repeat the #snmp-server host 10.1.4.16 FTE snmp command using the redundant server IP address (10.1.4.16 – is used as an example for the redundant server IP address).
16	To enable SNMP reads of the switch statistics, type: Snmp-server community FTE RO Note: Public is the default community name. Attention The Windows SNTP service does not provide the proper protocol for NTP that the switch expects. For this reason, you must configure an NTP timeserver in order to synchronize time with other switches and network nodes. Examples of NTP Time Servers: * Router * Dedicated NTP server node * GPS based NTP server If the NTP server is outside the FTE subnet, you need to establish a default gateway.
17	The following are used as examples in the switch displays: 10.1.4.1- Default gateway IP address 192.168.100.1 - NTP Time Server IP address If the timeserver is within the FTE subnet, go to the next step. Otherwise, type all values that appear in bold. Supply your own values for the text that appears in bold italic. <pre>Cisco_FTE4(config)#ip default-gateway 10.1.4.1</pre>
18	Configure the NTP timeserver. Type all values that appear in bold. Supply your own values for the text that appears in bold italic. <pre>Cisco_FTE4(config)#ntp server 192.168.100.1</pre>
19	Type exit. Type write. The switch configuration is complete. <pre>Cisco_FTE4(config)#exit Cisco_FTE4# 00:06:03: %SYS-5-CONFIG I: Configured from console by console <ENTER></pre>
20	The switch option configuration is complete. You are now ready to download the appropriate switch configuration file.

Using VLAN101 switch configuration files

If your system requires you to disable VLAN1 in the switches, you will need to use the alternate switch configuration files that are preceded with V101_. For example, instead of using 4u.text, you would use the V101_4u.text file. See Table 17-3 for a

list of all switch configuration files. The files with interface options for the VLAN1 are replicated with each interface attached to VLAN101. If a VLAN number other than 101 is needed, use a text editor to modify the current V101 file and replace all occurrences of 101 with the alternate VLAN number.

Loading the switch configuration file

The following procedure uses Hyperterm's Xmodem file transfer utility to transfer the correct switch configuration file from the installation media to the switch. After downloading the switch configuration file, you will write the configuration back to the switch memory.

You must familiarize with the base configuration files and modify them as necessary before loading them.

Press Enter after typing each value.

TIP

If you are not familiar with Xmodem, read Hyperterm's help and try a practice transfer before initiating the transfer in the switch. Read steps 1 through 4 in the following procedure before you begin.

Table 17-8 Loading the switch configuration file

Step	Action
1	Review Table 17-3 to determine the most appropriate switch configuration file for your system.
2	Initiate the transfer in the switch using the copy command. Type all values that appear in bold . <pre>Cisco_FTE4#copy xmodem: system:running-config Destination filename [running-config]?<ENTER></pre>
3	Initiate the transfer in Hyperterm and choose the appropriate switch configuration file: * From the Hyperterm menu bar, select Transfer > Send File. * Select Browse and navigate to the Switch Configuration folder in the following location: - C:\Program Files\Honeywell\ FTEDriver \Switch Configuration, or - Media Drive:\Packages\FTE_Driver\Switch_Configuration_Files, or - Location you saved the files to * Select the correct switch configuration file for your particular system and click OPEN. * Select Xmodem under Protocol. * Click Send to start the file transfer.

Table 17-8 Loading the switch configuration file

Step	Action
4	If there is a problem during the transfer, an error message displays by the switch. If this happens, retype the following command and press ENTER: <pre>Cisco FTE4#copy xmodem: system:running-config</pre> Result: The following is an example of the display when the transfer is complete. <pre>16256 bytes copied in 46.396 secs (353 bytes/sec)</pre>
5	Write the basic switch configuration file and the switch configuration file you downloaded back to the switch memory by typing all values that appear in bold. <pre>Cisco_FTE4#write</pre>
6	Display the new switch configuration options to the screen by typing all values that appear in bold. <pre>Cisco_FTE4#sho run</pre>

Table 17-8 Loading the switch configuration file

Step	Action
7	The following abridged example of the switch display uses options based on the switch configuration file you previously selected: * 4 uplink ports * 4 autospeed ports for FTEB * Remaining ports at 100 MB Press the space bar to advance the display when it pauses. <pre> Building configuration... Current configuration : 15560 bytes ! version 12.1 no service pad service timestamps debug uptime service timestamps log uptime no service password-encryption service sequence-numbers ! hostname Cisco_FTE4 ! enable secret 5 \$1\$JIQ6\$IJ3nKv2o82zCJZihoHEK1/ enable password Cisco FTE1 ! wrr-queue bandwidth 1 2 3 4 ! class-map match-all cda_medium match access-group 104 class-map match-all cda_urgent match access-group 102 class-map match-all cda_high match access-group 103 class-map match-all cda_low match access-group 105 ! ! policy-map cda_policy class cda_urgent set ip dscp 56 class cda_high set ip dscp 46 ! ip subnet-zero ip ftp username ps_user ip ftp password ps user local no ip igmp snooping ! spanning-tree extend system-id ! ! interface FastEthernet0/1 no ip address duplex full speed 100 service-policy input cda_policy storm-control broadcast level 20.00 18.00 storm-control multicast level 20.00 18.00 </pre>

Table 17-8 Loading the switch configuration file

Step	Action
	<pre> storm-control unicast level 20.00 18.00 storm-control action trap ! interface FastEthernet0/48 switchport trunk allowed vlan 1,1001-1005 no ip address duplex full speed 100 service-policy input cda_policy storm-control broadcast level 20.00 18.00 storm-control multicast level 20.00 18.00 storm-control unicast level 20.00 18.00 storm-control action trap spanning-tree portfast ! interface GigabitEthernet0/1 no ip address service-policy input cda_policy storm-control broadcast level 5.00 4.50 storm-control multicast level 5.00 4.50 storm-control unicast level 5.00 4.50 storm-control action trap ! interface GigabitEthernet0/2 no ip address service-policy input cda_policy storm-control broadcast level 5.00 4.50 storm-control multicast level 5.00 4.50 storm-control unicast level 5.00 4.50 storm-control action trap ! interface Vlan1 ip address 10.1.4.254 255.255.255.0 no ip route-cache shutdown ! ip http server ! access-list 102 permit tcp any any eq 55554 access-list 102 permit tcp any any eq 55555 access-list 103 permit tcp any any eq 55550 access-list 103 permit tcp any any eq 55551 access-list 103 permit tcp any any eq 55553 access-list 103 permit tcp any any eq 55552 access-list 103 permit tcp any any eq 55556 </pre>

Table 17-8 Loading the switch configuration file

Step	Action
	<pre> access-list 104 permit tcp any any eq 55557 access-list 104 permit tcp any any eq 55558 access-list 104 permit tcp any any eq 55559 access-list 104 permit udp any any eq 12321 access-list 104 permit tcp any any eq 55560 access-list 105 permit tcp any any eq 55560 access-list 105 permit udp any any eq 55560 access-list 105 permit tcp any any eq 55559 access-list 105 permit udp any any eq 12321 access-list 105 permit tcp any any eq 55556 access-list 105 permit tcp any any eq 55557 access-list 105 permit tcp any any eq 55558 ! line con 0 exec-timeout 0 0 line vty 0 4 password FTE1 login line vty 5 15 password FTE1 login ! end Cisco FTE4# </pre>
8	This is the end of the switch configuration. If you would like to archive the configuration file for future use, see Section 17.3 <i>Saving and Modifying Cisco Switch Configuration Files</i> .

17.3 Saving and Modifying Cisco Switch Configuration Files

Use the procedures in this section to save, modify and restore the switch configuration files. Following are the circumstances in which you may need to perform these tasks:

- A switch fails and you need to reload the switch configuration file.
- You add a new node type to the network
- You want to use an existing configuration file on another switch.

Downloading the switch configuration file (optional)

Use the following procedure to save a file containing the switch options you configured. This makes it easier to reconfigure the switch in case of a switch failure. Press Enter after entering each value.

Table 17-9 Downloading the switch configuration file

Step	Action
1	Open a telnet session from the command window on the FTE server node.
2	Click Start > Run and type cmd in the Run dialog box.

Table 17-9 Downloading the switch configuration file

Step	Action
3	At the command prompt type telnet followed by the IP address set in the switch configuration. 10.1.4.253 is used in the following example. <pre>cmd>telnet 10.1.4.253</pre>
4	If the switch connection is successful, you are asked to enter a password. Type the virtual terminal password you previously configured for the switch and press Enter. FTE4 is used in the following example. <pre>User Access Verification Password:FTE4</pre>
5	The enable command and the enable secret you previously configured allow you to access the switch configuration file in order to copy it: Cisco_FTE1 is used in the following example. Type all values that appear in bold. Supply your own values for the text that appears in bold italic. <pre>Cisco_FTE4#enable Password:Cisco_FTE1</pre>
6	Use the copy flash command followed by the name of the switch configuration file and ftp command to copy the switch configuration file from flash memory. Type all values that appear in bold. <pre>Cisco_FTE4#copy flash:config.text ftp:</pre>
7	Enter IP address of the FTP server to copy the switch configuration file from the switch memory to the FTP server node. 10.1.4.15 is used in the following example. Supply your own values for the text that appears in bold italic. <pre>Address or name of remote host []?10.1.4.15 Destination filename [config.text]?<ENTER> Writing config.text !!!! 15197 bytes copied in 4.240 secs (3799 bytes/sec) Cisco_FTE4#</pre>
8	The switch configuration file is saved in the inetpub\ftproot directory on the ftp server. Rename the config.text file to the switch host name. This allows you to download all your switch configuration files to this location. TIP If several switches are configured at once, the ftp archiving can be done at the same time.

Enabling Telnet on your system

The following is the procedure to enable Telnet on your system.

Table 17-10 Enabling Telnet on your system

Step	Action
1	Choose Start > Settings > Control Panel . The Control Panel window appears.
2	Click Classic View on the left pane, and choose Programs and Features . Or, Click Control Panel Home on the left pane, and choose Programs > Programs and Features .
3	Click Turn Windows features on or off .
4	Click Continue . A Windows Features dialog box appears.
5	Check the following options. • Telnet Client • Telnet Server
6	Click OK . The Telnet feature gets configured on your system.

Appendix – UCLM: CCP AND UCIM Redesigned

CCP (51402426) and UCIM (51402436) boards have been redesigned and are now known as UCLM-51306409. UCLM is a combination of CCP and UCIM.

Ordering Information

To order the boards use the Honeywell part numbers as given below:

Part Number	Board Name
51306409-100	CCP
51306409-200	UCIM

Switch Settings of UCLM board

Figure 1 illustrates the redesigned CCP and UCIM settings.

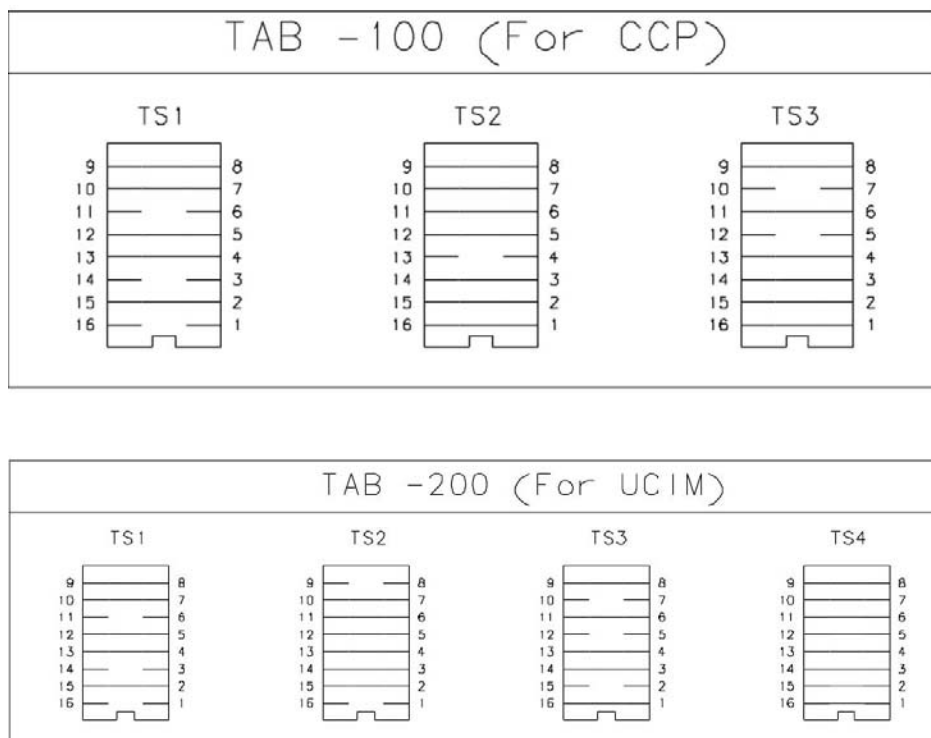


Figure 1 — Switch settings of UCLM board

There is no change in the procedure to load AM personality on UCLM. Procedure remains the same as it was in CCP and UCIM.

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